

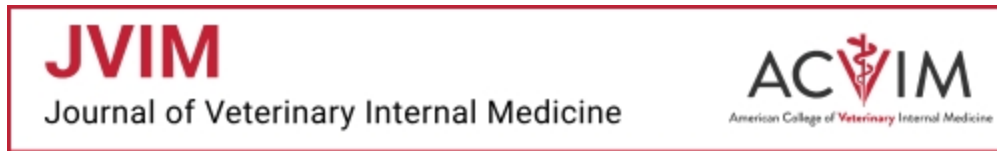
卷六·共病、生活质量与照护

老猫整体管理：老年猫筛查能查出什么、心血管与肾的共病、生活质量怎么评、照护者负担、居家监测新技术。决定'值不值得治、怎么治'的那一卷。

#	文献（中文标题 / 原文标题 / 要点）	来源	页
1	549只老年猫健康筛查的疾病患病率 Prevalence of disease in 549 older cats undergoing health screening at two United Kingdom-based veterinary clinics. 背景 随着年龄增长，某些猫的疾病风险增加，包括慢性肾病(CKD)和甲状腺功能亢进，但照顾者可能将临床症状忽视为与年龄相关的变化。假设/目标 量化老年猫疾病及其他主要临床发现的患病率。动物 数据来自在英国两家初级护理兽医诊所接受常规健康筛查的猫（≥8岁），时间跨度为52个月（2018年10月2日至2023年2月14日）。方法 在筛查过程中，进行了病史收集、体格检查以及血浆生物化学检查，有的还包括血浆总甲状腺素浓度测定。在可能的情况下，进行了收缩压（SBP）测量（97.1%的病例；多普勒法）和尿液分析（41.3%...	Journal of Veterinary Internal Med 2026	32
2	猫共病心血管疾病与肾脏疾病 Feline comorbidities: cardiovascular and kidney diseases. 实际意义：心血管和肾脏疾病在猫中非常常见，且经常同时存在，尤其是在高龄猫中。心脏和肾脏之间的多种相互作用途径可能导致在一个或两个系统发生疾病时功能逐渐下降。在猫中发现心脏或肾脏疾病，应促使临床医生考虑该疾病及其治疗可能如何影响另一个器官的功能，以及共存的另一个器官的原发性疾病可能如何影响病例管理和患者预后。临床挑战：在充血性心力衰竭、急性肾损伤或慢性肾病的管理中，一个常见的目标是维持最佳的体液平衡；然而，针对一个系统的许多治疗方法往往与另一个系统的最佳利益相冲突。例如，肾病的治疗通常包括肠外补液治疗，目的是维持平均...	Journal of Feline Medicine and Sur 2026	51
3	猫CKD健康相关生活质量评估 Evaluation of health-related quality of life in cats with chronic kidney disease 目标 本研究的目的是评估慢性肾病（CKD）猫的健康相关生活质量（HRQoL），并将这些结果与临床数据整合，以识别与HRQoL下降相关的因素。方法 照顾者完成了一份经过验证的HRQoL工具（VetMetrica），并收集了临床数据（病史、实验室检查、体格检查结果等）。使用非参数分析根据需要进行比较各组的活力、舒适度和情绪健康（EWB）评分。结果 收集了24只健康猫和68只CKD猫的数据。患有慢性肾病（CKD）的猫其活力、中度舒适感和情感幸福感（EWB）中位数评分显著低于健康猫（活力：41.7，范围4.3–64；舒适感...	Journal of Feline Medicine and Sur 2025	24
4	照护者视角下的猫CKD_葡萄牙问卷研究 Caregivers’ perspectives on feline chronic kidney disease in Portugal: a questionnaire-based study 目的 本研究旨在评估照顾者对猫慢性肾病（CKD）的看法。方法 邀请在葡萄牙居住并照顾患有CKD猫的人参与在线问卷调查。结果 共收集了405份有效问卷，其中大多数参与者（n = 309，76.3%）是首次处理猫CKD。国际肾脏兴趣学会（IRIS）CKD分期向260名（64.2%）受访者进行了说明，但仅有188名（46.4%）确认在诊断时进行了血压评估。387名（95.6%）受访者接收到肾脏饮食的建议，但仅有341名（84.2%）实际给予，其中139名（40.8%）最终每天至少提供一些...	Journal of Feline Medicine and Sur 2025	30
5	智能猫砂盆监测提升猫CKD检出 Enhancing Detection of Feline Chronic Kidney Disease Through Smart Litter Box Monitoring 慢性肾病（CKD）（CKD）在猫中很常见，并被认为是猫普遍死亡的主要原因之一，但猫的照顾者很难发现，尤其是在早期阶段。本回顾性研究旨在通过智能猫砂盆监测技术来增强猫照顾者对CKD的检测，该技术可追踪猫的排泄行为。我们建立了CKD猫和无已知疾病猫的队列，并分析了监测器捕捉的行为特征。混合效应模型识别出显著差异，然后将这些差异整合到机器学习框架中进行预测建模。该模型在使用交叉验证进行训练时实现了92.7%的加权F1分数，在验证时为89.9%，展示了对慢性肾病（CKD）预测的高精度。主要发现包括猫患CKD的行为特征独特，...	Animals : an Open Access Journal f 2026	31
6	猫CKD进展与血清胰脂肪酶浓度关联 Association between chronic kidney disease progression and serum feline pancreatic lipase concentrations in cats. 重要性 血清猫胰脂肪酶（fPL）用于诊断胰腺炎，但解释具有挑战性，因为fPL可能受到非胰腺疾病的影响，包括肾功能下降。目的 评估慢性肾病（CKD）分期与猫血清fPL的关联，并描述肾功能生物标志物与fPL的关系。方法 本回顾性研究评估了2024年2月至2025年3月在一家二级转诊兽医医院就诊猫的病历。纳入条件要求在同一天进行全血细胞计数、血清生化、血气分析、尿液分析、血清fPL及腹部超声检查。CKD分期遵循国际肾脏兴趣学会指南，使用血清肌酐、尿比重（USG）（USG）和超声检查进行评估。采用Kruska...	Journal of veterinary science 2026	23

As a library, NLM provides access to scientific literature. Inclusion in an NLM database does not imply endorsement of, or agreement with, the contents by NLM or the National Institutes of Health.

Learn more: [PMC Disclaimer](#) | [PMC Copyright Notice](#)



J Vet Intern Med. 2026 Aug 6;40(4):aalag157. doi: [10.1093/jvimsj/aalag157](https://doi.org/10.1093/jvimsj/aalag157) [↗](#)

Prevalence of disease in 549 older cats undergoing health screening at two United Kingdom-based veterinary clinics

[Jason P Bestwick](#)^{1,✉}, [Jonathan Elliott](#)², [Isabel R Orriss](#)³, [Nicola Lötter](#)⁴, [Rebecca F Geddes](#)⁵

[Author information](#) [Article notes](#) [Copyright and License information](#)

PMCID: PMC13446506 PMID: [42561144](#)

Abstract

Background

Risk of some diseases of cats increases with age, including chronic kidney disease (CKD) and hyperthyroidism, but clinical signs can be dismissed by caregivers as age-related changes.

Hypothesis/Objectives

Quantify the prevalence of disease and other major clinical findings in older cats.

Animals

Data from cats (≥8 years old) that underwent routine health screening at 2 United Kingdom (UK)-based primary care veterinary clinics over a 52-month period (October 2, 2018 to February 14,

2023) were analyzed.

Methods

At screening visits, history, physical examination, and blood tests for plasma biochemistry with or without plasma total thyroxine concentration were performed. When possible, systolic blood pressure measurement (97.1% of cases; Doppler method) and urinalysis (41.3% of cases; 98.2% by cystocentesis; and bacterial culture if indicated based on sediment examination) were performed.

Results

A total of 549 cats were screened with a median age of 13.5 (IQR: 11.3-15.8) years old. Diagnoses of hyperthyroidism, azotemic CKD, and hypertension were made in 89/464 (19.2%; 95% CI, 15.9%-23.0%), 62/542 (11.4%; 95% CI, 9.0%-14.4%), and 37/524 (7.1%; 95% CI, 5.2%-9.6%) of cats, respectively. Severe dental disease was present in 32/522 (6.1%; 95% CI, 4.4%-8.5%) and bacteriuria was identified in 22/227 (9.7%; 95% CI, 6.5%-14.2%) of urine samples. Other diagnoses included suspected neoplasia (8 cats), diabetes mellitus (8 cats), and suspected primary gastrointestinal disease (4 cats).

Conclusions and clinical importance

Routine medical health screening in older cats is useful in detecting diseases, many of which would benefit from prompt veterinary intervention. These results can assist veterinarians in explaining the value of health screening to cat caregivers.

Keywords: aging, chronic renal failure, feline, primary care practice, urinary tract

Introduction

The American Association of Feline Practitioners, based on expert panelist opinion, currently recommends examination of healthy cats aged 10-15 years biannually and every 4 months for cats > 15 years old, with laboratory diagnostic testing (including hematology, serum biochemistry, and urinalysis) and measurement of systolic blood pressure (SBP) advised at least annually and at increased frequency with advancing age.¹ The aim of health screening is to facilitate the detection of common diseases of older cats, including chronic kidney disease (CKD), hyperthyroidism, and hypertension, which appear to be of increased prevalence with advancing age.²⁻⁶ Other conditions

(eg, neoplasia, dental disease, and cardiomyopathy) may also be diagnosed through the routine diagnostic procedures carried out as part of health screening^{[1,7-9](#)}

However, data around the health screening recommendations provided by veterinarians, their uptake by cat caregivers, suitable intervals between screening events, and the utility of the diagnostic tests performed is challenging to obtain and likely variable among both individual veterinarians and practices. In the United Kingdom (UK), individual cats are less likely to be seen by veterinary practices than dogs.^{[10](#)} This difference could be a reflection of dogs experiencing a higher disease burden than cats, but could also indicate that cats are less likely to display clinical signs, or more likely to have them dismissed or unrecognized by caregivers. Supporting the latter hypothesis, in 206 middle-aged (7-10 years old) cats assessed by a university-based aging cat research clinic, although caregivers did report physical (53%), behavioral (47%), eating (41%), and activity changes (40%), many more cats (88%) were found to have evidence of at least 1 disease or abnormality.^{[7](#)} In a more recent study investigating the utility of repeated health screening in apparently healthy mature (7-10 years old) and senior (>10 years old) cats over a 2-year period in another university-based clinic, 21% of cats were not considered healthy at baseline and a further 28% of mature and 54% of senior cats initially considered healthy developed a new disease during follow-up.^{[9](#)}

Our aim was to report disease prevalence and other findings that warrant veterinary intervention in a large number of older cats presented for prospective health screening in a primary care practice (PCP) setting in the UK, using well-defined diagnostic criteria.

Materials and methods

Study design

Cats were health screened as part of a longitudinal observational project examining the health of aging cats (Royal Veterinary College Ethics and Welfare Committee [RVC-EWC] approved—URN20131258E/URN202322253). Data are from cats assessed for inclusion in a prospective diet trial (RVC-EWC approved—URN 201818453/URN 202321963) during a 52-month period (October 2, 2018 to February 14, 2023).

Cats were assessed by the RVC Aging Cat Clinic (London, UK) at 2 PCPs: Beaumont Sainsbury Animal Hospital (university-owned) and Bow People's Dispensary for Sick Animals Pet Hospital (charity-associated).

Cats newly presented during the study period and cats previously examined, lacking overt disease, and returning for re-assessment (qualifying every 6 months) were eligible for inclusion. Before screening, informed caregiver consent was obtained ([Figure S1](#); [Supplementary Appendix A](#)).

Study population and procedures

Cats were ≥ 8 years old and presented for routine health screening, and were excluded if they had prior diagnoses requiring long-term monitoring or management.

During screening, a clinical history was taken and a physical examination was performed ([Supplementary Appendix A](#)). Dental scoring was performed using a standardized subjective individual grading system for tartar and gingivitis: 0 = none, 1 = minimal, 2 = mild, 3 = moderate, and 4 = severe. Blood samples were taken for biochemistry (IDEXX Laboratories, Wetherby, UK), PCV, and plasma total protein concentrations.

Plasma total thyroxine (TT4) concentration (Siemens IMMULITE TT4 assay, Siemens Healthineers, Llanberis, Gwynedd, North Wales, UK) was measured if not evaluated in the previous 12 months or if there was clinical suspicion of hyperthyroidism. If the TT4 concentration was in the upper portion of the reference interval (RI) of 3.1-4.3 $\mu\text{g/dL}$ (40-55 nmol/L), clinicians advised repeated measurement (typically after 8 weeks) or immediate measurement of plasma thyroid-stimulating hormone using the canine assay (cTSH; Siemens IMMULITE Canine TSH assay, Siemens Healthineers, Llanberis, Gwynedd, North Wales, UK) based upon clinical suspicion.

Systolic blood pressure (SBP) was measured using the Doppler method ([Supplementary Appendix A](#); Doppler Flow Detector Model 811-B, Parks Medical Electronics, Aloha, OR, USA). If SBP was ≥ 160 mmHg, retinal examination (ClearView 1 or ClearView 2, Optibrand, Fort Collins, CO, USA) was performed to assess for hypertensive chorioretinopathy.¹¹ If present, antihypertensives were prescribed. If not present, it was recommended that cats be returned for repeated SBP measurement to assess for persistence. Urine was collected for urinalysis with bacterial culture if microscopic bacteriuria was present.

Criteria used to diagnose diseases and describe clinical findings are detailed in [Table 1](#).

Table 1.

Criteria used to define the diagnosis of a disease or finding in older cats undergoing health screening.

Diagnosis/Finding	Criteria
Azotemic CKD	Plasma creatinine concentration > 2 mg/dL (>177 µmol/L) with concurrent USG < 1.035; or, plasma creatinine > 2 mg/dL (>177 µmol/L) at the screening and consecutive follow-up visit recommended for approximately 2-4 weeks later.
Hyperthyroidism	Plasma TT4 > 4.3 µg/dL (>55 nmol/L); or, TT4 3.1-4.3 µg/dL (40-55 nmol/L) and cTSH < 0.03 ng/mL.
Hypertension	Mean SBP ≥ 160 mmHg with evidence of TOD (hypertensive chorioretinopathy) or repeatable mean SBP ≥ 160 mmHg on re-examination recommended for 1-2 weeks later.
Other diagnoses/findings:	
Increased ALT activity	Plasma ALT activity ≥ 2× RI (≥120 U/L)
Bacteriuria	Bacteria observed on urine sediment examination and positive urine culture
Dental disease	
Presence of tartar	Tartar score > 0
Presence of gingivitis	Gingivitis score > 0
Severe dental disease	Tartar or gingivitis score = 4
Suspected neoplasia	Palpable or visible mass lesion
Diabetes mellitus	Hyperglycemia and glucosuria with or without increased serum fructosamine concentration
Suspected primary GI disease	Chronic vomiting or diarrhea without an identified cause

[Open in a new tab](#)

Abbreviations: ALT, alanine transferase; CKD, chronic kidney disease; cTSH, canine thyroid-stimulating hormone; GI, gastrointestinal; RI, reference interval; SBP, systolic blood pressure; TOD, target organ damage; TT4, total thyroxine; USG, urine specific gravity.

Statistical analyses

Data were compiled and descriptive statistics performed using Microsoft Excel (Version 16.101.1). The 95% CI for disease prevalence was calculated using Wilson's method.^{[12](#)}

Results

Overall study population

A total of 549 eligible cats were health screened during the 52-month study period. The median age was 13.5 years (IQR, 11.3-15.8). Two hundred and sixty-eight (268/549; 48.8%) of the cats were male (2 intact, 266 neutered) and 281/549 (51.2%) were female (1 intact, 280 spayed). Seventy-eight percent (428/549) of the cats were domestic shorthair, 68/549 (12.4%) were domestic longhair, 43/549 (7.8%) were purebred, 7/549 (1.3%) were purebred crosses, and 3/549 (0.5%) did not have their breed recorded. Median body weight was 4.0 kg (IQR, 3.3-4.7).

For the majority of cats (468/549; 85.2%), the health screen was their first visit to the clinic. The remaining cats (81/549; 14.8%) had been examined ≥ 6 months previously and deemed healthy. Health screening was performed by 10 clinicians and a single veterinary nurse during the study period.

Health screening findings

Hyperthyroidism

At the time of health screening, hyperthyroidism was diagnosed in 19.2% of cats (89/464; [Figure 1A](#)). This estimate does not include cases without a plasma TT4 measurement at the time of health screening ($n = 64$). It does include cats diagnosed with hyperthyroidism on the basis of a plasma TT4 concentration $> 4.3 \mu\text{g/dL}$ ($>55 \text{ nmol/L}$; $n = 87$) and cats with hyperthyroidism excluded (plasma TT4 $< 3.1 \mu\text{g/dL}$ [$<40 \text{ nmol/L}$]; $n = 373$). For cats with plasma TT4 concentrations between

3.1 and 4.3 µg/dL (40-55 nmol/L; $n = 25$), it includes those that had cTSH measured ($n = 3$), resulting in either a diagnosis ($n = 1$) or non-diagnosis of hyperthyroidism ($n = 2$), as well as a single cat considered hyperthyroid outside of the diagnostic criteria specified in [Table 1](#). Regarding this cat, medical treatment commenced (PO thiamazole) without measurement of plasma cTSH, on the basis of compatible clinical signs and suspected sick euthyroid syndrome secondary to concurrent azotemic CKD. Treatment response was compatible with the diagnosis.

Figure 1.



[Open in a new tab](#)

Flow diagrams showing the routes to diagnosis (or non-diagnosis) in 549 health screened older cats for: (A) hyperthyroidism, (B) azotemic CKD, and (C) hypertension. Abbreviations: CKD = chronic kidney disease; cTSH = canine thyroid-stimulating hormone; LTFU = lost to follow-up; SBP = systolic blood pressure; TT4 = total thyroxine.

Of the remaining 21 cats with a plasma TT4 concentration between 3.1 and 4.3 µg/dL (40-55 nmol/L) and excluded from the prevalence estimate, 8 were lost to follow-up (ie, did not have cTSH or follow-up TT4 measured) and 13 had a later TT4 measurement available for review (Figure 1A): 6 had plasma TT4 concentrations ≤ 4.3 µg/dL (≤55 nmol/L; median follow-up

102 days; IQR, 19-215 days) and 7 received a delayed diagnosis of hyperthyroidism (TT4 > 4.3 µg/dL [>55 nmol/L]; median follow-up, 161 days; IQR, 98-186 days). In addition, the 2 cats with a plasma TT4 concentration between 3.1 and 4.3 µg/dL (40-55 nmol/L) and cTSH $\geq$ 0.03 ng/mL at the time of health screening also had later plasma TT4 concentrations available for review: one continued to have a TT4 concentration $\leq$ 4.3 µg/dL (≤ 55 nmol/L; 440 days after screening) and the other was diagnosed with hyperthyroidism (TT4 > 4.3 µg/dL [>55 nmol/L]; 196 days after screening).

Azotemic CKD

All cats had plasma creatinine concentration measured at the time of health screening; 227/549 (41.3%) cats had urinalysis simultaneously performed.

Overall, 62/542 (11.4%) cats were diagnosed with azotemic CKD ([Figure 1B](#)): 43/62 (69.4%) based on a concurrent plasma creatinine concentration > 2 mg/dL (>177 µmol/L) and USG < 1.035 and 19/62 (30.6%) based on plasma creatinine concentration > 2 mg/dL (>177 µmol/L) on 2 consecutive visits at least 2 weeks apart. An additional 16 cats were azotemic without a concurrent urine sample at the time of health screening, but were not diagnosed with azotemic CKD because 9/16 (56.3%) were non-azotemic at a subsequent visit and 7/16 (43.8%) were lost to follow-up. These latter cats were excluded from the overall azotemic CKD estimate because of missing follow-up data required to ascertain disease status.

Plasma symmetric dimethylarginine (SDMA) concentrations were available for 62/62 and 486/487 cats diagnosed with and without azotemic CKD, respectively. For cats diagnosed with azotemic CKD, SDMA was above the RI (>14 µg/dL) in 46/62 (74.2%) cats and ≥ 18 µg/dL in 30/62 (48.4%) cats. For cats not diagnosed with azotemic CKD, SDMA was above the RI (>14 µg/dL) in 57/486 (11.7%) cats and ≥ 18 µg/dL in 13/486 (2.7%) cats.

Hypertension

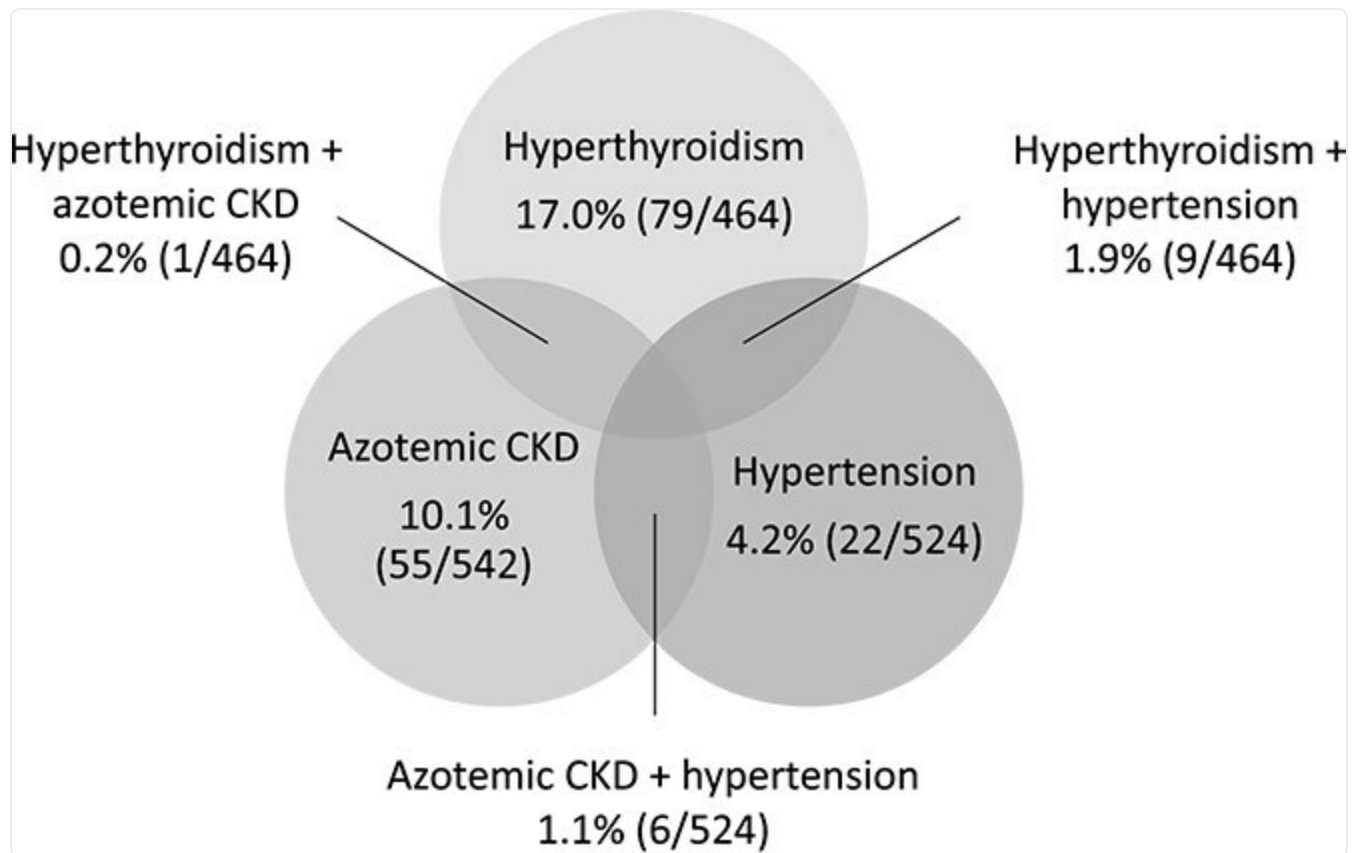
In total, 533/549 (97.1%) cats had SBP measured at the time of their health screening visit and 37/524 (7.1%) were diagnosed as hypertensive ([Figure 1C](#)). One (1/37) cat was diagnosed as hypertensive outside of the diagnostic criteria specified in [Table 1](#), with a mean SBP < 160 mmHg (148 mmHg), because of evidence of retinal edema (bullae) and tortuous retinal vessels in both eyes.

Of the 533 cats that had SBP measured, 32 cats had SBP of ≥ 160 mmHg but were not diagnosed with or treated for hypertension. Short-term follow-up was not available for 9 of these cats but, for all 9 cats, retinal examination was normal at the time of health screening. These 9 cats were excluded from the overall hypertension estimate because of missing follow-up data required to ascertain disease status. For the remaining 23 cats, mean SBP was < 160 mmHg at follow-up and the initial mean SBP of ≥ 160 mmHg was suspected to be a result of situational hypertension on the basis of lack of evidence for hypertensive chorioretinopathy (some had vessel tortuosity suspected but as a lone finding) and inconsistency of the finding on repeated examination.

Concurrent hyperthyroidism, azotemic CKD, and hypertension

Sixteen cats were diagnosed with more than 1 of the 3 most common medical conditions (hyperthyroidism, azotemic CKD, and hypertension; [Figure 2](#)).

Figure 2.



[Open in a new tab](#)

Venn diagram showing concurrent prevalence of the 3 most commonly diagnosed medical conditions of 549 older cats undergoing health screening. Abbreviation: CKD = chronic kidney disease.

Bacteriuria

Of the 549 cats, 227/549 (41.3%) had urine collected at the time of screening: 4/227 voided and 223/227 collected by palpation-guided cystocentesis. Bacteria were observed on sediment examination in 37/227 (16.3%) and urine samples therefore were submitted for bacterial culture and antibiotic sensitivity testing. Bacterial cultures were positive in 23 of 37 (62.2%) but one positive result was considered a contaminant based on the bacteria cultured (*Psychrobacter faecalis* and *Bacillus* spp.) and a voided sample being utilized ([Table 2](#)).

Table 2.

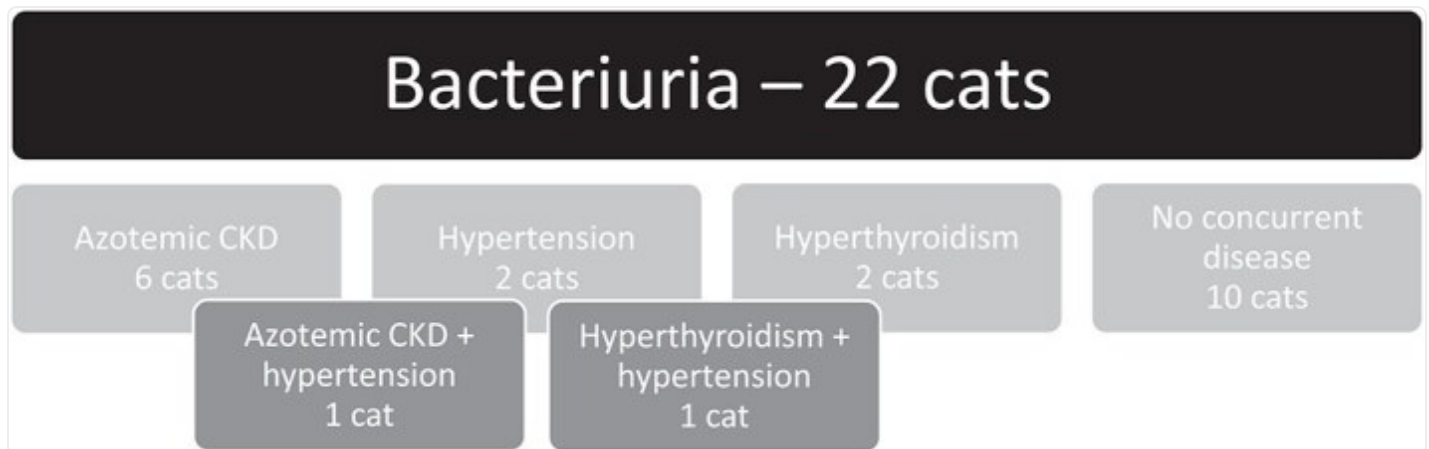
Frequency of bacterial isolates cultured from urine samples obtained from health screened older cats. All samples were acquired by palpation-guided cystocentesis. One positive culture is excluded because of suspected contamination (voided sample; *Psychrobacter faecalis* and *Bacillus* spp. cultured).

Bacteria cultured	Number of cats (n/22)
<i>Escherichia coli</i> (<i>E coli</i>) only	7
<i>Enterococcus faecalis</i> (<i>E faecalis</i>) only	7
<i>E coli</i> and <i>E faecalis</i>	2
<i>E coli</i> , <i>E faecalis</i> , and <i>Hafnia</i> spp.	1
<i>E coli</i> and <i>Klebsiella oxytoca</i>	1
<i>Staphylococcus</i>	
spp.	2
<i>pseudintermedius</i>	1
<i>felis</i>	1

[Open in a new tab](#)

Of the 22 cats (3 male neutered, 19 female spayed) with positive urine bacterial cultures considered not to be contaminants, *Escherichia coli* was the most common bacterial isolate either as the sole agent or in combination with other bacteria (11/22; 50.0%) and CKD was the most common concurrent disease diagnosed (7/22; 31.8%; [Figure 3](#)).

Figure 3.



[Open in a new tab](#)

Diagram showing concurrent disease diagnoses of the 22 cats with bacteriuria.
Abbreviation: CKD = chronic kidney disease.

Other diagnoses and findings

Diabetes mellitus was diagnosed in 8/549 (1.5%) cats: 5 based on hyperglycemia and glucosuria alone and 3 based on hyperglycemia, glucosuria, and a compatible serum fructosamine concentration (at the clinician's discretion). None of these cats had azotemic CKD, hyperthyroidism, hypertension, or bacteriuria (urine samples acquired by cystocentesis from 5/8 cats; 2/5 submitted for bacterial culture, both negative) concurrently diagnosed.

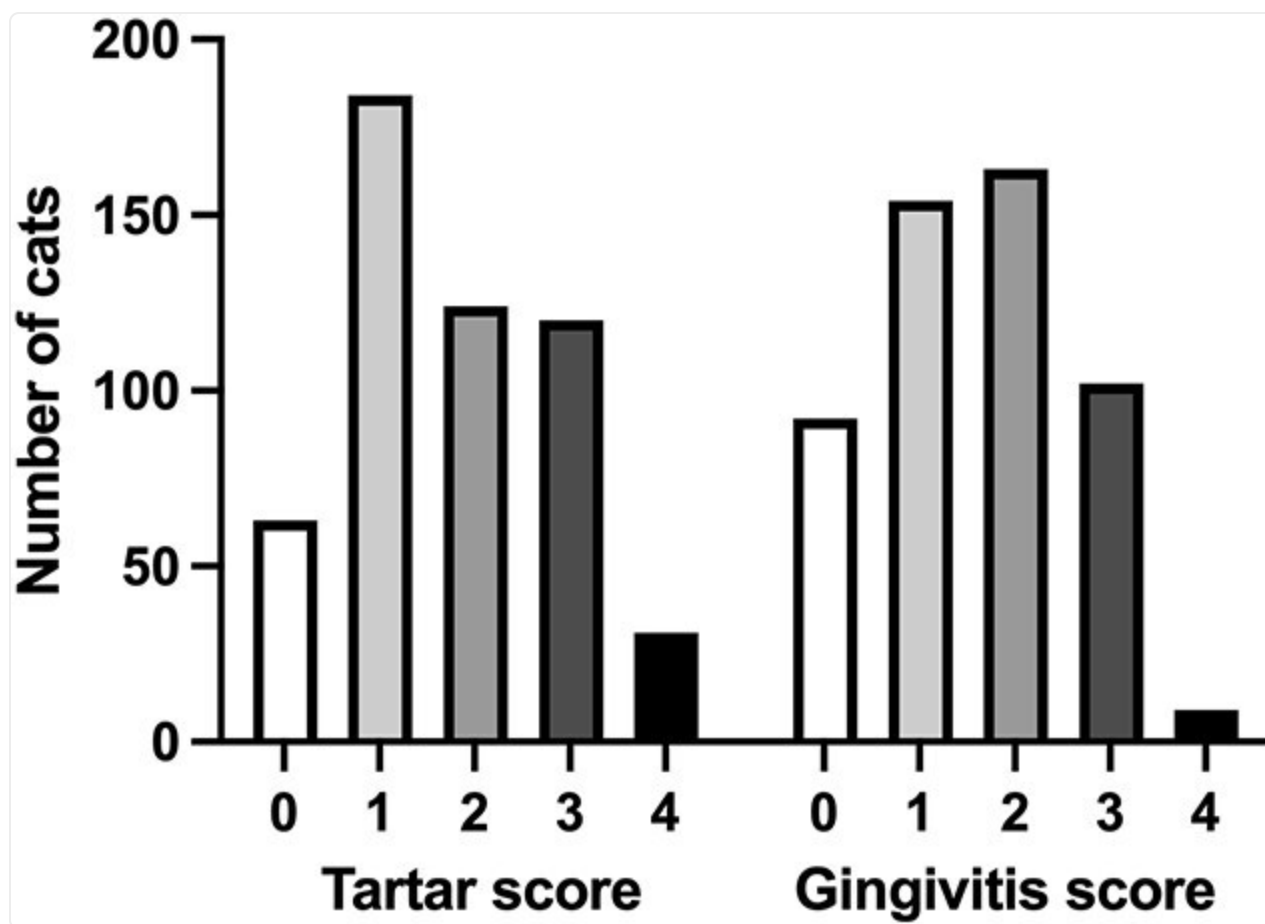
Neoplasia was suspected in 8/549 (1.5%) cats based on physical examination: 5 abdominal masses of uncertain origin, 1 bilateral renal enlargement (lymphoma suspected), 1 oral mass, and 1 facial mass. Of the 5 cats with abdominal masses of uncertain origin, one was also concurrently diagnosed as hypertensive and another as hyperthyroid.

Four cats (4/549; 0.7%) were suspected to have primary gastrointestinal (GI) disease on the basis of caregiver-reported chronic vomiting or diarrhea or both in the absence of an overt extra-GI cause on investigations performed for the health screen.

An increase in alanine aminotransferase (ALT) activity $\geq 2\times$ the RI (≥ 120 U/L) was found in 82/549 (14.9%) of cats. Of these, the increase was attributable to hyperthyroidism in 53/82 (64.6%) cats and diabetes mellitus in 4/82 (4.9%) cats. The cause of the ALT increase was unclear in the remaining 25 cats (25/82; 30.5%).

Five-hundred and twenty-two of 549 cats had a tartar score, and 520/549 cats had a gingivitis score, available for review ([Figure 4](#)). Of these, 459/522 (87.9%) and 428/520 (82.3%) had some degree of tartar and gingivitis (tartar or gingivitis score > 0), respectively, and among them 32/522 (6.1%) had severe tartar or gingivitis or both (tartar or gingivitis score = 4).

Figure 4.



[Open in a new tab](#)

A bar chart showing the frequency with which each tartar and gingivitis score (0 = no tartar or gingivitis, 1 = minimal tartar or gingivitis, 2 = mild tartar or gingivitis, 3 = moderate tartar or gingivitis, and 4 = severe tartar or gingivitis) was assigned in 549 health screened older cats.

One cat was found to be moderately anemic (PCV 17%) without an identifiable cause. Further investigations were advised but follow-up was not available.

Prevalence of diagnoses and findings

Derived from the results above, [Table 3](#) shows the calculated prevalence and 95% CI for major diseases and clinical findings from the 549 cats undergoing health screening during a 52-month period (October 2, 2018 to February 14, 2023).

Table 3.

Prevalence and 95% CIs for major diseases and clinical findings from the 549 older cats undergoing health screening during a 52-month period (October 2, 2018 to February 14, 2023).

Diagnosis/Finding	No. cats diagnosed	Prevalence (%) [95% CI]
Hyperthyroidism	89/464 ^a	19.2 [15.9-23.0]
Azotemic CKD	62/542 ^b	11.4 [9.0-14.4]
Hypertension	37/524 ^c	7.1 [5.2-9.6]
Other diagnoses/findings:		
Increased ALT activity $\geq 2 \times$ RI	82/549 ^d	14.9 [12.2-18.2]
Bacteriuria	22/227 ^e	9.7 [6.5-14.2]
Dental disease		
Tartar	459/522 ^f	87.9 [84.9-90.5]
Gingivitis	428/520 ^f	82.3 [78.8-85.3]
Severe dental disease	32/522 ^f	6.1 [4.4-8.5]
Suspected neoplasia	8/549 ^g	1.5 [0.7-2.8]
Diabetes mellitus	8/549 ^h	1.5 [0.7-2.8]
Suspected primary GI disease	4/549 ⁱ	0.7 [0.3-1.9]
Moderate or severe anemia of unknown cause	1/549 ^j	0.2 [0.0-1.0]

[Open in a new tab](#)

^aPlasma TT4 measured in 485/549 cats. Twenty-one cats with a plasma TT4 concentration between 3.1 and 4.3 $\mu\text{g/dL}$ (40-55 nmol/L) without concurrent measurement of cTSH were either lost to follow-up ($n = 8$) or had delayed diagnosis ($n = 7$) or non-diagnosis of hyperthyroidism ($n = 6$) and were excluded from the prevalence estimate.

^bPlasma creatinine and USG measured in 549/549 and 227/549 cats, respectively. Seven cats that were azotemic but without concurrent urinalysis and lacking follow-up were excluded from the prevalence estimate.

^cSBP measured in 533/549 cats. Nine cats with a mean SBP ≥ 160 mmHg at health screening had an absence of hypertensive chorioretinopathy but lacked follow-up blood pressure measurements, and were excluded from the prevalence estimate.

^dPlasma ALT measured in 549/549 cats.

^eUrinalysis performed in 227/549 cats; bacteria observed on sediment examination in 37/227 samples and sent for bacterial culture.

^fTartar and gingivitis scores recorded for 522/549 and 520/549 cats, respectively.

^gPhysical examination performed in 549/549 cats.

^hGlucosuria assessed for in all cats with an available urine sample (227/549); blood glucose measured in 344/549 cats, including all cats without an available urinalysis.

ⁱClinical history taken, physical examination, and plasma biochemistry performed for 549/549 cats.

^jPCV and total solids measured in 549/549 cats.

Abbreviations: ALT, alanine transferase; CKD, chronic kidney disease; cTSH, canine thyroid-stimulating hormone; GI, gastrointestinal; RI, reference interval; SBP, systolic blood pressure; TT4, total thyroxine; USG, urine specific gravity.

Findings in previously examined cats

Eighty-one cats (81/549) had been examined previously at the clinic (and found to be healthy) and were presented again for health screening during the period of the study. Sixty of these cats (60/81) were presented ≤ 24 months after their previous examination. Regarding these cats, 4/59 (6.8%) were diagnosed with azotemic CKD at the time of health screening. At the prior examination, all 4 cats had plasma creatinine concentration ≤ 2 mg/dL (≤ 177 μ mol/L) and 2 cats had plasma SDMA concentration > 14 μ g/dL, one of these had a concentration ≥ 18 μ g/dL. Hypertension was diagnosed in 2/58 (3.4%) cats and individual cases of hyperthyroidism (1/33; 3.0%) and diabetes mellitus (1/60; 1.7%) also were diagnosed, all in separate cats. None of these 60 cats had a positive urine culture and 9/60 cats had an ALT $\geq 2 \times$ RI, including the hyperthyroid, 2 hypertensive, and 1 of the azotemic CKD cats. One cat (1/60; 1.7%) had severe tartar or gingivitis or both (tartar or gingivitis score = 4).

Discussion

We report the prevalence of disease and major clinical findings for a large population of health-screened cats in a PCP setting. Forty-two percent (231/549) of cats aged 8 years and older had ≥ 1 condition that might benefit from veterinary intervention or further investigation, including hyperthyroidism, azotemic CKD and hypertension, severe dental disease, diabetes mellitus, suspected neoplasia, suspected primary GI disease, increased ALT activity ($\geq 2 \times$ RI), bacteriuria, and moderate to severe anemia of unknown cause. Some of these diseases are progressive and have management options shown to improve outcomes or ameliorate disease sequelae.¹³⁻¹⁷ Hence, routine health screening of older cats (≥ 8 years old) detects a high frequency of previously unrecognized abnormalities or conditions and, as a consequence may facilitate earlier intervention, which has been associated with improved outcomes.

Moreover, in a subpopulation of cats that were examined ≤ 24 months before their current health screening visit (and found to have an absence of overt disease at this prior assessment), diagnoses were still made. The prevalence of disease and notable findings within this subpopulation is less than that found in the entire population of health-screened cats. This finding was expected because these cats had known veterinary assessments that previously excluded overt disease, compared with those visiting the clinic for the first time with variable degrees of veterinary contact. However, the fact that diseases potentially benefiting from veterinary intervention were still found within this subpopulation supports the value of repeated health screening in cats ≥ 8 years old, as also was demonstrated in a recent study examining cats ≥ 7 years old in Belgium.⁹

Hyperthyroidism was the most common medical condition identified in our study, with a prevalence of 19.2%. This percentage falls within the wide prevalence rates for hyperthyroidism in cats reported previously (0%-21.1%),^{2,7-9,18-26} but interestingly is more than twice the prevalence of 8.7% reported for cats ≥ 10 years in a large-scale study using UK primary care practice (PCP) data from the VetCompass Animal Surveillance Project.² This disparity likely demonstrates the effect of ascertainment bias, namely, the difference between performing free-of-charge prospective health screening in our study vs evaluating prevalence in a multicenter PCP setting without standardized diagnostic and inclusion criteria where many cats will not have been screened for the disease. The former possibly allows for increased detection of disease compared to situations in which cats are most likely presented only when clinical signs have been detected by a caregiver. As might be expected, measurement of TT4 in asymptomatic cats could allow for earlier disease diagnosis and intervention before overt clinical signs arise (eg, cachexia, cardiovascular disease). Previous studies that have reported the occurrence of hyperthyroidism in apparently healthy cats, reported prevalences of 0%-5.5%, which is much lower than in our study.^{7-9,19,26} This difference could be related to the younger cat populations utilized in prior studies or increasing disease prevalence over time,²⁷ but also could be a result of different

diagnostic and inclusion criteria as well as geographic location. An earlier study performed at the same clinic as our current study reported a baseline prevalence rate of hyperthyroidism of 5.5% in health-screened cats ≥ 9 years old.¹⁹ This finding may suggest that the frequency of hyperthyroidism diagnosis in this UK-based PCP population has increased between the 2 study periods (2004-2006 compared to 2018-2023).

Azotemic CKD was diagnosed in 11.4% of the 549 cats screened in our study. No attempt was made to confirm a diagnosis of CKD in non-azotemic cats, but 11.7% of non-azotemic cats in our study had a plasma SDMA concentration above the RI, which may indicate early CKD. This 11.4% prevalence of azotemic CKD is consistent with 3 other studies (2 from the UK and 1 from Belgium) that reported on the health screening of older cats and described prevalences of 10.1%, 10.6%, and 7.7% for azotemia, International Renal Interest Society (IRIS) stage 2 CKD and IRIS stage ≥ 2 CKD, respectively.^{7,9,19} Interestingly, plasma SDMA concentrations were within the RI for 25.8% of the cats diagnosed with azotemic CKD in our current study. Whereas all of our azotemic cats were IRIS stage 2 or higher based on their plasma creatinine concentrations, 51.6% of them had SDMA concentrations < 18 $\mu\text{g/dL}$, consistent with IRIS stage 1. Additionally, of the 4 cats that developed azotemic CKD after a prior assessment at the clinic ≤ 24 months earlier, only 2 had plasma SDMA concentrations above the RI (14 $\mu\text{g/dL}$) at the earlier examination. These findings were unexpected given that SDMA reportedly increases above the RI earlier than creatinine does in 81% of cats ultimately diagnosed with azotemic CKD.²⁸ However, a recent study examining alterations in gut-derived uremic toxins before the onset of azotemic CKD in cats found that SDMA remained within the IRIS stage 1 interval (< 18 $\mu\text{g/dL}$) for all 22 cats in the 6 months before developing azotemic CKD, and although SDMA IRIS stages at the point of azotemic CKD diagnosis were not reported, the median SDMA concentration was 13.4 $\mu\text{g/dL}$ (ie, still within the RI).²⁹ It therefore appears that many cats diagnosed with azotemic CKD can have a discrepancy between their assigned IRIS stages based on plasma creatinine and SDMA concentrations.

We identified hypertension in 7.1% of the cats screened, which is within the range of prevalences (2.8%-12.7%) reported in other health-screened populations of older cats.^{5,7,9,19} No underlying cause of hypertension was identified in the majority of our cases (56.8%; 21/37 cats; all 21 cats had a plasma biochemistry panel performed including measurement of plasma creatinine, SDMA, and TT4 concentrations, and 9/21 had urinalysis results). In the remaining cats (43.2%; 16/37 cats), the most common cause of potential secondary hypertension was hyperthyroidism (9/37 cats) followed by azotemic CKD (6/37 cats), and 1 cat had an abdominal mass (1/37 cats). This predominance of hypertension without an overt cause identified on the health screening tests performed is an unanticipated observation given that previous studies report that idiopathic hypertension only accounts for 13%-23% of hypertensive cats (ie, most cats have secondary

hypertension).³⁰⁻³² This finding, however, may be attributable to differences in tests performed to screen for underlying disease and SBP thresholds used to diagnose hypertension among studies, resulting in disparate study cohorts. A mean SBP cut-off of 160 mmHg was used in our study, based on the American College of Veterinary Internal Medicine (ACVIM) consensus statement on systemic hypertension,³³ compared with 170 or 175 mmHg in earlier studies.³⁰⁻³² The hypertensive cats in our study also were older than those in previous studies, possibly leading to the expectation that the prevalence of underlying conditions, and therefore secondary hypertension, would be higher in our population. However, in humans, although essential (idiopathic) hypertension is more common, the prevalence of secondary hypertension is higher in younger patients compared with the entire population.³⁴ A lack of screening for primary hyperaldosteronism and, as previously mentioned, testing for non-azotemic CKD as potential causes of secondary hypertension are limiting factors for interpreting these data. Previous studies share these limitations and longitudinal follow-up would be required to assess if some cases of primary hypertension might in fact have an underlying cause that has not yet emerged or requires persistent documentation of mild clinicopathological abnormalities for diagnosis (eg, non-azotemic CKD). Interestingly, plasma SDMA concentrations were above the RI ($>14 \mu\text{g/dL}$) in 4/21 (19.0%) cats with hypertension and no underlying cause identified, and $\geq 18 \mu\text{g/dL}$ in 2 of these.

Bacteriuria was identified in 9.7% of cats that had urinalysis. In our protocol, urine culture was only performed if bacteria were observed on urinary sediment examination. Recently, a low prevalence (3.4%) of positive urine cultures with inactive sediment was reported in dogs.³⁵ Therefore, although our methodology may have resulted in a slight underreporting of bacteriuria prevalence, the effect is expected to be minimal. Unfortunately, based on available clinical records, it was not possible for the majority of cats to determine if a urinary tract infection (UTI) or subclinical bacteriuria was present, if the cat was symptomatic or asymptomatic, or if treatment was appropriate or effective as defined by recent guidelines.³⁶ More than 50% of cats with positive urine cultures had concurrent hyperthyroidism, azotemic CKD, hypertension, or some combination of these (most commonly CKD). Hyperthyroidism and CKD are predisposing factors for bacterial UTIs and subclinical bacteriuria in cats.³⁷⁻⁴⁰

In addition to hyperthyroidism, azotemic CKD, hypertension and bacteriuria, several other diseases, and important clinical findings were detected on routine health screening in our study. These included diabetes mellitus, neoplasia, and suspected primary GI disease. Diabetes mellitus was diagnosed in 1.5% of cats, a much higher prevalence than the 0.4%-0.6% reported previously in diverse populations of cats in the UK.^{41,42} We hypothesize that this higher prevalence could be in part related to the age of the respective study populations because one of the previous studies reported increased disease risk in cats > 6 years old. However, for diabetes mellitus and primary GI

disease, the observed prevalences in our study are surprising considering these conditions are expected to be associated with clinical signs and that cats were presented for routine health screening rather than veterinary attention being sought sooner for an unwell cat. A possible reason for this finding includes a potential lack of recognition of illness by caregivers. This situation reflects a real-world population where cats may not only display clinical signs inconsistently and be variably presented to a veterinarian. Several factors might account for this finding, including disease severity and rate of progression, cat and caregiver lifestyle (eg, indoor, outdoor, or semi-feral cat and level of caregiver and cat interaction), and availability and cost of veterinary care.^{[43,44](#)}

Dental disease, specifically gingivitis and tartar accumulation were present in the majority of cats, and 6.1% had severe tartar or gingivitis or both that would have resulted in a recommendation for further dental assessment, including examination, dental cleaning with or without tooth extractions under general anesthesia. As found here, dental disease consistently ranks among the 2 most common findings on physical examination of health-screened older cats^{[5,7-9,19,45](#)}; in one study, it was associated with a matted coat and increased sleeping, potentially suggesting the presence of chronic oral pain.^{[7](#)} Because a complete oral examination cannot routinely be performed in conscious cats and dental radiographs were not obtained, it is likely that the number of cats affected by periodontal disease and that would benefit from dental procedures is higher than estimated by the detection of severe gingivitis and tartar accumulation in our study.^{[46](#)}

The ALT activity was increased $\geq 2\times$ the RI in 14.9% of cats, but the majority (69.5%) of these increases could be explained by the concurrent diagnosis of either hyperthyroidism or diabetes mellitus. In the remaining 30.5% of cats, the cause of the increased ALT activity could not be determined but could have indicated primary hepatic disease (eg, cholangitis-cholangiohepatitis syndrome or undetected neoplasia) or an undiscovered secondary cause (eg, subclinical GI disease or pancreatitis).

Our study had some limitations. In a similar way that caregivers' ability to recognize and report illness will have impacted the study population, a subpopulation of cats will have been examined with caregivers motivated enough to seek health screening and enroll their cats into a prospective research study. Caregivers also may have been motivated to actively withhold information about their cats' histories that could have excluded them from the free-of-charge assessments. This factor may have been somewhat ameliorated with cats recruited from the charity practice, because some caregivers already would be eligible for free pet health care. Our study used data from cats that were health screened for the purpose of identifying eligibility for a prospective diet trial. This feature could have introduced bias because the study population and processes were not explicitly

designed to answer secondary questions about the utility of health screening. However, despite this limitation, some benefits may have been derived from recruitment criteria that were equally appropriate for both studies (eg, ≥ 8 years old, absence of chronic disease diagnoses) and the standardized clinical protocol and diagnostic criteria that were implemented. The fact that cats with pre-existing chronic conditions requiring long-term monitoring and management were excluded likely underestimates the prevalence of overall disease. However, it seems probable that older cats with ongoing conditions are more likely to have regular veterinary visits and therefore have comorbidities detected, whereas our study provides insight into a population of cats considered well by caregivers yet, as observed, a proportion are affected by disease. Of note, our study reports the value of health screening in a recruited and specific population (cats ≥ 8 years old) and prevalence estimates cannot be extrapolated to the general cat population. For some conditions, the denominator population used to calculate prevalence was adjusted (decreased) to account for cats where it could not be determined if a disease was present because of the absence of data at the time of health screening or required follow-up. Examples include azotemic patients for which urinalysis results or follow-up plasma creatinine concentration were not available (azotemic CKD), cats that had their plasma TT4 concentration measured in the 12 months before health screening rather than at the time of screening (hyperthyroidism), and cats with mean SBP ≥ 160 mmHg without evidence for hypertensive chorioretinopathy and not presented for follow-up blood pressure measurement. Depending on whether or not these cats had the disease in question, prevalence could have been overestimated or underestimated. Likewise, 2 cases (1 hyperthyroid and 1 hypertensive cat) were diagnosed outside of the stated diagnostic criteria but based upon reasonable real-world clinical reasoning.

In conclusion, routine clinical examination and diagnostic tests identified conditions requiring veterinary intervention or treatment in 42.1% of 549 cats 8 years or older in our study. These data support the need for health screening in aging cats and emphasize the importance of a veterinary examination to detect health conditions of concern. Veterinarians can utilize this information to make evidence-based recommendations on the utility and need for health screening in older cats.

Supplementary Material

Supplementary_material_aalag157

[supplementary_material_aalag157.zip](#) (641.9KB, zip)

Acknowledgments

Preliminary findings from these data were presented at the 33rd European College of Veterinary Internal Medicine-Companion Animal Congress, September 21-23, 2023, Barcelona, Spain. The authors extend their thanks and gratitude to the caregivers of the cats involved in the Royal Veterinary College's Aging Cat Clinic as well as past and present members of the group's Research and Clinical Team.

Abbreviations

ALT

alanine aminotransferase

cTSH

canine thyroid-stimulating hormone

CKD

chronic kidney disease

GI

gastrointestinal

PCP

primary care practice

RI

reference interval

SBP

systolic blood pressure

SDMA

symmetric dimethylarginine

TT4

total thyroxine

USG

urine specific gravity

UTI

urinary tract infection

Contributor Information

Jason P Bestwick, Clinical Science and Services, Royal Veterinary College, Royal College Street, London NW1 0TU, United Kingdom.

Jonathan Elliott, Comparative Biomedical Sciences, Royal Veterinary College, Royal College Street, London NW1 0TU, United Kingdom.

Isabel R Orriss, Comparative Biomedical Sciences, Royal Veterinary College, Royal College Street, London NW1 0TU, United Kingdom.

Nicola Lötter, Comparative Biomedical Sciences, Royal Veterinary College, Royal College Street, London NW1 0TU, United Kingdom.

Rebecca F Geddes, Clinical Science and Services, Royal Veterinary College, Royal College Street, London NW1 0TU, United Kingdom.

Author contributions

Jason Percy Bestwick (Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Visualization, Writing—original draft, Writing—review & editing), Jonathan Elliott (Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Supervision, Writing—review & editing), Isabel R. Orriss (Conceptualization, Methodology, Project administration, Resources, Supervision, Writing—review & editing), Nicola Lotter (Data curation, Investigation, Resources, Writing—review & editing), and Rebecca Frances Geddes (Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Supervision, Writing—review & editing)

Conflicts of Interest

J. P. Bestwick is currently undertaking a PhD studentship funded by Royal Canin SAS and previously completed an MPhil studentship funded by BSAVA PetSavers. J. Elliott has consultancy agreements with Elanco Ltd., CEVA Animal Health Ltd., Boehringer Ingelheim Ltd., MSD Animal Health Ltd., Orion Incorp, Idexx Ltd., Waltham Petcare Science Institute, Invetx Inc., and Zoetis Ltd., and has received grant funding from Elanco Ltd., Waltham Centre for Pet Nutrition, Royal Canin, Idexx Ltd., CEVA Animal Health, and Zoetis Ltd. He is a member of the International Renal Interest Society, which receives sponsorship from Zoetis. He is a co-applicant with Royal Canin on pending patents; these are not directly linked to this project/manuscript and garner no financial benefit to

the authors or their institution. R. F. Geddes has received funding from Petplan, Royal Canin, an RVC Internal Grant, The Academy of Medical Sciences, The Everycat Foundation, and Elanco; has previously had a consultancy agreement with Boehringer Ingelheim; and has received speaking honoraria from Boehringer Ingelheim, Idexx, and Royal Canin.

Funding

This project was conducted as part of J. P. Bestwick's PhD studentship, which is funded by Royal Canin (grant number 4061).

Off-label antimicrobial declaration

The authors declare no off-label use of antimicrobials.

Institutional animal care and use committee or other approval declaration

Approved by the Royal Veterinary College Ethics and Welfare Committee (RVC-EWC): URN 20131258E/URN 202322253 and URN 201818453/URN 202321963.

Human ethics approval declaration

The authors declare that human ethics approval was not needed for this study.

References

1. Ray M, Carney HC, Boynton B, et al. 2021 AAFP feline senior care guidelines. *J Feline Med Surg*. 2021;23:613-638. 10.1177/1098612x211021538 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
2. Stephens MJ, Neill DGO, Church DB, McGreevy PD, Thomson PC, Brodbelt DC. Feline hyperthyroidism reported in primary-care veterinary practices in England: prevalence, associated factors and spatial distribution. *Vet Rec*. 2014;175:458. 10.1136/vr.102431 [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
3. Brown CA, Elliott J, Schmiedt CW, Brown SA. Chronic kidney disease in aged cats: clinical features, morphology, and proposed pathogeneses. *Vet Pathol*. 2016;53:309-326.

10.1177/0300985815622975 [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

4. Bijsmans ES, Jepson RE, Chang YM, Syme HM, Elliott J. Changes in systolic blood pressure over time in healthy cats and cats with chronic kidney disease. *J Vet Intern Med*. 2015;29:855-861. 10.1111/jvim.12600 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
5. Jepson RE, Brodbelt D, Vallance C, Syme HM, Elliott J. Evaluation of predictors of the development of azotemia in cats. *J Vet Intern Med*. 2009;23:806-813. 10.1111/j.1939-1676.2009.0339.x [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
6. Bodey AR, Sansom J. Epidemiological study of blood pressure in domestic cats. *J Small Anim Pract*. 1998;39:567-573. 10.1111/j.1748-5827.1998.tb03710.x [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
7. Dowgray N, Pinchbeck G, Eyre K, Biourge V, Comerford E, German AJ. Aging in cats: owner observations and clinical finding in 206 mature cats at enrolment to the cat prospective aging and welfare study. *Front Vet Sci*. 2022;9:859041. 10.3389/fvets.2022.859041 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
8. Paepe D, Verjans G, Duchateau L, Piron K, Ghys L, Daminet S. Routine health screening: findings in apparently healthy middle-aged and old cats. *J Feline Med Surg*. 2012;15:8-19. 10.1177/1098612x12464628 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
9. Mortier F, Daminet S, Marynissen S, Smets P, Paepe D. Value of repeated health screening in 259 apparently healthy mature adult and senior cats followed for 2 years. *J Vet Intern Med*. 2024;38:2089-2098. 10.1111/jvim.17138 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
10. Sánchez-Vizcaíno F, Noble PJM, Jones PH, et al. Demographics of dogs, cats, and rabbits attending veterinary practices in Great Britain as recorded in their electronic health records. *BMC Vet Res*. 2017;13:218. 10.1186/s12917-017-1138-9 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
11. Carter J. Hypertensive ocular disease in cats: a guide to fundic lesions to facilitate early diagnosis. *J Feline Med Surg*. 2019;21:35-45. 10.1177/1098612x18818668 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

12. Wilson EB. Probable inference, the law of succession, and statistical inference. *J Am Stat Assoc.* 1927;22:209-212. 10.1080/01621459.1927.10502953 [[DOI](#)] [[Google Scholar](#)]
13. Elliott J, Rawlings JM, Markwell PJ, Barber PJ. Survival of cats with naturally occurring chronic renal failure: effect of dietary management. *J Small Anim Pract.* 2000;41:235-242. 10.1111/j.1748-5827.2000.tb03932.x [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
14. Young WM, Zheng C, Davidson MG, Westermeyer HD. Visual outcome in cats with hypertensive chorioretinopathy. *Vet Ophthalmol.* 2019;22:161-167. 10.1111/vop.12575 [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
15. Taylor SS, Sparkes AH, Briscoe K, et al. ISFM consensus guidelines on the diagnosis and management of hypertension in cats. *J Feline Med Surg.* 2017;19:288-303. 10.1177/1098612x17693500 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
16. Reynolds BS, Lefebvre HP. Feline CKD: pathophysiology and risk factors—what do we know? *J Feline Med Surg.* 2013;15:3-14. 10.1177/1098612x13495234 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
17. Carney HC, Ward CR, Bailey SJ, et al. 2016 AAFP guidelines for the management of feline hyperthyroidism. *J Feline Med Surg.* 2016;18:400-416. 10.1177/1098612x16643252 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
18. Miyamoto T, Miyata I, Kurobane K, et al. Prevalence of feline hyperthyroidism in Osaka and the Chugoku region. *J Jpn Vet Méd Assoc.* 2011;55:289. 10.12935/jvma1951.55.289 [[DOI](#)] [[Google Scholar](#)]
19. Wakeling J, Elliott J, Syme H. Evaluation of predictors for the diagnosis of hyperthyroidism in cats. *J Vet Intern Med.* 2011;25:1057-1065. 10.1111/j.1939-1676.2011.00790.x [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
20. Köhler I, Ballhausen BD, Stockhaus C, Hartmann K, Wehner A. Prevalence of and risk factors for feline hyperthyroidism among a clinic population in southern Germany. *Tierärztliche Prax Ausg K Kleintiere Heimtiere.* 2015;44:149-157. 10.15654/tpk-150590 [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
21. Bree L, Gallagher BA, Shiel RE, Mooney CT. Prevalence and risk factors for hyperthyroidism in Irish cats from the greater Dublin area. *Ir Vet J.* 2018;71:2. 10.1186/s13620-017-0113-x [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

22. Domínguez AP, Tostado RS, Bernabe LF, Corredor AP, Prat JP. Prevalence of feline hyperthyroidism in a laboratory-based sample of 27,888 cats in Spain. *J Feline Med Surg.* 2024;26:1098612X241303304. 10.1177/1098612x241303304 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
23. O'Neill DG, Gunn-Moore D, Sorrell S, et al. Commonly diagnosed disorders in domestic cats in the UK and their associations with sex and age. *J Feline Med Surg.* 2023;25:1098612X231155016. 10.1177/1098612x231155016 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
24. Wet CSD, Mooney CT, Thompson PN, Schoeman JP. Prevalence of and risk factors for feline hyperthyroidism in Hong Kong. *J Feline Med Surg.* 2008;11:315-321. 10.1016/j.jfms.2008.08.001 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
25. O'Neill DG, Church DB, McGreevy PD, Thomson PC, Brodbelt DC. Prevalence of disorders recorded in cats attending primary-care veterinary practices in England. *Vet J.* 2014;202:286-291. 10.1016/j.tvjl.2014.08.004 [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
26. Dell'Osa D, Jaensch S. Prevalence of clinicopathological changes in healthy middle-aged dogs and cats presenting to veterinary practices for routine procedures. *Aust Vet J.* 2016;94:317-323. 10.1111/avj.12481 [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
27. Peterson M. Hyperthyroidism in cats. *J Feline Med Surg.* 2012;14:804-818. 10.1177/1098612x12464462 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
28. Hall JA, Yerramilli M, Obare E, Yerramilli M, Jewell DE. Comparison of serum concentrations of symmetric dimethylarginine and creatinine as kidney function biomarkers in cats with chronic kidney disease. *J Vet Intern Med.* 2014;28:1676-1683. 10.1111/jvim.12445 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
29. Mulders LV, Broecke EV, Paepe ED, Mortier F, Vanhaecke L, Daminet S. Alterations in gut-derived uremic toxins before the onset of azotemic chronic kidney disease in cats. *J Vet Intern Med.* 2025;39:e17289. 10.1111/jvim.17289 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
30. Elliott J, Barber PJ, Syme HM, Rawlings JM, Markwell PJ. Feline hypertension: clinical findings and response to antihypertensive treatment in 30 cases. *J Small Anim Pract.* 2001;42:122-129. 10.1111/j.1748-5827.2001.tb02008.x [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

31. Maggio F, DeFrancesco TC, Atkins CE, Pizzirani S, Gilger BC, Davidson MG. Ocular lesions associated with systemic hypertension in cats: 69 cases (1985–1998). *J Am Vet Méd Assoc*. 2000;217:695-702. 10.2460/javma.2000.217.695 [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
32. Jepson RE, Elliott J, Brodbelt D, Syme HM. Effect of control of systolic blood pressure on survival in cats with systemic hypertension. *J Vet Intern Med*. 2007;21:402-409. 10.1111/j.1939-1676.2007.tb02982.x [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
33. Acierno MJ, Brown S, Coleman AE, et al. ACVIM consensus statement: guidelines for the identification, evaluation, and management of systemic hypertension in dogs and cats. *J Vet Intern Med*. 2018;32:1803-1822. 10.1111/jvim.15331 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
34. de Freminville JB, Gardini M, Cremer A, et al. Prevalence and risk factors for secondary hypertension in young adults. *Hypertension*. 2024;81:2340-2349. 10.1161/hypertensionaha.124.22753 [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
35. Strachan NA, Hales EN, Fischer JR. Prevalence of positive urine culture in the presence of inactive urine sediment in 1049 urine samples from dogs. *J Vet Intern Med*. 2022;36:629-633. 10.1111/jvim.16378 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
36. Taylor S, Boysen S, Buffington T, et al. 2025 iCatCare consensus guidelines on the diagnosis and management of lower urinary tract diseases in cats. *J Feline Med Surg*. 2025;27:1098612X241309176. 10.1177/1098612x241309176 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
37. Teichmann-Knorrn S, Reese S, Wolf G, Hartmann K, Dorsch R. Prevalence of feline urinary tract pathogens and antimicrobial resistance over five years. *Vet Rec*. 2018;183:21. 10.1136/vr.104440 [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
38. Mayer-Roenne B, Goldstein RE, Erb HN. Urinary tract infections in cats with hyperthyroidism, diabetes mellitus and chronic kidney disease. *J Feline Med Surg*. 2006;9:124-132. 10.1016/j.jfms.2006.09.004 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
39. Dorsch R, von Vopelius-Feldt C, Wolf G, Mueller RS, Straubinger RK, Hartmann K. Urinary tract infections in cats. Prevalence of comorbidities and bacterial species, and determination of antimicrobial susceptibility to commonly used antimicrobial agents.

Tierärztliche Prax Ausg K Kleintiere Heimtiere. 2016;44:227-236. 10.15654/tpk-150604 [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

40. White JD, Stevenson M, Malik R, Snow D, Norris JM. Urinary tract infections in cats with chronic kidney disease. J Feline Med Surg. 2013;15:459-465.

10.1177/1098612x12469522 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

41. O'Neill DG, Gostelow R, Orme C, et al. Epidemiology of diabetes mellitus among 193,435 cats attending primary-care veterinary practices in England. J Vet Intern Med. 2016;30:964-972. 10.1111/jvim.14365 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

42. McCann TM, Simpson KE, Shaw DJ, Butt JA, Gunn-Moore DA. Feline diabetes mellitus in the UK: the prevalence within an insured cat population and a questionnaire-based putative risk factor analysis. J Feline Med Surg. 2007;9:289-299.

10.1016/j.jfms.2007.02.001 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

43. Ma GC, McLeod LJ, Zito SJ. Characteristics of cat semi-owners. J Feline Med Surg. 2023;25:1098612X231194225. 10.1177/1098612x231194225 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

44. Cats Protection . CATS Report: Cats and Their Stats UK. 2025. Accessed November 5, 2025. https://www.cats.org.uk/media/onnnmi1h/cats-report-uk-2025_digital.pdf

45. Finch NC, Syme HM, Elliott J. Risk factors for development of chronic kidney disease in cats. J Vet Intern Med. 2016;30:602-610. 10.1111/jvim.13917 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

46. Perry R, Tutt C. Periodontal disease in cats: back to basics—with an eye on the future. J Feline Med Surg. 2015;17:45-65. 10.1177/1098612x14560099 [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

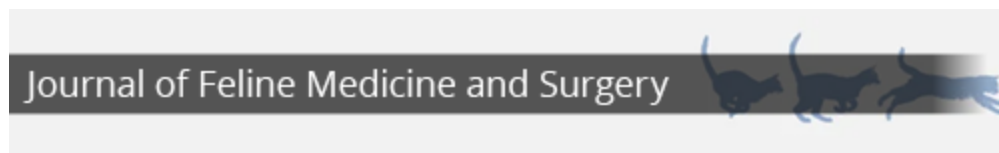
Supplementary_material_aalag157

[supplementary_material_aalag157.zip](#) (641.9KB, zip)

Articles from Journal of Veterinary Internal Medicine are provided here courtesy of **Oxford University Press**

As a library, NLM provides access to scientific literature. Inclusion in an NLM database does not imply endorsement of, or agreement with, the contents by NLM or the National Institutes of Health.

Learn more: [PMC Disclaimer](#) | [PMC Copyright Notice](#)



J Feline Med Surg. 2026 Mar 21;28(3):1098612X251407521. doi: [10.1177/1098612X251407521](https://doi.org/10.1177/1098612X251407521) [↗](#)

Feline comorbidities: cardiovascular and kidney diseases

[Amanda E Coleman](#)^{1,*}, [Bianca N Lourenço](#)¹

[Author information](#) [Article notes](#) [Copyright and License information](#)

PMCID: PMC13009769 PMID: [41863293](#)

Abstract

Practical relevance:

Cardiovascular and kidney diseases are highly prevalent and frequently coexistent in cats, particularly those of advanced age. Several pathways of interaction between the heart and kidneys likely contribute to progressive functional decline of these systems during disease of one or both. Identification of heart or kidney disease in a cat should prompt the clinician to consider how the disease and its treatment might impact the function of the other organ and how coexisting primary disease of the other might impact case management and patient prognosis.

Clinical challenges:

A common goal in the management of congestive heart failure, acute kidney injury or chronic kidney disease is to maintain optimal body fluid balance; however, many therapies for disease of one system are at odds with the best interests of the other. For example, treatment of kidney

disease often includes parenteral fluid therapy with the goal of maintaining mean arterial pressure and adequate renal blood flow, while diuretics are almost always necessary to reduce capillary hydrostatic pressure in cats with heart failure. Overly aggressive volume expansion or reduction during the management of kidney or heart failure, respectively, can lead to injury to or excessive stress on the other organ system.

Evidence base:

Recommendations are based on available peer-reviewed literature and expert consensus guidelines when available.

Audience:

This article is aimed at all veterinary practitioners who work with cats.

Keywords: Feline cardiomyopathy, cardiovascular–renal syndromes, cardiorenal syndrome, renocardiac syndrome

Introduction

Heart disease is common in domestic cats. The most frequently diagnosed of these diseases – idiopathic primary hypertrophic cardiomyopathy (HCM) – affects 15–16% of apparently healthy cats^{[1,2](#)} and results in cardiovascular death in 28% of cases.^{[3](#)} In one large study of cats presented to primary care practices in the UK, cardiac disease was the attributed cause of death in 4.6% of those aged 5 years or older.^{[4](#)}

Kidney disease is also one of the most common causes of morbidity and mortality in cats, particularly those of advanced age. The reported prevalence of chronic kidney disease (CKD) in cats depends on the population studied and diagnostic criteria used, ranging from 2–50% in cats of all ages^{[5,6](#)} and up to 80% in cats aged 15 years or more.^{[5](#)} In the large UK study referenced above, ‘renal disorder’ was the most common cause of death in cats aged 5 years or older, cited in 13.6%.^{[4](#)} In addition, cats with kidney disease are frequently diagnosed with systemic arterial hypertension, which has important pathophysiologic consequences for both the kidneys and the heart.

Given the relatively high prevalence of heart and kidney diseases in cats, the intimate relationship between these systems in health and disease, and the fact that disease (or treatment) of one can

cause direct or indirect injury or dysfunction of the other, it is unsurprising that clinicians commonly encounter cats with cardiovascular and renal comorbidities. Indeed, in a recent study, 51 (44%) of 116 cats treated for congestive heart failure (CHF) were azotemic prior to initiation of parenteral diuretics.⁷ Likewise, 'cardiovascular events' were recorded in 104 (62%) of 168 cats admitted at a different institution for urinary tract obstruction.⁸

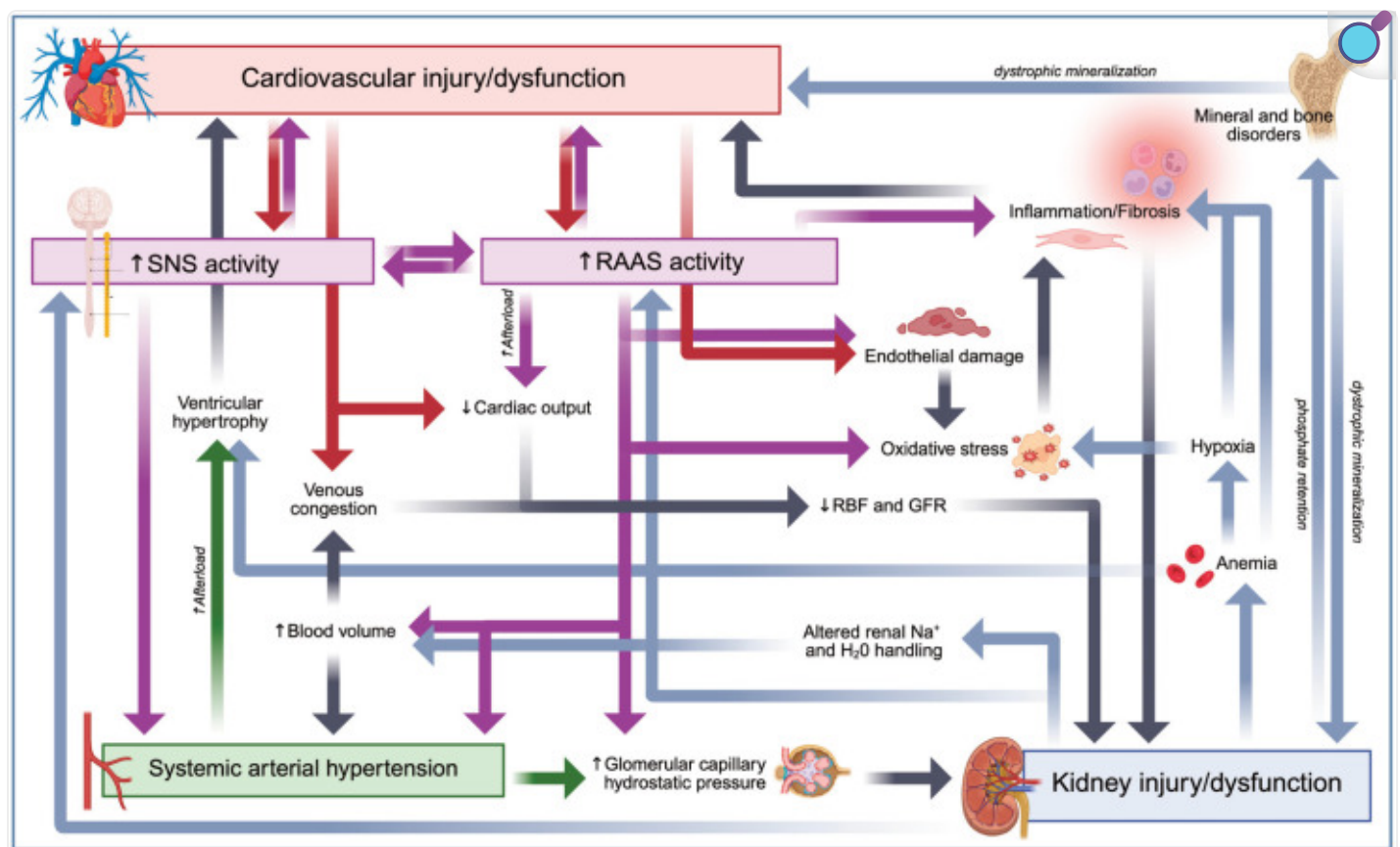
In this article, we review the relationships between diseases of the heart and kidneys and present some considerations for the management of cats affected by both.

The cardiorenal axis and cardiovascular–renal disorders

The functions of the heart, blood vessels and kidneys are highly interrelated. Consideration of an integrated 'cardiorenal axis' is a useful lens through which to view these complex relationships. As defined by the veterinary Cardiovascular–Renal Axis Disorders Consensus Group, cardiovascular–renal disorders of dogs and cats are 'disease, toxin or drug-induced structural and/or functional damage to the kidney and/or cardiovascular system, leading to disruption of the normal interaction between these systems, to the ongoing detriment of one or both'.⁹ This approach encourages clinicians to consider connections between the cardiovascular system and kidneys during disease of either or both, and their interactions with other disease conditions. Further, it emphasizes the pathways, often bidirectional, by which disease progression in either or both might be mediated.

Although little is known regarding whether and how frequently primary disease of the kidneys or heart can initiate disease in the other organ, several pathways of interaction between the heart and kidneys likely contribute to progressive functional decline during disease of one or both ([Figure 1](#)).

Figure 1.



[Open in a new tab](#)

Proposed hemodynamic, neurohormonal and inflammatory pathways linking the heart and kidneys during injury or dysfunction of either. GFR = glomerular filtration rate; RAAS = renin-angiotensin-aldosterone system; RBF = renal blood flow; SNS = sympathetic nervous system. Created in BioRender. Lourenço, B. (2025) <https://BioRender.com/z0jjvq>

How might heart disease and failure, or the treatment of these, impact kidney function?

Cats with severe heart disease can experience reductions in cardiac output and are at risk of developing systemic arterial hypotension, CHF and arterial thromboembolic complications.

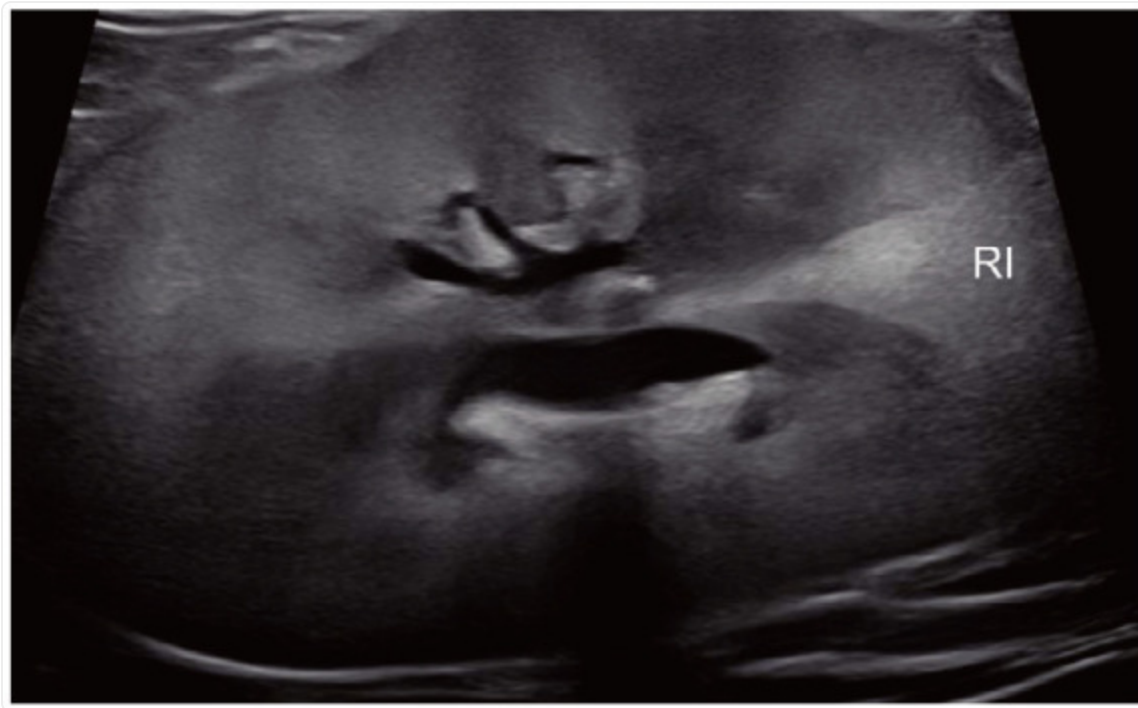
RAAS and SNS activation

While initially compensatory and supportive of glomerular filtration rate (GFR), activation of the renin–angiotensin–aldosterone system (RAAS) and sympathetic nervous system (SNS) leads to chronic myocardial and kidney stress.¹⁰ Heart failure can be viewed as an inflammatory condition,^{11–13} with the production of reactive oxygen species and pro-inflammatory mediators leading to endothelial dysfunction and tissue fibrosis. These effects can impair renal blood flow, oxygen delivery and GFR; damage corticomedullary renal tubular cells; and promote renal interstitial inflammation and fibrosis, both part of a final common pathway of disease progression in CKD.¹⁴ In human patients with severe heart disease, the major pathophysiologic role of renal venous congestion and increased intra-abdominal pressure in kidney injury is also well recognized;^{15,16} whether this is a factor in cats is currently unstudied. Finally, the pattern (eg, whether sustained, low-level insults or more severe, episodic injuries are to blame) and time course (eg, at which point in the course of cardiac disease or its treatment) of the above effects on the kidney are unclear.¹⁷

Arterial thromboembolism leading to renal infarction

Cats with heart disease frequently experience arterial thromboembolism (ATE).³ While this most commonly manifests as distal aortic thromboembolism (ie, ‘saddle thrombus’), the kidneys can also be a direct target.¹⁸ The prevalence of renal artery thromboembolism in cats with heart disease is undetermined. However, cats with renal infarction ([Figure 2](#)) appear more likely to have echocardiographic evidence of heart disease than those without.^{19,20}

Figure 2.



[Open in a new tab](#)

Two-dimensional transabdominal ultrasound image taken in a sagittal plane through the right kidney of a cat with chronic kidney disease. There are undulant cortical margins, wedge-shaped hyperechoic renal cortical infarcts (RI) and a diffuse decrease in corticomedullary distinction

Pharmacologic agents used for cardiovascular disease and heart failure

Drugs commonly used to treat heart disease and failure (eg, loop diuretics, angiotensin-converting enzyme inhibitors [ACEis]) or systemic arterial hypertension (eg, angiotensin receptor blockers [ARBs]) can affect kidney perfusion, renal autoregulation and glomerular hemodynamics, particularly when administered at high dosages alone or in combination, or when given in the setting of decreased circulating blood volume.²¹⁻²³ While they represent a cornerstone of therapy for both acute and chronic life-threatening CHF in cats,^{24,25} loop diuretics can impair renal autoregulation and reduce renal blood flow under certain conditions by interfering with tubuloglomerular feedback.^{22,26}

How common is kidney injury in cats with primary heart disease?

Azotemia is common in cats with apparently asymptomatic cardiomyopathy²⁷ and in those with acute CHF even before CHF has been treated.^{7,12,27,28} Serum creatinine concentrations >2.1 mg/dl (186 µmol/l) or >1.6 mg/dl (140 µmol/l) were documented prior to furosemide therapy in 11 (24%) of 45 cats²⁸ and 51 (44%) of 116 cats⁷ hospitalized for acute CHF, respectively. In the larger of these studies, only 21 (18%) of 116 cats had known CKD at intake, suggesting that kidney injury due to primary heart disease might have explained pretreatment azotemia in some of the cases, although interpretation is complicated by the cats' advanced age (median [range], 10 years [0.8–19.3]) and challenges in diagnosing early-stage CKD. In the same study, 75% of cats experienced a serum creatinine concentration >1.6 mg/dl (140 µmol/l) at some point during hospitalization, and the frequency of serum creatinine concentration >1.6 mg/dl remained high (ie, 45–69% of surveilled cases) post-discharge.⁷ Increases ≥ 0.3 mg/dl in serum creatinine concentration, consistent with the study's definition of kidney injury, were experienced by 49% of cats during hospitalization,⁷ in keeping with previous studies.^{28,29} While, in general, acute increases in serum creatinine have been linked to worse outcomes in cats,³⁰ kidney injury (as defined above) coincident with acute CHF treatment was not associated with poorer long-term prognosis in the study performed by Rogg and colleagues.⁷

How might kidney disease and its treatment impact cardiovascular function?

The kidney regulates extracellular fluid volume and composition (including electrolyte concentrations) and systemic arterial blood pressure (BP). Due to renal functional derangements or drugs and therapies used in their treatment, cats with kidney disease can experience excesses or deficits of blood volume and pressure that affect cardiovascular function. Aside from the established effects of systemic arterial hypertension on the heart, whether these other derangements can induce heart disease or failure de novo is unclear; however, for cats with underlying heart disease, they are likely relevant.

Hypo- and hypervolemia

Uremic cats can experience hypovolemia due to obligatory polyuria if not met by sufficient intake, or fluid losses associated with vomiting or diarrhea, leading to decreases in cardiac output. Conversely, kidney disease can be associated with hypervolemia due to aggressive parenteral fluid administration, abnormal renal sodium and water handling, activation of the RAAS and chronic

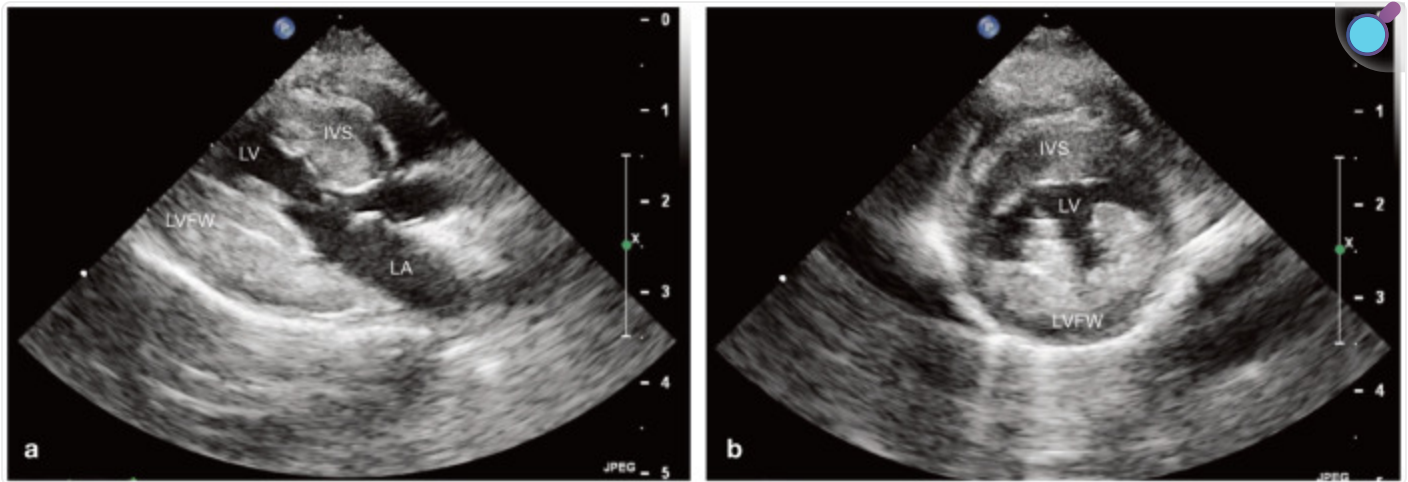
anemia, all of which can exacerbate heart disease and failure by increasing the workload of the heart.

Cats with kidney disease frequently require parenteral fluid therapy to address excessive renal and (when uremic) gastrointestinal fluid losses. While supplemental fluids can restore euolemia and euhydration, their use – particularly of sodium-rich, isotonic crystalloids – can promote excessive extracellular fluid volume if not carefully optimized to patient needs.³¹ This can be particularly problematic when administration is for long periods or at high rates to cats with underlying heart disease. Similarly, blood transfusion for severe anemia can cause transfusion-associated circulatory overload.³²

Systemic arterial hypertension

Systemic arterial hypertension is noted in 40–61% of cats with CKD^{33,34} and 28% of those with acute kidney injury (AKI).³⁵ It is the most well established pathophysiologic connection between kidney and heart diseases. The relationship between kidney disease and systemic arterial hypertension,³⁶ and the importance of the heart, vasculature and kidneys as targets of hypertensive end-organ damage, are detailed elsewhere.^{37,38} In small cross-sectional studies, 48–85% of systemically hypertensive cats have echocardiographic evidence of hypertensive cardiomyopathy ([Figure 3](#)).^{39–42} However, the clinical relevance of these changes is unclear. Although a cardiovascular cause of death or complication at the time of euthanasia was assigned for 4 (21%) of 19 hypertensive cats who died in one study,⁴³ the authors of a separate report found no significant difference in mean survival time between hypertensive cats with or without echocardiographic abnormalities.³⁹ It is uncommon for a cat to develop CHF or arterial thromboembolism secondarily to systemic arterial hypertension alone.⁴⁴ However, its consequences are relevant in patients with pre-existing heart disease or those undergoing additional hemodynamic stressors (eg, blood transfusion, intravenous fluid therapy), in whom CHF might be more easily precipitated.

Figure 3.



[Open in a new tab](#)

Two-dimensional (a) long- and (b) short-axis transthoracic echocardiographic images taken from a 10-year-old spayed female cat with chronic kidney disease and untreated systemic arterial hypertension (average indirect systolic blood pressure, 200 mmHg). Each image was taken at end-diastole from a right parasternal view. There is moderate-to-severe concentric hypertrophy of the left ventricular wall. IVS = interventricular septum; LA = left atrium; LV = left ventricle; LVFW = left ventricular free wall

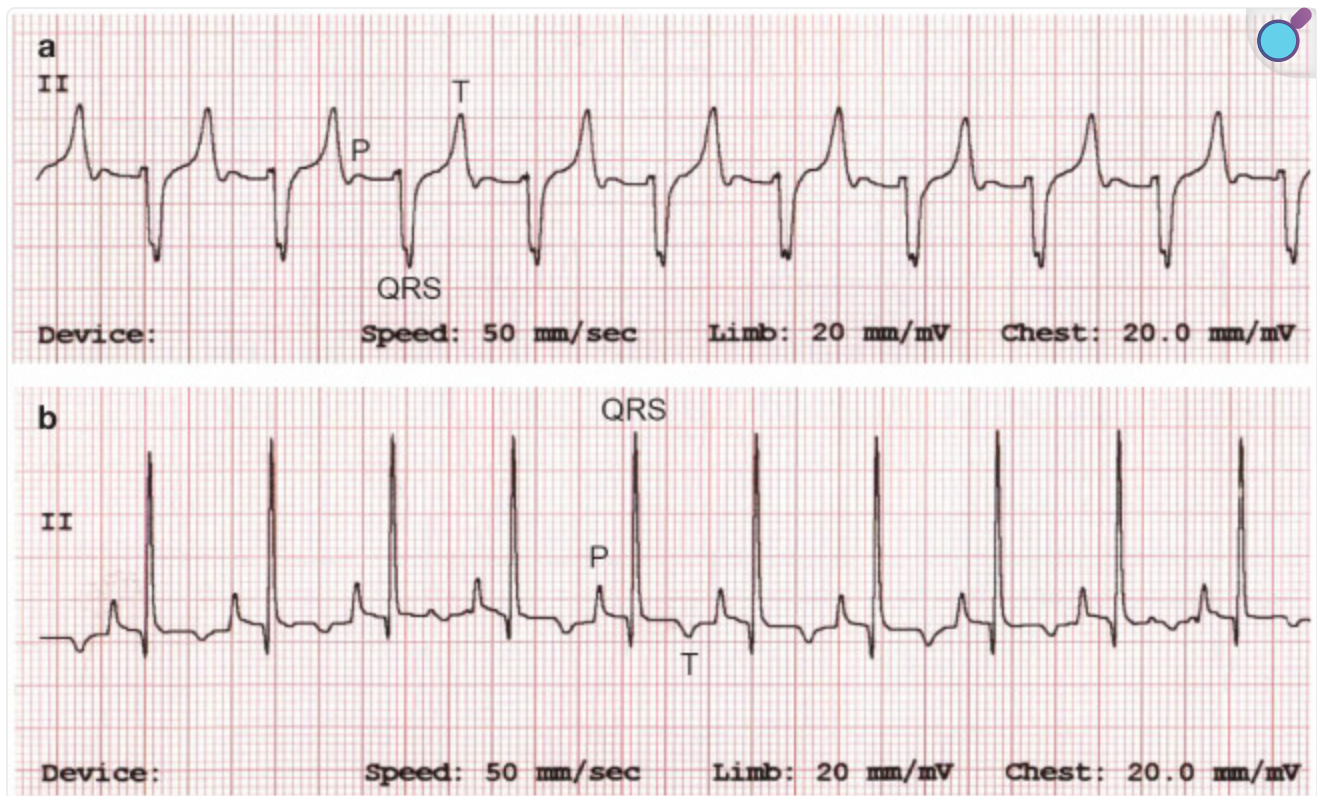
Electrolyte derangements

Kidney disorders are frequently accompanied by electrolyte derangements. These include hypo- and hyperkalemia, hypo- and hypercalcemia, and hyperphosphatemia, which are common in cats with AKI^{[45-48](#)} or CKD^{[45,49-51](#)} and can affect cardiovascular function and structure.

Potassium abnormalities can alter cardiac excitability and lead to brady- or tachyarrhythmias (eg, sinus bradycardia, transient atrial standstill, sinoventricular rhythm, ventricular tachyarrhythmias, asystole)^{[52,53](#)} or abnormalities of cardiac conduction ([Figure 4](#)). In one retrospective study of 96 azotemic cats with acute ureteral or urethral obstruction, hyperkalemia was reported in 45 (46%) and serum potassium concentration was significantly associated with the occurrence of cardiovascular events – 34% of which were arrhythmias – based on multivariate analysis.^{[8](#)} Conversely, hypokalemia is more common with polyuria and is therefore most likely in

cats with CKD who are not fed a potassium-supplemented therapeutic renal diet,⁴⁹ especially those with mild or moderate decreases in GFR (ie, International Renal Interest Society [IRIS] CKD stages 2 or 3) or those with diuresis following relief of urethral obstruction.⁵⁴

Figure 4.



[Open in a new tab](#)

Lead II surface electrocardiograms (paper speed, 50 mm/s; sensitivity, 20 mm/mV) taken (a) before and (b) after treatment of severe hyperkalemia (serum potassium concentration, 8.18 mmol/l [reference interval (RI), 3.41–4.76]) in a 7-year-old castrated male cat with aortic and renal arterial thromboembolism. In both recordings, the rhythm is sinus. After treatment, there is normalization of P wave amplitude (0.05–0.2 mV), PR interval (0.10–0.07 s [RI, 0.05–0.09]), QRS duration (0.07–0.03 s [RI <0.04 s]) and T wave morphology and amplitude (RI $\pm$ 0.3 mV)

Calcium is critical for normal cardiomyocyte contraction, cardiac automaticity and nodal conduction, and excitation–contraction coupling.^{55,56} In humans, low blood calcium

concentrations can cause systemic arterial hypotension due to clinically relevant compromises in myocardial and vascular smooth muscle contract-ility.^{57,58} Conversely, sustained hypercalcemia, which can occur in CKD– mineral and bone disorder (CKD– MBD), can promote vascular mineralization and is associated with increased cardiovascular morbidity and mortality in people.^{59–62} Although vascular mineralization is recognized in cats with CKD,^{63,64} the impact of calcium disorders on cardiovascular outcomes remains understudied.

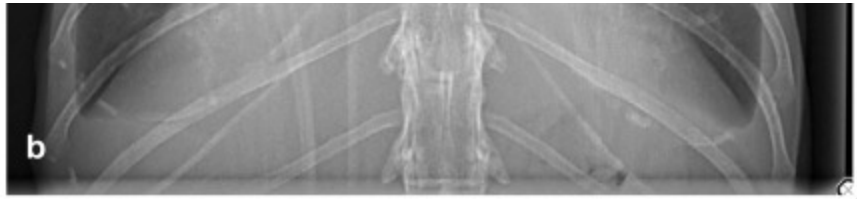
Finally, decreased renal excretion of phosphate stimulates the production of fibroblast growth factor 23 (FGF-23), which is central to the pathophysiology of CKD–MBD.⁶⁵ Increased circulating phosphate and FGF-23 concentrations are important risk factors for CKD progression and mortality in cats,^{66–68} and strongly predict worse cardiovascular outcomes in humans, likely due in part to the dystrophic mineralization they promote.^{69–71}

CKD-associated anemia

The anemia of CKD increases in prevalence and severity with disease stage.⁵⁰ In addition to exacerbating renal hypoxia and oxidative stress and accelerating CKD progression in cats,⁷² chronic anemia triggers detrimental adaptations (eg, RAAS and SNS activation) that expand extracellular fluid volume, increase cardiac workload, and promote left ventricular and left atrial hypertrophy.⁷³ These effects can be clinically relevant ([Figure 5](#)): in one report of eight severely anemic (ie, hematocrit $\leq 18\%$) cats without underlying heart disease or systemic arterial hypertension, left heart enlargement and CHF were noted in 100% and 71%, respectively.⁷⁴ While these data are almost certainly influenced by selection bias – included cats had to have undergone echocardiography – they illustrate the potentially serious cardiovascular consequences of anemia.

Figure 5.





[Open in a new tab](#)

(a) Left lateral and (b) ventrodorsal thoracic radiographs taken from a 10-year-old spayed female cat with chronic kidney disease (International Renal Interest Society stage 3, normotensive) and chronic non-regenerative anemia (packed cell volume, 25%), receiving once daily treatment with subcutaneous fluids. There is mild-to-moderate cardiomegaly, a mild amount of pleural effusion and a multifocal interstitial coalescing-to-patchy alveolar pulmonary pattern, consistent with congestive heart failure

Erythropoiesis-stimulating agents (eg, darbepoetin) are used in cats with declining quality of life due to CKD-associated anemia,⁷⁵ potentially benefitting the kidneys and heart in addition to improving clinical signs. However, systemic arterial hypertension is a well-known potential adverse effect of this drug class,^{76,77} so careful monitoring of BP is recommended during therapy. Molidustat, a hypoxia-inducible factor prolyl hydroxylase inhibitor that stimulates endogenous production of erythropoietin, and for which currently available data do not support an effect on BP in cats,⁷⁸ might represent a potentially viable alternative for this indication. At the time of writing, molidustat has conditional approval in the USA for the treatment of non-regenerative anemia in cats with CKD. However, drugs of the same class are associated with the development of systemic arterial hypertension in humans,⁷⁹ suggesting that BP monitoring is advisable in treated cats.

RAAS and SNS activation

As in heart disease, excessive activation of the RAAS and SNS occurs in kidney disease independent of anemia ([Figure 1](#)). As mentioned above, the chronic hemodynamic (eg, increases in blood volume and pressure) and non-hemodynamic (eg, pro-inflammatory, pro-fibrotic, pro-oxidative stress) effects can harm the kidneys and cardiovascular system.⁸⁰ While evidence for activation of the circulating RAAS in cats with kidney disease is inconsistent, activation of the local, independently regulated, intrarenal RAAS is likely.^{81–85}

Reduced renal clearance of cardiovascular pharmacologic agents

Abnormal kidney function can reduce the renal clearance of various pharmacologic agents. This is potentially clinically relevant for cardiovascular drugs whose parent compound or active metabolites (eg, atenolol, enalapril and its active metabolite, enalaprilat) are primarily eliminated by the kidneys ([Table 1](#)), as this would raise concerns for drug accumulation and increased risk of dose-dependent adverse effects.^{86,91} Therefore, dose reductions or the selection of other drugs in the same class with no or less renal elimination might be necessary in cats with kidney function impairment. For example, the clearances of benazepril and benazeprilat, which largely undergo biliary elimination, are unaffected by reduced kidney function in cats.⁹²

Table 1.

Mode of endogenous elimination for drugs commonly used in the treatment of cardiovascular disease of cats

Indication	Drug class	Drug	Primary mode(s) of elimination
Arrhythmia or cardiomyopathy	✦ Beta (3)-adrenergic receptor blocker	✦ Atenolol [*]	✦ Renal excretion (humans) 86
	✦ Potassium channel- and 3-adrenergic receptor-blocker	✦ Sotalol [*]	✦ Renal excretion 87 ‡
Congestive heart failure	✦ Loop diuretic	✦ Furosemide [*]	✦ Renal excretion 88 ‡
		✦ Torsemide	✦ 37% eliminated in urine as unchanged drug 89 ‡ ✦ Substantial hepatic elimination (humans) 90
	✦ Angiotensin-converting enzyme inhibitor	✦ Enalapril [*]	✦ Renal excretion of the active metabolite, enalaprilat (dogs) 91
		✦ Benazepril	✦ Unpublished data: hepatobiliary excretion (85%) 92 ‡ ✦ Biliary (55%) and renal (45%) excretion of the active metabolite, benazeprilat (dogs) 91
	✦ Inodilator	✦ Pimobendan	✦ Hepatobiliary/fecal excretion (dogs) 93

Indication	Drug class	Drug	Primary mode(s) of elimination
Thrombosis prophylaxis	✚ Platelet aggregation inhibitor	✚ Dobutamine [*]	✚ Renal (67%) and fecal (20%) excretion (dogs) ⁹⁴
	✚ Factor Xa inhibitor	✚ Clopidogrel [*]	✚ Renal (50%) and fecal (46%) excretion (humans) ⁹⁵
Systemic arterial hypertension		✚ Rivaroxaban	✚ Renal excretion of unchanged drug (33%), hepatic metabolism to inactive metabolites (66%; humans) ⁹⁶
	✚ Angiotensin receptor blocker	✚ Telmisartan	✚ Hepatobiliary/fecal excretion (dogs, humans) ⁹⁷
	Calcium channel blocker	Amlodipine [*]	✚ Glucuronidation ⁹⁸⁷
			Hepatobiliary/fecal (40-75%) and renal (25-60%) excretion, depending on the species ⁹⁹

[Open in a new tab](#)

*Drugs for which greater than half of the dose is eliminated by the kidneys in cats (if data for this species are available) or in at least one other species (if data for cats are not available)

†Pharmacokinetic data generated in cats are available. In all other instances, information was extrapolated from other species

Suboptimization of heart failure protocols

Finally, there is a general lack of evidence to guide heart failure therapies in cats with advanced kidney disease, which might impact cardiovascular function and outcomes. An easily imagined effect is the underprescription and suboptimization of heart failure protocols – an idea that is supported by experience in human medicine. ¹⁰⁰ For example, in one retrospective study of 20,902 elderly patients in heart failure, prescription of ACEis, beta-adrenergic blockers or both, was associated with similar reductions in 1-year mortality risk in patients with and without azotemic CKD (ie, serum creatinine >2.0 mg/dl) compared with patients treated with neither drug. ¹⁰¹

However, in that study and another,¹⁰² these drugs were less often prescribed to patients with azotemic CKD compared with those without.

Impacts of other organ system diseases on cardiovascular and renal function

Several extrarenal and extracardiac conditions, including infectious (eg, sepsis,¹⁰³ feline infectious peritonitis¹⁰⁴), neoplastic (eg, lymphosarcoma¹⁰⁵) and endocrine (eg, hyperthyroidism,^{106,107} hypersomatotropism,¹⁰⁸ primary hyperaldosteronism^{109,110}) diseases can have deleterious effects on cardiovascular and renal function, structure or both. Discussion of these conditions is beyond the scope of this review; however, clinicians should consider their interactions with the cardiovascular–renal axis in affected cats.



[Open in a new tab](#)

Recommendations for diagnostic testing in patients with cardiovascular or kidney disease

Identification of heart or kidney disease in a cat should prompt the clinician to consider how the disease and its treatment might impact the function of the other organ and how coexisting primary disease of the other might impact case management and patient prognosis. This might include complete disease staging and screening for other diseases and conditions (eg, hyperthyroidism, systemic arterial hypertension) that might explain or exacerbate disease in one or both systems.

Assessment of kidney function in patients with heart disease

The IRIS classification systems for AKI and CKD⁷⁵ are useful for the characterization and grading (AKI; [Table 2](#)) or staging (CKD; [Table 3](#)) of kidney function in any patient, including those with heart disease.

Table 2.

International Renal Interest Society (IRIS) grading and subgrading criteria for acute kidney injury (AKI) [109](#)

AKI grading			
Grade	Blood creatinine		Clinical description
	μmol/l	mg/dl	
Grade I	<140	<1.6	Non-azotemic AKI AKI evidenced by the history, presence of oligoanuria (UOP <1 ml/kg/h), clinicopathologic data (eg, SDMA >reference interval, renal glucosuria, cylindruria, renal proteinuria, inflammatory sediment, microalbuminuria) or urinary tract imaging Includes cats with: – Progressive, non-azotemic increases in blood Cr ≥ 26.4 μmol/l (≥ 0.3 mg/dl) over 48 h – Volume-responsive AKI (ie, UOP >1 ml/kg/h within 6 h; decrease in Cr to baseline within 48 h)
Grade II	140 [*] –220	1.6 [*] –2.5	Mild AKI: Documented AKI (as defined for grade I) with mild azotemia Includes cats with: – Progressive, azotemic increases in blood Cr ≥ 26.4 μmol/l (≥ 0.3 mg/dl) over 48 h – Volume-responsive AKI (as defined for grade I)
Grade III	221–439	2.6–5.0	Moderate to severe AKI:
Grade IV	440–880	5.1–10.0	Documented AKI (as defined for grade I) with progressively greater degrees of parenchymal damage and functional failure (uremia)
Grade V	>880	>10.0	
AKI subgrading			

AKI grading

Grade	Blood creatinine		Clinical description
	μmol/l	mg/dl	
Characteristic			Clinical description
UOP	Non-oliguric		UOP >1 ml/kg/h
	Oligoanuric		UOP <1 ml/kg/h or anuria (ie, no urine produced) over 6 h
Need for RRT	Not required		Standard therapy sufficient
	Required		RRT is required to eliminate nephrotoxins or correct life-threatening iatrogenic or clinical consequences of AKI (eg, severe azotemia, hyperkalemia, acid–base disorders, overhydration, oligoanuria)

[Open in a new tab](#)

*Lower limits differ from published values (ie, 141 μmol/l and 1.7 mg/dl); changed here after personal communication with an IRIS board member Cr = creatinine; RRT = renal replacement therapy; SDMA = symmetric dimethylarginine; UOP = urine output

Table 3.

International Renal Interest Society (IRIS) staging and substaging of chronic kidney disease (CKD) in cats [109](#)

CKD staging				
Stage	Blood creatinine		BloodSDMA(µg/dl)	Clinical description
	µmol/l	mg/dl		
Stage 1	<140	<1.6	<18	Normal Cr; normal or mildly increased blood SDMA Other renal abnormalities noted (eg, inadequate urinary concentrating ability in the absence of a non-renal cause; abnormal renal palpation or imaging; renal proteinuria; SDMA persistently >14 µg/dl, increasing Cr or SDMA in serial samples; abnormal renal histopathology)
Stage 2	140–250	1.6–2.8	18–25	Normal or mildly increased creatinine, mild renal azotemia Mildly increased SDMA Clinical signs usually absent
Stage 3	251–440	2.9–5.0	26–38	Moderate renal azotemia Extrarenal signs may be present, but their extent and severity may vary
Stage 4	>440	>5.0	>38	Increasing risk of systemic clinical signs and uremic crises

CKD substaging based on proteinuria

UPC	Proteinuria substage
<0.2	Non-proteinuric
0.2–0.4	Borderline proteinuric

CKD staging				
Stage	Blood creatinine		BloodSDMA(μg/dl)	Clinical description
	μmol/l	mg/dl		
>0.4			Proteinuric	

CKD substaging based on blood pressure

Systolic BP (mmHg)	Blood pressure substage	Risk of TOD
<140	Normotensive	Minimal
140–159	Prehypertensive	Low
160–179	Hypertensive	Moderate
≥180	Severely hypertensive	High

[Open in a new tab](#)

BP = blood pressure; Cr = creatinine; SDMA = symmetric dimethylarginine; TOD = target organ damage; UPC = urinary protein:creatinine ratio

Evaluation of routine blood and urinary biomarkers – including markers of GFR (ie, serum creatinine, symmetric dimethylarginine [SDMA] and urea nitrogen concentrations), serum electrolyte and albumin concentrations, urine specific gravity, urinary protein:creatinine ratio, and presence of renal glucosuria or cylindruria – allows the clinician to screen for kidney injury with reasonable sensitivity. However, functional markers like serum creatinine and SDMA concentrations are poorly sensitive and specific for mild injury characterized by little (or current) impact on glomerular filtration. Furthermore, the influences of heart function and medications for cardiovascular disease on renal blood flow and GFR can substantially impact the utility of these markers to diagnose intrinsic kidney disease in cats with concurrent heart disease ([Table 4](#)). Repeated sampling and documentation of progressive increases in serum creatinine or SDMA concentrations within the normal reference interval are likely to increase the sensitivity of these markers for the detection of decreased kidney function.

Table 4.

Potential impacts of kidney or heart dysfunction on tests used to monitor or diagnose disease of the other organ

Diagnostics used for assessment of kidney function or disease

Test	Assesses	Considerations in patients with heart dysfunction
Creatinine BUN SDMA	GFR	✦ GFR is impacted by cardiac output and renal perfusion ✦ Circulating concentrations can be increased (signaling decreased GFR) in cats with heart failure, even in the absence of intrinsic renal disease
Urine output	Glomerular filtration and tubular function	✦ Filtration and tubular function are impacted by cardiac output and renal perfusion ✦ Urine output can be decreased in cats with heart failure

Diagnostics used for assessment of heart function or disease

Test	Assesses	Considerations in patients with kidney dysfunction
Cardiac auscultation	Heart sounds	✦ Heart murmurs or gallop sounds can be caused by chronic CKD-associated anemia or systemic arterial hypertension alone ✦ Electrolyte derangements can cause cardiac arrhythmias even in the absence of primary cardiac disease
NT-proBNP	Cardiac stretch or stress	✦ These markers undergo renal clearance
Cardiac troponin I	Cardiomyocyte damage	✦ Circulating concentrations can be increased in cats with kidney dysfunction,

Diagnostics used for assessment of kidney function or disease

Test	Assesses	Considerations in patients with heart dysfunction
Echocardiography	Cardiac structure and function	even in the absence of primary cardiac disease Even in the absence of primary heart disease: ✦ Hypovolemia from urinary fluid losses can reduce cardiac chamber size and lead to concentric 'pseudohypertrophy' of the ventricular walls, mimicking an HCM phenotype ✦ Chronic CKD-associated anemia can cause cardiac chamber dilatation ✦ Chronic CKD-associated systemic arterial hypertension can cause concentric left ventricular hypertrophy (HCM phenotype)
Thoracic radiography	Cardiac size and whether CHF is likely present	✦ Hypovolemia from urinary fluid losses can reduce heart size ✦ Chronic CKD-associated anemia can cause or exacerbate cardiomegaly or heart failure ✦ Chronic CKD-associated systemic arterial hypertension can cause (if severe and long-standing) or exacerbate cardiomegaly or heart failure

[Open in a new tab](#)

BUN = blood urea nitrogen; CKD = chronic kidney disease; CHF = congestive heart failure; GFR = glomerular filtration rate; HCM = hypertrophic cardiomyopathy; NT-proBNP = N-terminal prohormone of B-type natriuretic peptide; SDMA = symmetric dimethylarginine

For cats with heart disease whose treatment includes a diuretic, it can be challenging to differentiate among volume-responsive (previously, 'prerenal') azotemia, azotemia due to acute kidney injury and chronic cardiovascular disease leading to progressive CKD (ie, type 2 cardiovascular–renal syndrome). Importantly, because of the urine-diluting effects of diuretics, urine specific gravity measurements cannot be used to help determine whether azotemia is most likely due to extrarenal causes. Ultrasonographic changes to the kidneys (see below), senior or geriatric patient age, and the presence of non-regenerative anemia increase concern for intrinsic renal azotemia. There is growing interest in the use of urinary biomarkers of kidney injury (eg, urinary cystatin B, clusterin, neutrophil gelatinase-associated lipocalin) to detect early or active/ongoing damage.^{[111–114](#)} Of these, only a test for urinary cystatin B is commercially available at the time of writing. The effects of extrarenal conditions, including heart and vascular diseases, on these markers deserve future study.

In cats with heart disease, radiographic or ultrasonographic evaluation of kidney structure can be useful to diagnose concurrent kidney disease. Imaging should be considered for cats whose kidney health status is questioned but unclear based on clinicopathologic findings. Small, irregularly shaped kidneys, greater than normal echogenicity of the renal cortex and medulla, decreased cortico-medullary distinction, loss of normal internal architecture or a combination of these supports a diagnosis of CKD.^{[115,116](#)} An intrinsic acute nephropathy is suggested by enlarged kidneys and increased cortical echogenicity with preserved corticomedullary distinction, whereas marked pyelectasia or hydronephrosis, hydroureter or ureteral calculi can signal obstructive nephropathy.^{[115,116](#)}

Assessment of cardiac structure and function in cats with kidney disease

Imaging of the cardiopulmonary structures is necessary for accurate and comprehensive assessment of heart structure and function, and is recommended for cats with kidney disease who also have historical or clinical findings suggestive of heart disease or failure. The American College of Veterinary Internal Medicine feline cardiomyopathy staging system can be used to characterize the clinical impact of cardiomyopathy in cats ([Table 5](#)).^{[25](#)}

Table 5.

American College of Veterinary Internal Medicine staging of feline cardiomyopathy [25](#)

Stage		Clinical description
A	Predisposed	✦ Cats predisposed to cardiomyopathy without evidence of myocardial disease
B1	Subclinical (low risk)	✦ Cats with cardiomyopathy but without clinical signs ✦ Normal atrial size or mild atrial enlargement ✦ Low risk of imminent CHF or ATE
B2	Subclinical (high risk)	✦ Cats with cardiomyopathy but without clinical signs ✦ Moderate or severe atrial enlargement ✦ Higher risk of imminent CHF or ATE ✦ Additional risk factors for CHF or ATE: presence of gallop sound or arrhythmia; decreased left atrial function; extreme left ventricular hypertrophy; left ventricular systolic dysfunction; spontaneous echogenic contrast or thrombus; regional wall motion abnormalities
C	Current or previous CHF or ATE	✦ Cats who have developed signs of CHF or ATE, even if these resolve with treatment
D	Refractory CHF	✦ Cats with signs of CHF refractory to high oral dosages (>6 mg/kg/day) of furosemide

[Open in a new tab](#)

ATE =arterial thromboembolism; CHF = congestive heart failure

In an ideal world, and assuming no practical or financial limitations, all cats with known kidney disease would undergo assessment of cardiac structure and function using complementary information from echocardiography and thoracic radiography. This is because in this species, physical examination findings (eg, presence of a heart murmur or gallop) and thoracic radiography

alone have poor sensitivity for the identification of subclinical heart disease and heart enlargement, respectively.^{1,117,118} Instead, the clinical gold standard for assessment of cardiac structure and function is echocardiography, which allows for the diagnosis, specific phenotyping and staging of heart disease. The findings of thoracic radiography complement echocardiographic data by determining whether signs of CHF are present.

Echocardiography is not feasible in all cases or for all owners, however. The most practical reason to screen for heart disease in cats with kidney disease is to identify individuals who, compared with cats without heart disease, are at meaningfully increased risk for complications (eg, CHF, ATE, cardiac arrhythmias) when challenged with certain interventions or stressors. This largely equates to cats with moderate or severe heart disease/injury and enlargement.¹¹⁹

The accuracy of widely available circulating cardiac biomarker (eg, N-terminal prohormone B-type natriuretic peptide [NT-proBNP], cardiac troponin I) concentrations to establish the presence of subclinical heart disease or injury in cats with kidney disease is limited by several factors. Among the most relevant is that both biomarkers undergo renal clearance, which decreases their specificity for cardiac injury in cats with abnormal kidney function ([Table 4](#)).^{120,121} In addition, little is known about their performance compared with echocardiography in cats with AKI or CKD. Therefore, the use of cardiac biomarker concentrations in isolation to rule in heart disease or injury in this situation is not advised. However, NT-proBNP concentration might be a useful component of screening protocols in certain cases, as discussed below.¹²²

In the authors' practice, echocardiography is most strongly recommended for cats with kidney disease who have one or more of the following, particularly when treatments (ie, parenteral fluids, blood transfusion) that increase the workload of the heart are planned: (1) historical or clinical findings suggestive of current or previous heart disease or failure; (2) thoracic radiographic findings suggestive of cardiomegaly, cardiogenic edema or effusions, or both; or (3) physical examination findings that, when present, imply increased risk of cardiomyopathy (eg, loud [$\geq$ grade IV/VI] heart murmur, gallop sound or arrhythmia).^{25,119} The information provided can allow for the formulation of a rational treatment plan that minimizes the risk of excess hemodynamic stressors on a diseased heart. This approach is supported by the findings of one study in which 61 (51%) of 120 cats with CHF experienced antecedent events (eg, intravenous fluid administration, anesthesia, surgery, corticosteroid administration) that might have precipitated the episode.¹²³

In cats for whom echocardiography is not feasible, the combined results of serum NT-proBNP concentrations and thoracic radiographs can help to identify cases that are unlikely to have moderate or severe heart disease or failure. Importantly, this approach will misclassify some cats

with mild forms of disease (false negative) or normal hearts (false positive), although ‘clinically relevant’ disease is less likely in older cats with normal findings in both tests. ¹²⁴ However, this approach has not been rigorously tested in cats with kidney disease.

Assessment of systemic arterial BP in cats with heart or kidney disease

Given the link between BP abnormalities and heart and kidney disease, the authors, as well as a consensus panel of veterinary cardiologists and nephrologists, ⁹ recommend routine monitoring of systemic arterial BP in cats with diseased kidneys, with the goal of identifying and correcting pathologic systemic arterial hypertension or hypotension. In the authors’ practice, routine BP monitoring is also performed in cats with moderate or severe heart disease who have risk factors for systemic hypertension, clinical signs of CHF or are receiving medications known to impact systemic BP.

Considerations for the management of patients with concurrent cardiovascular and renal diseases

A common goal in the management of CHF, AKI or CKD is to achieve and maintain optimal body fluid balance. However, many therapies for disease of one system are fundamentally at odds with the best interest of the other. For example, the treatment of kidney disease often includes parenteral fluid therapy with the goal of maintaining mean arterial pressure and adequate renal blood flow, while diuretics are almost always necessary to reduce capillary hydrostatic pressure in cats with CHF. Overly aggressive volume expansion or reduction during the management of kidney or heart failure, respectively, can lead to injury to or excessive stress on the other system. Two commonly encountered clinical scenarios are described on the following pages to illustrate the balance required to manage cats with concurrent underlying disease of both systems. In general, the approach should be to treat the system whose decompensation is responsible for the patient’s clinical signs in a manner that minimizes risk to the other system and the cardiovascular–renal axis.



Cats with active CHF and concurrent CKD or azotemia

Fluid balance can be difficult to achieve in cats with CHF and coexisting kidney disease. A primary goal should be to relieve and prevent recurrence of life-threatening cardiogenic edema or effusions; however, excessive reduction in circulating blood volume can exacerbate hemodynamic azotemia or cause acute or acute on chronic kidney injury and uremia. In humans, that risk is greatest in those with diastolic heart failure,⁵ which is very common in cats. Some suggestions that might minimize the risk of renal decompensation are listed below.

Use the lowest effective dose of loop diuretics

In almost all cases of active CHF, the use of a loop diuretic (eg, furosemide) is necessary for the relief of life-threatening cardiogenic edema or to slow or stop the re-accumulation of cavity effusions.²⁵ Cats with cardiomyopathies characterized by impaired ventricular diastolic function can be particularly sensitive to aggressive diuresis: congestion does not always correspond to intravascular volume overload,¹²⁵ and even small decreases in blood volume can lead to systemic hypotension because of poor ventricular compliance and a steep relationship between preload and ventricular stroke volume.^{125,126}

The use of furosemide constant rate infusion (CRI) was the only examined variable that predicted kidney injury during early hospitalization in one study of cats, although in the same study kidney injury did not affect survival.⁷ In another, cats treated with furosemide CRI experienced greater

increases in serum creatinine concentration than those treated with intermittent bolus dosing.²⁹ Therefore, use of the lowest furosemide dose that effectively controls cardiogenic edema or effusions, and caution when using furosemide as a CRI in cats with pre-existing kidney disease is advisable. Diuretic dosing should be most heavily guided by the severity of clinical signs. In the authors' practice, diuretic dosing is also guided – among other factors – by echocardiographic assessment of left atrial size, radiographic severity of edema and whether the CHF episode appears to have been precipitated by acute changes in loading conditions (eg, acute ATE, iatrogenic fluid overload). The milder the left atrial enlargement, the milder the radiographic signs of cardiogenic edema and the higher the likelihood that acute increases in cardiac workload precipitated the episode, the more conservative the diuretic dose used should be.

Use of the lowest furosemide dose that effectively controls cardiogenic edema or effusions, and caution when using furosemide as a CRI in cats with pre-existing kidney disease is advisable.



Quickly identify and remove any clinically relevant cavity effusions

For cats in whom the primary manifestation of CHF is pleural effusion of sufficient volume to impact normal breathing, fluid removal should improve renal and myocardial oxygen delivery by improving ventilation.¹²⁷ Although large volume

ascites is uncommon in cats with CHF due to the most common cardiomyopathies, if present, its removal should lessen the contribution of increased intra-abdominal pressure to renal venous congestion. As diuretics alone are ineffective at quickly resolving large existing effusions, thoraco- or abdominocentesis is indicated.

Consider positive inotropic support

Although the primary abnormality of an HCM phenotype is diastolic dysfunction, longitudinal systolic (ie, contractile) performance can also be impaired in cats with HCM, especially those with CHF,¹²⁸ implying that drugs that support systolic function (eg, pimobendan, dobutamine at dosages that do not cause or worsen tachycardia) might be useful. Compared with diuretics, these drugs should have less of a negative impact on renal

blood flow, although this is unstudied in cats. Some cardiologists recommend that pimobendan is reserved for cats with overt ventricular systolic dysfunction (eg, dilated cardiomyopathy phenotype or end-stage HCM) or for cats with HCM phenotype without dynamic left ventricular outflow tract obstruction for fear of worsening outflow tract obstruction.²⁵ However, others have reported its safety and tolerability in cats with HCM phenotype with or without obstruction.¹²⁹

Resist the temptation to administer intravenous fluids in azotemic patients with active cardiogenic edema

Administration of parenteral fluids might seem attractive in cats with active CHF and azotemia. However, this will only worsen cardiogenic edema. There is no published evidence to suggest that cats treated for heart failure who are

azotemic or develop worsening of azotemia during hospitalization have poorer long-term outcomes compared with cats who do not.⁷ While common sense dictates that severe AKI is avoided when possible, intravenous fluid administration is contraindicated in patients who have symptomatic cardiogenic pulmonary edema.

Continued on next page

[Open in a new tab](#)



Continued from previous page

Identify and address systemic arterial hypotension

In one study, 16% of cats hospitalized for acute CHF were hypotensive (ie, systolic <90 mmHg) on presentation.²⁸ Systemic arterial BP should be assessed at intake and monitored frequently in cats with active CHF. Hypotension,

which can cause both kidney and cardiovascular injury through hypoperfusion, can then be addressed using positive inotropes (eg, pimobendan or dobutamine) and medications that might exacerbate hypotension discontinued, when possible.

Measure indicators of hydration status and kidney function frequently and adjust medication dosages accordingly

Azotemia in isolation does not indicate that diuretic therapy should be withheld in a cat with clinical signs of congestion.²⁵ However, markers of kidney function (ie, serum creatinine and urea nitrogen concentration), serum electrolyte concentrations and indicators of hydration status (ie, packed cell volume [PCV]/total solids [TS], body weight, physical examination) should be evaluated regularly and the results used to carefully adjust therapy. In the authors' practice, serum creatinine, urea nitrogen, electrolyte concentrations and PCV/TS are measured once daily, and body weight measurement and physical examination are

performed twice daily, at a minimum. Urine production should also be monitored, as development of oligoanuria might signal AKI. In addition, RAAS blockers (eg, ACEis, ARBs) are temporarily discontinued in most cats with acute CHF necessitating hospitalization, especially those with known azotemic CKD (ie, IRIS stages 2, 3 or 4) or systemic hypotension. Once CHF is controlled, hydration status has stabilized, and normal drinking and eating behaviors have resumed, these inhibitors are reinitiated in most patients. In a recent retrospective study, chronic use of ACEis in cats with CHF was associated with fewer lifetime episodes of AKI, although whether this finding reflected a renoprotective effect or simply prescription bias could not be determined.⁷

Optimize nutritional intake

Cardiac cachexia is associated with reduced survival in cats with CHF,¹³⁰ and prioritization of caloric and high-protein intake has been advocated in these cases.^{25,131} This could result in the long-term withholding of a diet optimized for kidney disease, for which moderate protein restriction can improve nitrogenous and acid-base balance, reduce uremic toxin formation and the risk for development of uremic crises, ameliorate glomerular hyperfiltration and reduce oxidative stress, among other benefits.^{132–134} Moderate protein restriction (ie, 7–8 g/100 kcal¹³⁵) achieved by feeding a therapeutic renal diet is recommended for cats with

azotemic CKD, especially those with advanced stages.⁷⁵ In cats with concurrent kidney and heart disease, an intermediate- or long-term goal should be to encourage caloric and protein intake that minimizes both uremia and loss of lean body mass. This can be achieved in many cases by offering a therapeutic renal diet and encouraging consumption by feeding and environmental modifications or pharmacologic interventions (eg, appetite stimulants), if necessary. Importantly, caloric intake should be prioritized over an ideal nutritional profile in the acute setting, and therapeutic renal diets should be avoided during hospitalization to lessen the risk of food aversion, unless they are preferred by the affected cat.

Cats with AKI or acute-on-chronic kidney disease and concurrent heart disease or stable CHF

Achieving appropriate fluid balance in cats with AKI or acute-on-chronic kidney disease and concurrent heart disease or stable CHF can also be challenging. Here, the primary goal should be to maintain renal perfusion and support GFR while addressing the underlying cause of the uremic episode and preventing a pathologic decrease or increase in circulating blood volume. Suggestions for assisting with this are listed below.

For patients with mild azotemia who are receiving chronic oral therapy for CHF, consider a diuretic 'vacation' alone (ie, before parenteral fluid administration) as a first step in optimizing hydration and circulating volume

Depending on the degree of azotemia, clinical signs and whether a correctable precipitating factor (eg, iatrogenic

diuretic overdose, systemic or organ illness leading to excessive decreases in circulating blood volume) is identified, this might mean temporary discontinuation of diuretics for one to two days or longer, with reintroduction at a lower dose or frequency once it is considered safe to do so.

Consider hospitalization

Hospitalization could facilitate careful monitoring of respiratory rate and effort during fluid therapy.

Continued on next page

[Open in a new tab](#)

Remember that the goal of fluid therapy in kidney disease is to maintain euvolesmia and euhydration

In any cat, but particularly those with concurrent heart disease, fluid administration should be adjusted to the individual's needs. Overhydration, which is often justified by an effort to 'diurese' or 'flush the kidneys', must be avoided. In practical terms, once normal hydration and circulating volume have been restored, enteral and parenteral fluid administration must not exceed losses. Some suggestions to achieve this goal include the following:

❖ **Carefully assess patient volume status prior to fluid therapy.** In the authors' practice, cats with known or suspected heart disease frequently undergo echocardiographic evaluation of heart size as one indicator of circulating blood volume and for disease staging. For cats with severe left atrial dilatation, enteral fluids – administered via nasoesophageal or nasogastric tube – are part (or often the sole source) of fluid support.

❖ **Adjust total volume, rate and route of fluid administration to avoid iatrogenic fluid overload.** If fluid administration is necessary in a patient with volume-responsive azotemia, correction of deficits should be performed over a longer period than might be typical for a patient without heart disease or controlled CHF.¹³⁶ In some cases, placement of a feeding tube to facilitate enteral water administration, which is virtually sodium free and less likely to result in fluid overload, can be useful. For cats with acute-on-chronic kidney disease who are unable to maintain normal hydration, esophagostomy tubes can be considered for chronic administration of water in lieu of higher sodium-containing subcutaneous fluids. When required, blood

transfusions should be given as slowly as possible (ie, over 4–6 h) to balance the risks of circulatory overload and bacterial contamination.³²

In cats with severe volume-responsive azotemia and uremia, parenteral fluid administration is often necessary. Depending on the feasibility of hospitalization, this might involve subcutaneous administration for patients who must be managed at home or intravenous administration alongside close monitoring of respiratory rate for cats who can be hospitalized. The authors prefer the intravenous route over the subcutaneous route in cats with substantial heart disease or controlled heart failure, as the former allows for administration of lower sodium fluids (eg, 0.45% saline or 5% dextrose solutions), more precise dose titration and the ability to stop administration altogether if clinical or radiographic signs suggest iatrogenic fluid overload.

❖ **Measure indicators of hydration status frequently and adjust medication dosages and rate of fluid administration accordingly.** Regular assessment of body weight, PCV/TS, serum albumin concentration, and respiratory rate and effort is imperative in cats with moderate or severe heart disease who are being treated with parenteral fluids. Sleeping respiratory rates that increase acutely or exceed 30 breaths per minute^{137,138} should prompt discontinuation of fluid therapy. If persistent tachypnea or increased respiratory effort occur, point-of-care ultrasound and thoracic radiography should be used to examine for signs of congestion and its severity. In cats with volume overload and respiratory signs, thoracocentesis (for treatment of pleural effusion) or cautious use of diuretics (for treatment of pulmonary edema) might be necessary for a short period.

Overhydration, which is often justified by an effort to 'diurese' or 'flush the kidneys', must be avoided.



Identify and address systemic arterial hypotension

Systemic arterial BP should be assessed at intake and monitored frequently. In systemically hypotensive cats, tempo-

rary discontinuation of medications that might exacerbate hypotension (ie, ACEis, ARBs) should be considered, when possible.

Avoid sodium overload

Low-sodium fluids (eg, 0.45% saline solution) should be used when parenteral fluid administration is necessary; alternatively, enteral administration of salt-free water can be con-

sidered. Administration of normal (0.9%) saline solution is discouraged in cats with AKI, regardless of whether heart disease coexists, due to its potentially negative effect on GFR and acid-base status.¹³⁶

Avoid serum potassium derangements and treat any cardiac arrhythmias

Both hyper- and hypokalemia can cause or exacerbate car-

diac arrhythmias. Cats with known hyper- or hypokalemia and those with oliguria or anuria should undergo continuous electrocardiogram monitoring during hospitalization.

Consider referral for renal replacement therapy in certain cases

Renal replacement therapy can be of benefit to patients with oligoanuria, uremia, metabolic derangements (eg, hyper-

kalemia) or when iatrogenic fluid overload is refractory to standard therapy. Prompt identification and correction of these abnormalities can improve cardiovascular function and overall patient status.

Key Points

- ❖ Heart and kidney diseases frequently coexist in the same cat.

There are several pathways of interaction between the heart and kidneys during health and disease, and it is helpful to think of these systems as belonging to an integrated 'cardiovascular–renal axis'.

- ❖ One of the most well-described links between heart and kidney diseases is systemic arterial hypertension, which is frequently associated with CKD in cats, and can lead to progressive damage to both. Given the link between BP abnormalities and heart and kidney diseases, routine monitoring of systemic arterial BP and the use of antihypertensive therapy (eg, amlodipine, telmisartan) in cats with diseased kidneys is recommended.

- ❖ The general approach to the treatment of cats with both conditions should be to treat the system whose decompensation is responsible for the patient's clinical signs in a manner that minimizes risk to the other system and the cardiovascular–renal axis.

Footnotes

Amanda Coleman declares no potential conflicts of interest with respect to the research, authorship, and/or publication of this article. Bianca Lourenço has received research funding, speaker honoraria, and consulting fees from Ceva Animal Health and Boehringer Ingelheim Vetmedica, and research funding from Elanco Animal Health.

Funding: The authors of this commissioned Clinical Spotlight review received an honorarium; as for all JFMS articles, this Clinical Spotlight article went through peer review.

Ethical approval: This work did not involve the use of animals and therefore ethical approval was not specifically required for publication in JFMS.

Informed consent: This work did not involve the use of animals (including cadavers) and therefore informed consent was not required. No animals or people are identifiable within this publication, and

therefore additional informed consent for publication was not required.

ORCID iD: Amanda E Coleman  <https://orcid.org/0000-0001-5476-5963> 

Bianca N Lourenço  <https://orcid.org/0000-0001-5249-4645> 

References

1. Paige CF, Abbott JA, Elvinger F. et al. Prevalence of cardiomyopathy in apparently healthy cats. J Am Vet Med Assoc 2009; 234: 1398–1403. [[DOI](#) ] [[PubMed](#)] [[Google Scholar](#) ]
2. Payne JR, Brodbelt DC, Luis Fuentes V. Cardiomyopathy prevalence in 780 apparently healthy cats in rehoming centres (the CatScan study). J Vet Cardiol 2015; 17 Suppl 1: 244–257. [[DOI](#) ] [[PubMed](#)] [[Google Scholar](#) ]
3. Fox PR, Keene BW, Lamb K. et al. International collaborative study to assess cardiovascular risk and evaluate long-term health in cats with preclinical hypertrophic cardiomyopathy and apparently healthy cats: the REVEAL study. J Vet Intern Med 2018; 32: 930–943. [[DOI](#) ] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#) ]
4. O'Neill DG, Church DB, McGreevy PD. et al. Longevity and mortality of cats attending primary care veterinary practices in England. J Feline Med Surg 2015; 17: 125–133. [[DOI](#) ] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#) ]
5. Marino CL, Lascelles BD, Vaden SL. et al. Prevalence and classification of chronic kidney disease in cats randomly selected from four age groups and in cats recruited for degenerative joint disease studies. J Feline Med Surg 2014; 16: 465–472. [[DOI](#) ] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#) ]
6. O'Neill DG, Church DB, McGreevy PD. et al. Prevalence of disorders recorded in cats attending primary-care veterinary practices in England. Vet J 2014; 202: 286–291. [[DOI](#) ] [[PubMed](#)] [[Google Scholar](#) ]
7. Rogg S, Mochel JP, Kundu D. et al. Frequency and progression of azotemia during acute and chronic treatment of congestive heart failure in cats. J Vet Intern Med 2025; 39. DOI: 10.1111/jvim.17254. [[DOI](#) ] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#) ]

8. Duperrier-Simond C, Pouzot-Nevoret C, Allaouchiche B. et al. Occurrence of cardiovascular events in 168 cats with acute urinary tract obstruction. *Can Vet J* 2024; 65: 67–74. [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
9. Pouchelon JL, Atkins CE, Bussadori C. et al. Cardiovascular–renal axis disorders in the domestic dog and cat: a veterinary consensus statement. *J Small Anim Pract* 2015; 56: 537–552. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
10. Adams KF, Jr. Pathophysiologic role of the renin–angiotensin–aldosterone and sympathetic nervous systems in heart failure. *Am J Health Syst Pharm* 2004; 61 Suppl 2: 4–13. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
11. Mohebi R, Liu Y, van Kimmenade R. et al. Inflammation across universal definition of heart failure stages: the CASABLANCA study. *Eur J Heart Fail* 2023; 25: 152–160. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
12. Liu M, Köster LS, Fosgate GT. et al. Cardiovascular–renal axis disorder and acute phase proteins in cats with congestive heart failure caused by primary cardiomyopathy. *J Vet Intern Med* 2020; 34: 1078–1090. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
13. Li Q, Homilius M, Achilles E. et al. Metabolic abnormalities and reprogramming in cats with naturally occurring hypertrophic cardiomyopathy. *ESC Heart Failure* 2025; 12: 1256–1270. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
14. Mimura I, Nangaku M. The suffocating kidney: tubulointerstitial hypoxia in endstage renal disease. *Nat Rev Nephrol* 2010; 6: 667–678. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
15. Dupont M, Mullens W, Tang WH. Impact of systemic venous congestion in heart failure. *Curr Heart Fail Rep* 2011; 8: 233–241. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
16. Lo KB, Rangaswami J. Mechanistic insights in cardiorenal syndrome. *NEJM Evid* 2022; 1. DOI: 10.1056/EVIDra2200053. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
17. Cowgill LD, Polzin DJ, Elliott J. et al. Is progressive chronic kidney disease a slow acute kidney injury? *Vet Clin North Am Small Anim Pract* 2016; 46: 995–1013. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
18. Smith SA, Tobias AH, Jacob KA. et al. Arterial thromboembolism in cats: acute crisis in 127 cases (1992–2001) and long-term management with low-dose aspirin in 24 cases. *J Vet Intern Med/ACVIM* 2003; 17: 73–83. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

19. Gibson SD, Lourenco BN, De la Rosa MB. et al. Diagnostic value of echocardiography in cats with and without ultrasonographic evidence of renal infarction. *J Vet Intern Med* 2025; 39. DOI: 10.1111/jvim.70107. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
20. Hickey MC, Jandrey K, Farrell KS. et al. Concurrent diseases and conditions in cats with renal infarcts. *J Vet Intern Med* 2014; 28: 319–323. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
21. Lee Y, Baek M, Lee D. et al. Retrospective evaluation of risk factors for worsening renal function after angiotensin-converting enzyme inhibitor treatment in dogs. *J Vet Intern Med* 2025; 39. DOI: 10.1111/jvim.17252. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
22. Duchin KL, Peterson LN, Burke TJ. Effect of furosemide on renal autoregulation. *Kidney Int* 1977; 12: 379–386. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
23. Kaloyanides GJ, Bastron RD, DiBona GF. Impaired autoregulation of blood flow and glomerular filtration rate in the isolated dog kidney depleted of renin. *Circ Res* 1974; 35: 400–412. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
24. Rishniw M, Pion PD. Is treatment of feline hypertrophic cardiomyopathy based in science or faith? A survey of cardiologists and a literature search. *J Feline Med Surg* 2011; 13: 487–497. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
25. Luis Fuentes V, Abbott J, Chetboul V. et al. ACVIM consensus statement guidelines for the classification, diagnosis, and management of cardiomyopathies in cats. *J Vet Intern Med* 2020; 34: 1062–1077. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
26. Just A, Wittmann U, Ehmke H. et al. Autoregulation of renal blood flow in the conscious dog and the contribution of the tubu-loglomerular feedback. *J Physiol* 1998; 506 (Pt 1): 275–290. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
27. Gouni V, Chetboul V, Pouchelon JL. et al. Azotemia in cats with feline hypertrophic cardiomyopathy: prevalence and relationships with echocardiographic variables. *J Vet Cardiol* 2008; 10: 117–123. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
28. Goutal CM, Keir I, Kenney S. et al. Evaluation of acute congestive heart failure in dogs and cats: 145 cases (2007–2008). *J Vet Emerg Crit Care (San Antonio)* 2010; 20: 330–337. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
29. Ohad DG, Segev Y, Kelmer E. et al. Constant rate infusion vs. intermittent bolus administration of IV furosemide in 100 pets with acute left-sided congestive heart failure: a

retrospective study. *Vet J* 2018; 238: 70–75. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

30. Harison E, Langston C, Palma D. et al. Acute azotemia as a predictor of mortality in dogs and cats. *J Vet Intern Med* 2012; 26: 1093–1098. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

31. Langston C, Gordon D. Effects of IV fluids in dogs and cats with kidney failure. *Front Vet Sci* 2021; 8: DOI: 10.3389/fvets.2021.659960. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

32. Taylor S, Spada E, Callan MB. et al. 2021. ISFM consensus guidelines on the collection and administration of blood and blood products in cats. *J Feline Med Surg* 2021; 23: 410–432. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

33. Bijsmans ES, Jepson RE, Chang YM. et al. Changes in systolic blood pressure over time in healthy cats and cats with chronic kidney disease. *J Vet Intern Med* 2015; 29: 855–861. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

34. Kobayashi DL, Peterson ME, Graves TK. et al. Hypertension in cats with chronic renal failure or hyperthyroidism. *J Vet Intern Med* 1990; 4: 58–62. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

35. Cole L, Jepson R, Humm K. Systemic hypertension in cats with acute kidney injury. *J Small Anim Pract* 2017; 58: 577–581. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

36. Lawson JS, Jepson RE. Feline comorbidities: the intermingled relationship between chronic kidney disease and hypertension. *J Feline Med Surg* 2021; 23: 812–822. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

37. Coleman AE, Brown SA. Hypertension and the heart and vasculature. In: Elliott J, Syme HM, Jepson RE. (eds). *Hypertension in the dog and cat*. Springer, 2020, pp 187–215. [[Google Scholar](#)]

38. Elliott J, Brown CA. Hypertension and the kidney. In: Elliott J, Syme HM, Jepson RE. (eds). *Hypertension in the dog and cat*. Springer, 2020, pp 171–186. [[Google Scholar](#)]

39. Chetboul V, Lefebvre HP, Pinhas C. et al. Spontaneous feline hypertension: clinical and echocardiographic abnormalities, and survival rate. *J Vet Intern Med* 2003; 17: 89–95. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

40. Henik RA, Stepien RL, Bortnowski HB. Spectrum of M-mode echocardiographic abnormalities in 75 cats with systemic hypertension. *J Am Anim Hosp Assoc* 2004; 40: 359–

363. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

41. Littman MP. Spontaneous systemic hypertension in 24 cats. J Vet Intern Med 1994; 8: 79–86. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

42. Snyder PS, Sadek D, Jones GL. Effect of amlodipine on echocardiographic variables in cats with systemic hypertension. J Vet Intern Med 2001; 15: 52–56. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

43. Elliott J, Barber PJ, Syme HM. et al. Feline hypertension: clinical findings and response to antihypertensive treatment in 30 cases. J Small Anim Pract 2001; 42: 122–129. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

44. Flora Z, Tang PK, Smith K. et al. Cardiac pathology associated with hypertension and chronic kidney disease in aged cats. J Comp Pathol 2025; 216: 40–49. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

45. Hoehne SN, Hopper K, Epstein SE. Retrospective evaluation of the severity of and prognosis associated with potassium abnormalities in dogs and cats presenting to an emergency room (January 2014–August 2015): 2441 cases. J Vet Emerg Crit Care (San Antonio) 2019; 29: 653–661. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

46. Coady M, Fletcher DJ, Goggs R. Severity of ionized hypercalcemia and hypocalcemia is associated with etiology in dogs and cats. Front Vet Sci 2019; 6: 276. DOI: 10.3389/fvets.2019.00276. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

47. Savary KC, Price GS, Vaden SL. Hypercalcemia in cats: a retrospective study of 71 cases (1991–1997). J Vet Intern Med 2000; 14: 184–189. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

48. Lee YJ, Chan JP, Hsu WL. et al. Prognostic factors and a prognostic index for cats with acute kidney injury. J Vet Intern Med 2012; 26: 500–505. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

49. DiBartola SP, Rutgers HC, Zack PM. et al. Clinicopathologic findings associated with chronic renal disease in cats: 74 cases (1973– 1984). J Am Vet Med Assoc 1987; 190: 1196–1202. [[PubMed](#)] [[Google Scholar](#)]

50. Elliott J, Barber PJ. Feline chronic renal failure: clinical findings in 80 cases diagnosed between 1992 and 1995. J Small Anim Pract 1998; 39: 78–85. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

51. Barber PJ, Elliott J. Feline chronic renal failure: calcium homeostasis in 80 cases diagnosed between 1992 and 1995. *J Small Anim Pract* 1998; 39: 108–116. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
52. Weiss JN, Qu Z, Shivkumar K. Electrophysiology of hypokalemia and hyperkalemia. *Circ Arrhythm Electrophysiol* 2017; 10. DOI: 10.1161/CIRCEP.116.004667. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
53. Tag TL, Day TK. Electrocardiographic assessment of hyperkalemia in dogs and cats. *J Vet Emerg Crit Care* 2008; 18: 61–67. [[Google Scholar](#)]
54. DiBartola SP. Management of hypokalaemia and hyperkalaemia. *J Feline Med Surg* 2001; 3: 181–183. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
55. Fabiato A, Fabiato F. Calcium and cardiac excitation–contraction coupling. *Ann Rev Physiol* 1979; 41: 473–484. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
56. Szedlak P, Steele DS, Hopkins PM. Cardiac muscle physiology. *BJA Educ* 2023; 23: 350–357. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
57. Ghent S, Judson MA, Rosansky SJ. Refractory hypotension associated with hypocalcemia and renal disease. *Am J Kidney Dis* 1994; 23: 430–432. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
58. Lang RM, Fellner SK, Neumann A. et al. Left ventricular contractility varies directly with blood ionized calcium. *Ann Intern Med* 1988; 108: 524–529. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
59. Fang Y, Ginsberg C, Sugatani T. et al. Early chronic kidney disease-mineral bone disorder stimulates vascular calcification. *Kidney Int* 2014; 85: 142–150. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
60. Kovesdy CP, Kuchmak O, Lu JL. et al. Outcomes associated with serum calcium level in men with non-dialysis-dependent chronic kidney disease. *Clin J Am Soc Nephrol* 2010; 5: 468–476. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
61. Shanahan CM, Crouthamel MH, Kapustin A. et al. Arterial calcification in chronic kidney disease: key roles for calcium and phosphate. *Circ Res* 2011; 109: 697–711. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

62. Tentori F, Blayney MJ, Albert JM. et al. Mortality risk for dialysis patients with different levels of serum calcium, phosphorus, and PTH: the Dialysis Outcomes and Practice Patterns Study (DOPPS). *Am J Kidney Dis* 2008; 52: 519–530. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
63. Foster JD. Update on mineral and bone disorders in chronic kidney disease. *Vet Clin North Am Small Anim Pract* 2016; 46: 1131–1149. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
64. Tang PK, Geddes RF, Jepson RE. et al. A feline-focused review of chronic kidney diseasemineral and bone disorders – part 2: pathophysiology of calcium disorder and extraosseous calcification. *Vet J* 2021; 275. DOI: 10.1016/j.tvjl.2021.105718. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
65. Salera D, Merkel N, Bellasi A. et al. Pathophysiology of chronic kidney diseasemineral bone disorder (CKD-MBD): from adaptive to maladaptive mineral homeostasis. *Clin Kidney J* 2025; 18 Suppl 1: i3–i14. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
66. King JN, Tasker S, Gunn-Moore DA. et al. Prognostic factors in cats with chronic kidney disease. *J Vet Intern Med* 2007; 21: 906–916. [[PubMed](#)] [[Google Scholar](#)]
67. Geddes RF, Elliott J, Syme HM. Relationship between plasma fibroblast growth factor-23 concentration and survival time in cats with chronic kidney disease. *J Vet Intern Med* 2015; 29: 1494–1501. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
68. Chakrabarti S, Syme HM, Elliott J. Clinicopathological variables predicting progression of azotemia in cats with chronic kidney disease. *J Vet Intern Med* 2012; 26: 275–281. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
69. Hruska KA, Mathew S, Lund R. et al. Hyperphosphatemia of chronic kidney disease. *Kidney Int* 2008; 74: 148–157. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
70. Seiler S, Reichart B, Roth D. et al. FGF-23 and future cardiovascular events in patients with chronic kidney disease before initiation of dialysis treatment. *Nephrol Dial Transplant* 2010; 25: 3983–3989. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
71. Kendrick J, Cheung AK, Kaufman JS. et al. FGF-23 associates with death, cardiovascular events, and initiation of chronic dialysis. *J Am Soc Nephrol* 2011; 22: 1913–1922. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
72. Elliott J. Therapeutics of managing reduced red cell mass associated with chronic kidney disease – is there a case for earlier intervention? *J Vet Pharmacol Ther* 2023; 46: 145–157. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

73. Anand IS. Anemia and chronic heart failure implications and treatment options. *J Am Coll Cardiol* 2008; 52: 501–511. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
74. Wilson HE, Jasani S, Wagner TB. et al. Signs of left heart volume overload in severely anaemic cats. *J Feline Med Surg* 2010; 12: 904–909. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
75. International Renal Interest Society. IRIS guidelines. www.iris-kidney.com/iris-guidelines-1 (2023, accessed 3 June 2025).
76. Chalhoub S, Langston CE, Farrelly J. The use of darbepoetin to stimulate erythropoiesis in anemia of chronic kidney disease in cats: 25 cases. *J Vet Intern Med* 2012; 26: 363–369. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
77. Cowgill LD, James KM, Levy JK. et al. Use of recombinant human erythropoietin for management of anemia in dogs and cats with renal failure. *J Am Vet Med Assoc* 1998; 212: 521–528. [[PubMed](#)] [[Google Scholar](#)]
78. Charles S, Sussenberger R, Settje T. et al. Use of molidustat, a hypoxia-inducible factor prolyl hydroxylase inhibitor, in chronic kidney disease-associated anemia in cats. *J Vet Intern Med* 2024; 38: 197–204. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
79. Zhang W, Li Y, Wang JG. Hypertension induced by hypoxia-inducible factor prolyl hydroxylase inhibitors in treating anemia in patients with chronic kidney disease: a minireview. *J Clin Hypertens (Greenwich)* 2024; 26: 1375–1383. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
80. Brewster UC, Setaro JF, Perazella MA. The renin–angiotensin–aldosterone system: cardiorenal effects and implications for renal and cardiovascular disease states. *Am J Med Sci* 2003; 326: 15–24. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
81. Levens NR, Peach MJ, Carey RM. Role of the intrarenal renin–angiotensin system in the control of renal function. *Circ Res* 1981; 48: 157–167. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
82. Carey RM. The intrarenal renin–angiotensin system in hypertension. *Adv Chronic Kidney Dis* 2015; 22: 204–210. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
83. Mitani S, Yabuki A, Sawa M. et al. Intrarenal distributions and changes of angiotensin-converting enzyme and angiotensin-converting enzyme 2 in feline and canine chronic kidney disease. *J Vet Med Sci* 2014; 76: 45–50. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

84. Mitani S, Yabuki A, Taniguchi K. et al. Association between the intrarenal reninangiotensin system and renal injury in chronic kidney disease of dogs and cats. *J Vet Med Sci* 2013; 75: 127–133. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
85. Lourenco BN, Coleman AE, Berghaus RD. et al. Characterization of the intrarenal renin-angiotensin system in cats with naturally occurring chronic kidney disease. *J Vet Intern Med* 2022; 36: 647–655. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
86. Kirch W, Kohler H, Mutschler E. et al. Pharmacokinetics of atenolol in relation to renal function. *Eur J Clin Pharmacol* 1981; 19: 65–71. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
87. Salmon SJ, Coleman AE, Lynn CR. et al. Single-and multiple-dose pharmacokinetics of sotalol hydrochloride in healthy cats. *J Vet Cardiol* 2024; 51: 86–96. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
88. Sleeper MM, O'Donnell P, Fitzgerald C. et al. Pharmacokinetics of furosemide after intravenous, oral and transdermal administration to cats. *J Feline Med Surg* 2019; 21: 882–886. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
89. Roche-Catholy M, Paepe D, Devreese M. et al. Pharmacokinetic and pharmacodynamic properties of orally administered torasemide in healthy cats. *J Vet Intern Med* 2022; 36: 1782–1791. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
90. Gehr TW, Rudy DW, Matzke GR. et al. The pharmacokinetics of intravenous and oral torsemide in patients with chronic renal insufficiency. *Clin Pharmacol Ther* 1994; 56: 31–38. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
91. Lefebvre HP, Laroute V, Concordet D. et al. Effects of renal impairment on the disposition of orally administered enalapril, benazepril, and their active metabolites. *J Vet Intern Med* 1999; 13: 21–27. [[PubMed](#)] [[Google Scholar](#)]
92. King JN, Strehlau G, Wernsing J. et al. Effect of renal insufficiency on the pharmacokinetics and pharmacodynamics of benazepril in cats. *J Vet Pharmacol Ther* 2002; 25: 371–378. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
93. Boehringer Ingelheim Vetmedica I. Vetmedin (pimobendan) chewable tablets [package insert]. St Joseph, MO: Boehringer Ingelheim Vetmedica, 2012. [[Google Scholar](#)]
94. Murphy PJ, Williams TL, Kau DL. Disposition of dobutamine in the dog. *J Pharmacol Exp Ther* 1976; 199: 423–431. [[PubMed](#)] [[Google Scholar](#)]

95. Lins R, Broekhuysen J, Necciari J. et al. Pharmacokinetic profile of C-labeled clopidogrel. *Semin Thromb Hemost* 1999; 25: 29–33. [[PubMed](#)] [[Google Scholar](#)]
96. Kreutz R. Pharmacodynamic and pharmacokinetic basics of rivaroxaban. *Fundam Clin Pharmacol* 2012; 26: 27–32. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
97. Wienen W, Entzeroth M, van Meel JCA. et al. A review on telmisartan: a novel, long-acting angiotensin II-receptor antagonist. *Cardiovasc Drug Rev* 2000; 18: 127–156. [[Google Scholar](#)]
98. Ebner T, Schanzle G, Weber W. et al. In vitro glucuronidation of the angiotensin II receptor antagonist telmisartan in the cat: a comparison with other species. *J Vet Pharmacol Ther* 2013; 36: 154–160. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
99. Stopher DA, Beresford AP, Macrae PV. et al. The metabolism and pharmacokinetics of amlodipine in humans and animals. *J Cardio vasc Pharmacol* 1988; 12 Suppl 7: 55–59. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
100. Ortiz A, Navarro-Gonzalez JF, Nunez J. et al. The unmet need of evidence-based therapy for patients with advanced chronic kidney disease and heart failure: position paper from the Cardiorenal Working Groups of the Spanish Society of Nephrology and the Spanish Society of Cardiology. *Clin Kidney J* 2022; 15: 865–872. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
101. Shlipak MG, Browner WS, Noguchi H. et al. Comparison of the effects of angiotensin converting-enzyme inhibitors and beta blockers on survival in elderly patients with reduced left ventricular function after myocardial infarction. *Am J Med* 2001; 110: 425–433. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
102. McAlister FA, Ezekowitz J, Tonelli M. et al. Renal insufficiency and heart failure: prognostic and therapeutic implications from a prospective cohort study. *Circulation* 2004; 109: 1004–1009. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
103. Klainbart S, Agi L, Bdolah-Abram T. et al. Clinical, laboratory, and hemostatic findings in cats with naturally occurring sepsis. *J Am Vet Med Assoc* 2017; 251: 1025–1034. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
104. Muller TR, Penninck DG, Webster CR. et al. Abdominal ultrasonographic findings of cats with feline infectious peritonitis: an update. *J Feline Med Surg* 2023; 25. DOI: 10.1177/1098612X231216000. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

105. Carter TD, Pariaut R, Snook E. et al. Multicentric lymphoma mimicking decompensated hypertrophic cardiomyopathy in a cat. *J Vet Intern Med* 2008; 22: 1345–1347. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
106. Sangster JK, Panciera DL, Abbott JA. et al. Cardiac biomarkers in hyperthyroid cats. *J Vet Intern Med* 2014; 28: 465–472. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
107. Geddes R, Aguiar J. Feline comorbidities: balancing hyperthyroidism and concurrent chronic kidney disease. *J Feline Med Surg* 2022; 24: 641–650. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
108. Scudder C, Church D. Feline comorbidities: hypersomatotropism-induced diabetes in cats. *J Feline Med Surg* 2024; 26. DOI: 10.1177/1098612X241226690. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
109. Javadi S, Djajadiningrat-Laanen SC, Kooistra HS. et al. Primary hyperaldosteronism, a mediator of progressive renal disease in cats. *Domest Anim Endocrinol* 2005; 28: 85–104. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
110. Ash RA, Harvey AM, Tasker S. Primary hyperaldosteronism in the cat: a series of 13 cases. *J Feline Med Surg* 2005; 7: 173–182. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
111. Chen H, Avital Y, Peterson S. et al. Urinary Cystatin B as a marker of acute kidney injury in cats. *Vet J* 2024; 308: DOI: 10.1016/j.tvjl.2024.106262. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
112. Hall JA, Forman FJ, Bobe G. et al. The impact of periodontal disease and dental cleaning procedures on serum and urine kidney biomarkers in dogs and cats. *PloS One* 2021; 16. DOI: 10.1371/journal.pone.0255310. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
113. KuKanich K, George C, Roush JK. et al. Effects of low-dose meloxicam in cats with chronic kidney disease. *J Feline Med Surg* 2021; 23: 138–148. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
114. Wang IC, Hsu WL, Wu PH. et al. Neutrophil gelatinase-associated lipocalin in cats with naturally occurring chronic kidney disease. *J Vet Intern Med* 2017; 31: 102–108. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

115. Debruyne K, Haers H, Combes A. et al. Ultrasonography of the feline kidney: technique, anatomy and changes associated with disease. *J Feline Med Surg* 2012; 14: 794–803. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
116. Griffin S. Feline abdominal ultrasonography: what's normal? what's abnormal? The kidneys and perinephric space. *J Feline Med Surg* 2020; 22: 409–427. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
117. Schober KE, Maerz I, Ludewig E. et al. Diagnostic accuracy of electrocardiography and thoracic radiography in the assessment of left atrial size in cats: comparison with transthoracic 2-dimensional echocardiography. *J Vet Intern Med* 2007; 21: 709–718. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
118. Guglielmini C, Diana A. Thoracic radiography in the cat: identification of cardiomegaly and congestive heart failure. *J Vet Cardiol* 2015; 17 Suppl 1: 87–101. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
119. Payne JR, Borgeat K, Connolly DJ. et al. Prognostic indicators in cats with hypertrophic cardiomyopathy. *J Vet Intern Med* 2013; 27: 1427–1436. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
120. Porciello F, Rishniw M, Herndon WE. et al. Cardiac troponin I is elevated in dogs and cats with azotaemia renal failure and in dogs with non-cardiac systemic disease. *Aust Vet J* 2008; 86: 390–394. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
121. Lalor SM, Connolly DJ, Elliott J. et al. Plasma concentrations of natriuretic peptides in normal cats and normotensive and hypertensive cats with chronic kidney disease. *J Vet Cardiol* 2009; 11 Suppl 1: 71–79. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
122. Rishniw M. Should we be screening cats for cardiomyopathy? If so, how? *J Am Vet Med Assoc* 2022; 260: 1626–1630. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
123. Rush JE, Freeman LM, Fenollosa NK. et al. Population and survival characteristics of cats with hypertrophic cardiomyopathy: 260 cases (1990–1999). *J Am Vet Med Assoc* 2002; 220: 202–207. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
124. Cote E, Edwards NJ, Ettinger SJ. et al. Management of incidentally detected heart murmurs in dogs and cats. *J Vet Cardiol* 2015; 17: 245–261. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

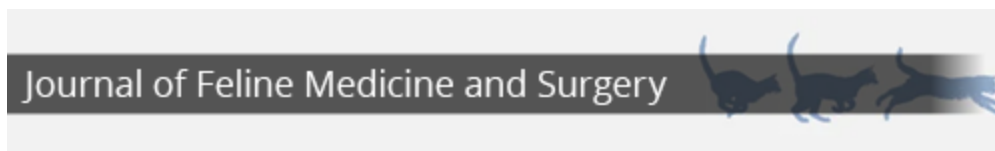
125. Kumric M, Kurir TT, Bozic J. et al. Pathophysiology of congestion in heart failure: a contemporary review. *Card Fail Rev* 2024; 10. DOI: 10.15420/cfr.2024.07. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
126. Patel RN, Sharma A, Prasad A. et al. Heart failure with preserved ejection fraction with CKD: a narrative review of a multispecialty disorder. *Kidney Med* 2023; 5. DOI: 10.1016/j.xkme.2023.100705. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
127. DeBiasi EM, Feller-Kopman D. Physiologic basis of symptoms in pleural disease. *Semin Respir Crit Care Med* 2019; 40: 305–313. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
128. Spalla I, Payne JR, Borgeat K. et al. Mitral annular plane systolic excursion and tricuspid annular plane systolic excursion in cats with hypertrophic cardiomyopathy. *J Vet Intern Med* 2017; 31: 691–699. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
129. Ward JL, Kussin EZ, Tropf MA. et al. Retrospective evaluation of the safety and tolerability of pimobendan in cats with obstructive vs nonobstructive cardiomyopathy. *J Vet Intern Med* 2020; 34: 2211–2222. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
130. Santiago SL, Freeman LM, Rush JE. Cardiac cachexia in cats with congestive heart failure: prevalence and clinical, laboratory, and survival findings. *J Vet Intern Med* 2020; 34: 35–44. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
131. Saker KE. Nutritional concerns for cancer, cachexia, frailty, and sarcopenia in canine and feline pets. *Vet Clin North Am Small Anim Pract* 2021; 51: 729–744. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
132. Elliott J, Rawlings JM, Markwell PJ. et al. Survival of cats with naturally occurring chronic renal failure: effect of dietary management. *J Small Anim Pract* 2000; 41: 235–242. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
133. Ko GJ, Obi Y, Tortorici AR. et al. Dietary protein intake and chronic kidney disease. *Curr Opin Clin Nutr Metab Care* 2017; 20: 77–85. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
134. Ross SJ, Osborne CA, Kirk CA. et al. Clinical evaluation of dietary modification for treatment of spontaneous chronic kidney disease in cats. *J Am Vet Med Assoc* 2006; 229: 949–957. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
135. Churchill JA, Eirmann L. Senior pet nutrition and management. *Vet Clin North Am Small Anim Pract* 2021; 51: 635–651. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

136. Segev G, Cortellini S, Foster JD. et al. International Renal Interest Society best practice consensus guidelines for the diagnosis and management of acute kidney injury in cats and dogs. *Vet J* 2024; 305. DOI: 10.1016/j.tvjl.2024.106068. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
137. Ljungvall I, Rishniw M, Porciello F. et al. Sleeping and resting respiratory rates in healthy adult cats and cats with subclinical heart disease. *J Feline Med Surg* 2014; 16: 281–290. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
138. Porciello F, Rishniw M, Ljungvall I. et al. Sleeping and resting respiratory rates in dogs and cats with medically-controlled left-sided congestive heart failure. *Vet J* 2016; 207: 164–168. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
-

Articles from *Journal of Feline Medicine and Surgery* are provided here courtesy of **SAGE Publications**

As a library, NLM provides access to scientific literature. Inclusion in an NLM database does not imply endorsement of, or agreement with, the contents by NLM or the National Institutes of Health.

Learn more: [PMC Disclaimer](#) | [PMC Copyright Notice](#)



J Feline Med Surg. 2025 Sep 12;27(9):1098612X251367535. doi: [10.1177/1098612X251367535](https://doi.org/10.1177/1098612X251367535) [↗](#)

Evaluation of health-related quality of life in cats with chronic kidney disease

[Sarah K Lorbach](#)¹, [Jessica M Quimby](#)^{1,✉}, [Eline Nijveldt](#)¹, [Rene E Paschall](#)¹, [E Marian Scott](#)², [Jacqueline Reid](#)²

[Author information](#) [Article notes](#) [Copyright and License information](#)

PMCID: PMC12432299 PMID: [40937630](#)

Abstract

Objectives

The aim of the present study was to evaluate health-related quality of life (HRQoL) in cats with chronic kidney disease (CKD) and integrate these results with clinical data to identify factors associated with decreased HRQoL.

Methods

Caregivers completed a validated HRQoL tool (VetMetrica) and clinical data (history, laboratory work, physical examination findings, etc) were collected. Vitality, comfort and emotional wellbeing (EWB) scores were compared between groups using non-parametric analyses as appropriate.

Results

Data from 24 healthy cats and 68 cats with CKD were collected. Cats with CKD had significantly lower median vitality, comfort and EWB scores (vitality: 41.7, range 4.3–64; comfort: 36.1, range 21.2–59.6; EWB: 44.6, range 1.6–58.8) than healthy cats (vitality: 56.5, range 40.7–64; comfort: 59.6, range 34.2–59.6; EWB: 58.7, range 45.1–58.8) ($P < 0.0001$ for all analyses). Cats with International Renal Interest Society (IRIS) stage 3 and 4 CKD had lower median EWB scores (30.8, range 1.6–54.4) than IRIS stage 1 and 2 CKD (47.3, range 11.2–58.8) ($P = 0.04$) and lower vitality scores (38.2, range 4.3–54.0) in comparison with IRIS stage 1 and 2 CKD (43.1, range 12.8–64.0) ($P = 0.04$). Cats with anaemia had lower median EWB scores (25.9, range 10.4–54.4) than those without (46.6, range 1.6–58.8) ($P = 0.005$), and haematocrit was positively correlated with EWB ($P = 0.005$, $r = 0.33$). Cats with abnormal appetite had lower median EWB scores (29.1, range 1.6–56.7) than cats with normal appetite (48.1, range 11.2–58.8) ($P = 0.001$). Cats with constipation had lower median comfort scores 29.5 (range 21.2–59.5) ($P = 0.0003$) and lower median EWB scores (22, range 1.6–58.8) ($P = 0.008$) than those without (comfort: 37.1, range 25.3–59.6; EWB: 47.9, range 11.2–58.8).

Conclusions and relevance

Cats with CKD have decreased HRQoL. Several modifiable clinical factors are associated with this decreased HRQoL.

Keywords: Caregiver burden, anaemia, constipation, appetite, polypharmacy, wellbeing

Introduction

Quality of life (QoL) is a broad concept that is impacted by many features of an individual's life, including physical health, psychological state, social relationships and salient features of their environment. Health-related quality of life (HRQoL) is concerned with the effects of illness and the impact treatment may have on QoL.^{1,2} In human medicine, assessment of HRQoL in addition to biomedical outcomes has become an integral part of the understanding of disease management. In the veterinary setting, the assessment of HRQoL is determined by caregiver observation and report of specific behaviours that indicate how the animal might 'feel' about its circumstances. The importance of HRQoL to the caregiver is likely a significant component in decisions made regarding the sick pet. For example, in a study of the perception of HRQoL in cats with heart disease, 93% of caregivers were willing to trade survival time for comfortable QoL.³

In human patients with chronic kidney disease (CKD), HRQoL is decreased in comparison with healthy individuals and also decreases according to severity and progression of disease.⁴⁻⁸ Children with CKD have lower HRQoL scores in comparison with healthy children, and children who have more severe clinical symptoms have lower HRQoL scores in comparison with those who do not.^{9,10} In cats with CKD, QoL scores related to eating (eg, liking food and appetite) and management aspects, such as administration of medication and accessing veterinary care, are lower in comparison with healthy cats.¹¹ The potential impact of disease stage and specific clinical and physiologic abnormalities associated with CKD has not been studied in cats.

Conditions such as hypertension, anaemia, electrolyte abnormalities, dysrexia, dehydration and constipation commonly occur as sequelae of CKD and should be identified and treated.¹² These pathologic changes may have a negative effect on the patient and thus alter HRQoL.¹¹ Administration of medications or treatments in cats that are not amenable to these interactions could adversely affect the human-animal bond and detract from HRQoL.^{3,11} Identification of modifiable factors that might be associated with a decreased HRQoL may aid clinicians in tailoring treatment recommendations for cats with CKD. Improving our understanding of HRQoL has the potential to better address the emotional health of our patients, understand the impact of disease on the patient and caretaker, and provide a tool to help assess the efficacy and impact of treatments.

The purpose of this study was to assess HRQoL in cats with CKD using a validated HRQoL instrument (VetMetrica)¹³⁻¹⁵ completed online by the caregiver, and integrate these results with medical information from the patient in order to identify factors associated with decreased HRQoL. We hypothesised that cats with CKD would have decreased HRQoL scores, and specific clinical factors (eg, anaemia, poor appetite) would negatively impact their score. The above represents an example of 'known-groups' construct validation, and assessing this in the feline CKD patient population was a secondary aim of the study.

Materials and methods

Cats

The study was approved by the Institutional Animal Care and Use Committee (IACUC#2019A00000128) at The Ohio State University. Cats were screened for prospective enrolment in 1/2 groups: healthy cats or cats with CKD. All cats enrolled in the study were required to have informed consent signed by the animal's caregiver.

Healthy cats Client-owned healthy cats with normal systemic blood pressure, serum biochemistry profile, complete blood count (CBC), urinalysis and serum total thyroxine measurement were enrolled. Healthy control cats were defined as those with no clinical abnormalities, normal physical examination, a serum creatinine concentration less than 1.6 mg/dl and a urine specific gravity greater than 1.035.

Cats with CKD Cats with stable International Renal Interest Society (IRIS) stage 1–4 CKD were enrolled. Diagnostic tests required before enrolment included systemic blood pressure, serum biochemistry profile, CBC, urinalysis and serum total thyroxine measurement. The diagnosis of CKD was confirmed, and the affected cats were staged by a board-certified small animal internal medicine specialist according to IRIS guidelines. In cats with stage 1 CKD and early stage 2 CKD, diagnosis was based on either ultrasound findings consistent with degenerative renal disease and/or persistent inadequate urinary concentrating ability without an identifiable non-renal cause. Based on the clinical experience of the investigators, cats with CKD were considered to have stable CKD if serum creatinine had not changed by more than 20% on at least two measurements taken at least 2 weeks apart. Exclusion criteria included other uncontrolled systemic illnesses, pyelonephritis, ureteral obstruction, decompensation of CKD (eg, sudden negative change in clinical status associated with an increase in azotaemia) or any condition requiring hospitalisation and intravenous fluid therapy. Patients receiving dietary management or various therapies, such as potassium supplementation, antihypertensive medications and subcutaneous fluids, for CKD management were not excluded.

HRQoL instrument and clinical data collection

A previously published web-based, generic HRQoL instrument (VetMetrica) was used to measure the impact of chronic disease in cats (see [Appendix in the supplementary material](#) ^{13–15}). Caregivers were asked to participate in the study at the time of the veterinary visit and to complete the survey within 1 week. This instrument consists of 20 behaviour-based items to which the cat caregiver responds using a 7-point Likert-type scale (0 = could not be less to 6 = could not be more), resulting in the generation of scores in three HRQoL domains: vitality, comfort and emotional wellbeing (EWB). A summary score for physical wellbeing (PWB) was obtained by averaging vitality and comfort scores. To aid interpretability of the tool, raw domain scores (0–6) are normalised such that 50 on a scale of 0–100 represents the score for the average healthy cat, and 70% of healthy cats will score 44.8 or higher.^{13,14} The VetMetrica instrument was supplemented with additional questions (see [Appendix in the supplementary material](#) ¹³) to gather information about the patient's clinical history, disease status and management regimen, but these did not impact the manner in which the HRQoL scores were calculated.

Clinical parameters collected for analysis included the following: anaemia (haematocrit or packed cell volume <27%; yes/no); muscle condition score (normal/mild vs moderate/severe); body condition score (BCS; 1–3, 4–5, 6–9); adequate hydration on physical examination (yes/no); decreased or picky appetite since last visit (yes/no); history of hypertension based on IRIS guidelines (yes/no); constipation in the past 6 months (difficulty defecating, smaller/drier stools or decreased frequency of bowel movements (yes/no); active urinary tract infection (UTI) in the past year (clinical signs of lower urinary tract disease plus positive urine culture; yes/no); and number of medications administered (0–1, 2–4, 5–7). Continuous variables collected for analysis included patient age, body weight, serum creatinine concentration, serum potassium concentration, serum phosphorus concentration, haematocrit, number of medications and systolic blood pressure.

Statistical analysis

Statistical analysis was performed using Prism version 9 (GraphPad Software) and Minitab version 20 (Minitab). Data were assessed for normality using the D'Agostino – Pearson test and visual inspection of quantile–quantile plots, and analysed using non-parametric methods accordingly. A Mann–Whitney U-test was used to compare each of the HRQoL domains between healthy cats and those with CKD aged 10 years or younger. All other analyses were performed within the CKD cat group. A Mann–Whitney U-test was used to compare each of the HRQoL domains between IRIS stages (1 and 2 vs 3 and 4), anaemia, muscle condition score, hydration, decreased or picky appetite, hypertension, constipation and active UTI. The Kruskal–Wallis test was used to compare each of the HRQoL domains between healthy cats, those with IRIS stage 1 and 2 CKD and those with IRIS stage 3 and 4 CKD, between BCS categories and between the number of medications administered. Spearman's rank analysis was used to assess the correlation between each of the HRQoL domains and age, serum creatinine concentration, haematocrit, body weight, number of medications, serum potassium concentration, serum phosphorus concentration, systolic blood pressure and serum creatinine vs haematocrit. Significance was determined by $P < 0.05$ for all comparisons.

Results

Cats

The 68 cats with CKD had a median age of 14 years (range 0.8–20) and consisted of 47 domestic shorthairs (DSHs), eight domestic longhairs (DLHs), seven domestic medium hairs (DMHs), three

Maine Coons, two Siamese and one Tonkinese. Of them, 31 cats were castrated males and 37 cats were spayed females. The 24 healthy cats had a median age of 6 years (range 2–10), which was significantly different from the CKD cats ($P < 0.0001$). They consisted of 18 DSHs, five DLHs and one DMH. Of them, 14 cats were castrated males and 10 cats were spayed females. CKD cats had a median serum creatinine concentration of 2.2 mg/dl (range 1.1–6.3). Distribution of disease stage was as follows: IRIS stage 1, $n = 8$, IRIS stage 2, $n = 41$, IRIS stage 3, $n = 15$ and IRIS stage 4, $n = 4$. Healthy cats had a median serum creatinine concentration of 1.3 mg/dl (range 0.9–1.5).

HRQoL scores

A summary of median (range) HRQoL scores is presented in [Table 1](#). Cats with IRIS stage 1 and 2 CKD and those with IRIS stage 3 and 4 CKD had significantly lower vitality, comfort, EWB and PWB scores than healthy cats ($P < 0.0001$ for all analyses) ([Figure 1](#)). When only cats with CKD were considered, cats with IRIS stage 3 and 4 CKD had lower EWB scores ($P = 0.04$) and lower vitality scores ($P = 0.04$) than cats with IRIS stage 1 and 2 CKD ([Figure 2](#)).

Table 1.

Normalised health-related quality of life (HRQoL) domain score results and physical domain summary score (physical wellbeing) for cats in each study population and clinical sign, treatment or clinicopathologic category

	Vitality	Comfort	EWB	PWB
Cat group				
Healthy	56.5 (40.7–64) ^{a,A}	59.6 (34.2–59.6) ^{a,A}	47.3 (11.2–58.8) ^{a,A}	56.1 (44.3–61.8) ^{a,A}
IRIS stage 1/2	43.0 (12.8–64) ^b	37.1 (21.2–59.6) ^b	47.3 (11.2–58.8) ^b	39.6 (17.0–61.8) ^b
IRIS stage 3/4	38.2 (4.3–54) ^c	34.4 (23.9–54.8) ^c	30.8 (1.6–54.4) ^c	37.5 (14.1–53.8) ^c
CKD cats aged ≤10 years	44.2 (27.8–64) ^B	37.5 (27.3–59.6) ^B	48.9 (11.7–58.8) ^B	40.6 (31.1–61.8) ^B
Anaemia <27%				
Yes	35.2 (4.3–58.4)	34.3 (23.9–59.6)	25.9 (10.4–54.4) ^a	34.6 (14.1–59.0)
No	42.8 (12.8–64.0)	36.6 (21.2–59.6)	46.6 (1.6–58.8) ^b	39.5 (17.0–61.8)
Abnormal appetite				
Yes	39.5 (4.3–59.6)	34.7 (21.2–59.6) ^a	29.1 (1.6–56.7) ^a	36.2 (14.1–59.6) ^a
No	42.8 (26.2–64.0)	37.6 (25.3–59.6) ^b	48.1 (11.2–58.8) ^b	40.0 (28.6–61.8) ^b
Constipation				
Yes	38.4 (4.3–56.9)	29.5 (21.2–59.5) ^a	22 (1.6–58.8) ^a	34.6 (14.1–58.2) ^a
No	42.9 (26.1–64.0)	37.1 (25.3–59.6) ^b	47.9 (11.2–58.8) ^b	40.4 (28.6–61.8) ^b
Muscle loss				

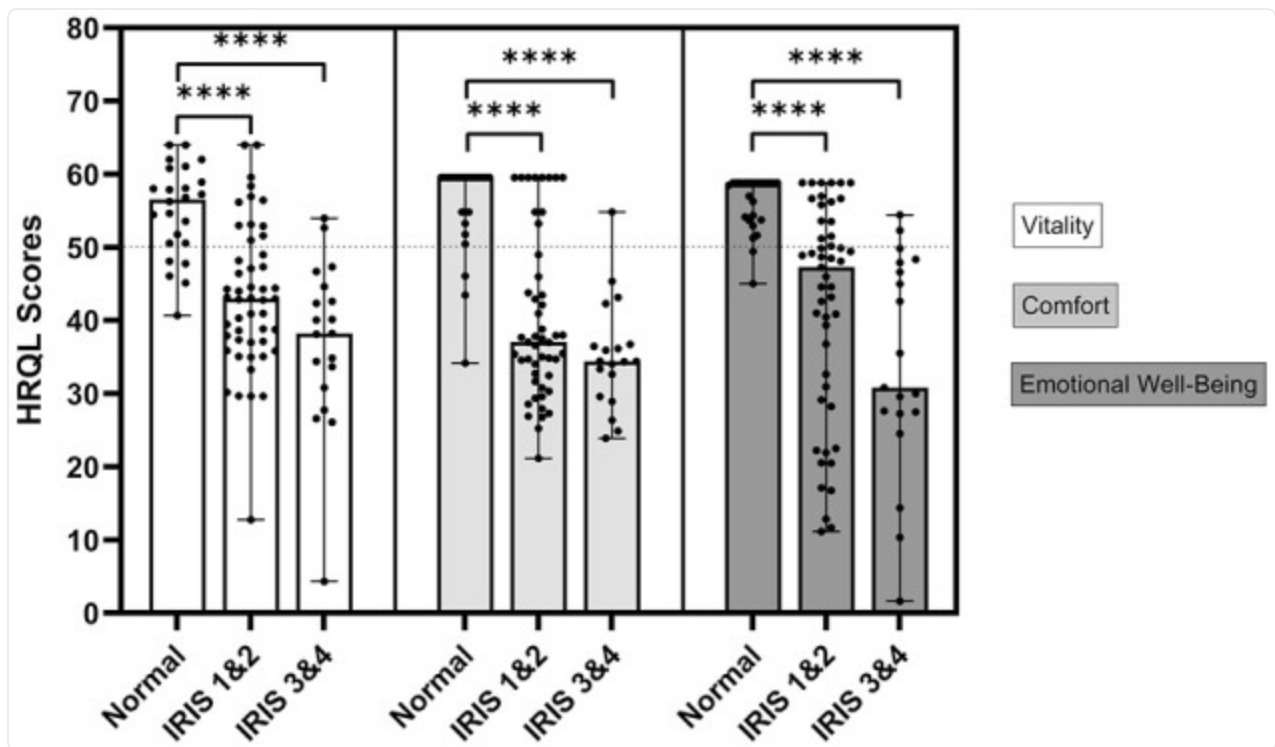
	Vitality	Comfort	EWB	PWB
Normal/mild	43.1 (26.1–64.0)	36.9 (25.3–59.6)	48 (11.2–58.8) <a>^	40.2 (28.6–61.8)
Moderate/severe	39.5 (4.3–59.6)	35.4 (21.2–59.6)	35.5 (1.6–56.7) <a>^	36.2 (14.1–59.6)
Medications/day				
0–1	43.1 (26.1–64.0) <a>^	37.3 (26.8–59.6) <a>^	48.4 (11.2–58.8) <a>^	40.3 (28.6–61.8) <a>^
2–4	38.1 (12.8–64.0)	34.4 (21.2–59.6)	29.6 (1.6–58.8)	38.5 (17.0–61.8)
5–7	37.7 (4.3–53.0) <a>^	34.5 (23.9–59.6) <a>^	33.9 (10.4–58.8) <a>^	35.7 (14.1–56.3) <a>^

[Open in a new tab](#)

Data are median (range). Differing superscript letters indicate significantly different HRQoL scores between groups or factors

CKD = chronic kidney disease; EWB = emotional wellbeing; IRIS = International Renal Interest Society; PWB = physical wellbeing

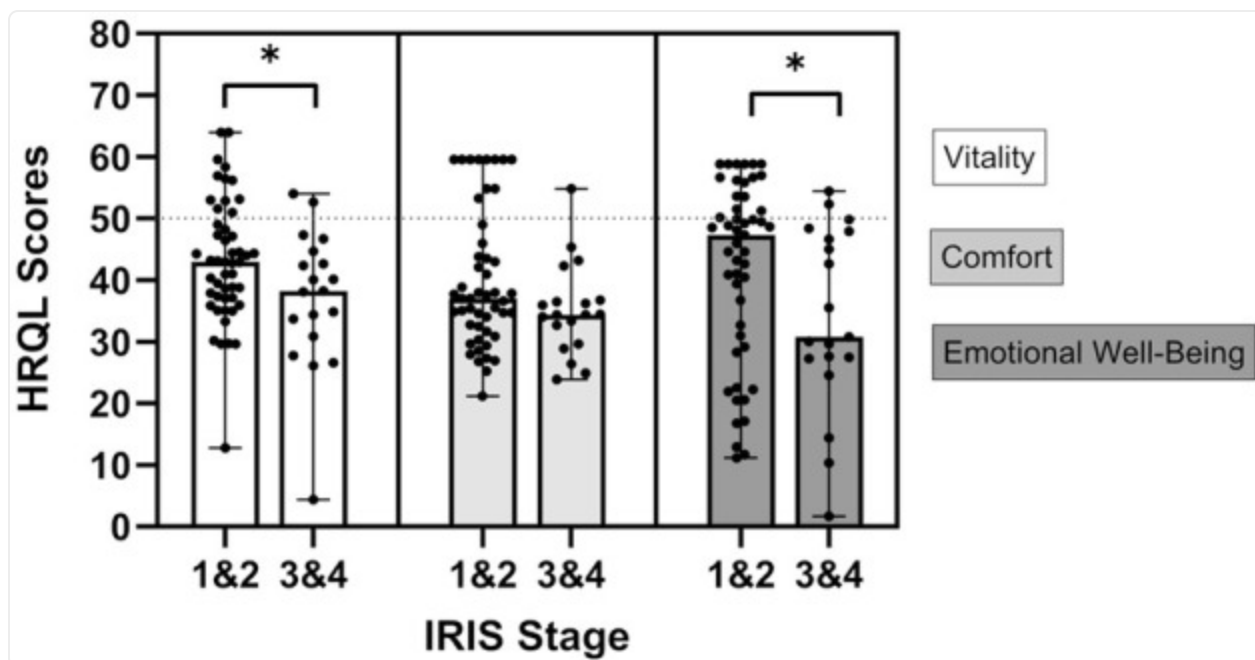
Figure 1.



[Open in a new tab](#)

Health-related quality of life (HRQoL) scores in healthy cats vs cats with International Renal Interest Society stage 1/2 and 3/4 chronic kidney disease (CKD). Cats with both early- and late-stage CKD have significantly lower HRQoL scores in all three domains than healthy cats. The dot and whisker plot portrays median and range, with dots representing individual cats for each domain. The dotted line represents the mean score for healthy cats. **** $P < 0.0001$. IRIS = International Renal Interest Society

Figure 2.



[Open in a new tab](#)

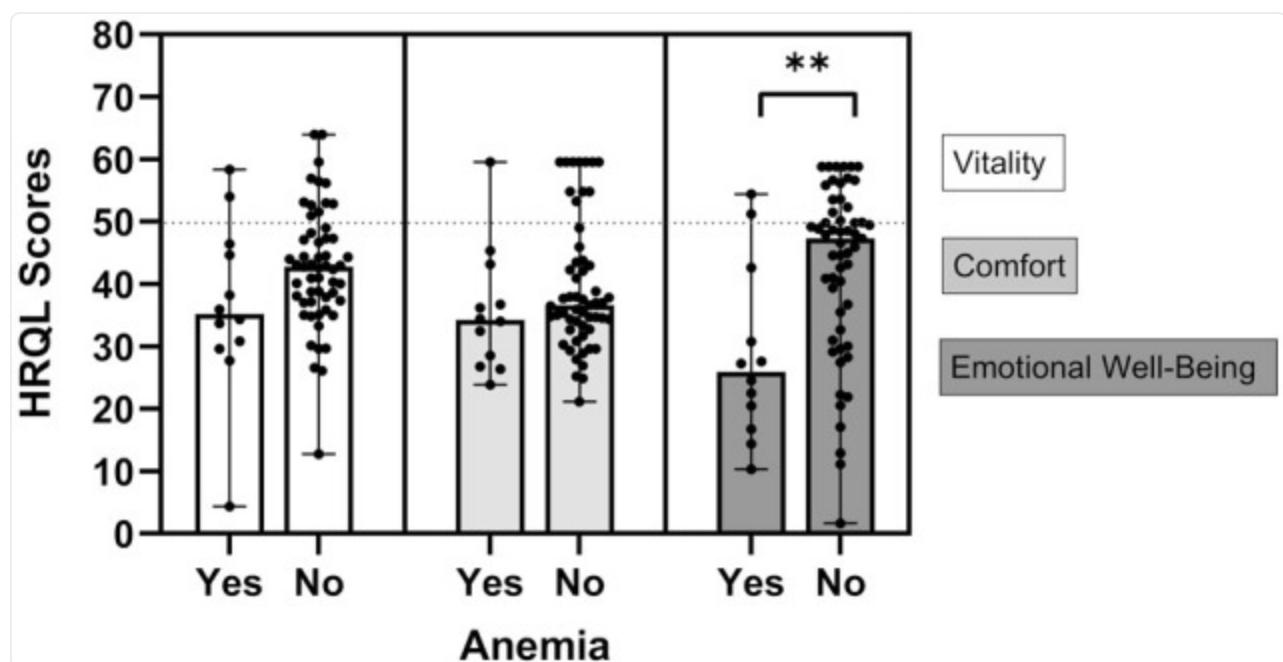
Health-related quality of life (HRQoL) scores in early and late International Renal Interest Society (IRIS) stage chronic kidney disease (CKD). Cats with IRIS stage 3 and 4 CKD ($n = 19$) had lower vitality and emotional wellbeing scores than IRIS stage 1 and 2 ($n = 49$) ($*P = 0.04$ for both comparisons). The dot and whisker plot portrays median and range, with dots representing individual cats for each domain. The dotted line represents the mean score for healthy cats

When only cats with CKD aged 10 years or younger ($n = 18$) were compared with healthy cats, those with CKD still had significantly lower vitality, comfort, EWB and PWB scores ($P = 0.0008$, $P < 0.0001$, $P = 0.0001$, $P < 0.0001$, respectively). Age within the CKD population was negatively correlated with vitality ($P = 0.0006$, $r = -0.40$), comfort ($P < 0.0001$, $r = -0.47$) and EWB ($P = 0.004$, $r = -0.47$).

Cats with anaemia $< 27\%$ had lower EWB scores than those with haematocrit of 27% or greater ($P = 0.005$) (Figure 3). Haematocrit was positively correlated with EWB ($P = 0.005$, $r = 0.33$). Cats with abnormal appetite had lower EWB ($P = 0.001$), lower comfort ($P = 0.04$) and lower PWB scores ($P = 0.03$) than cats with normal appetite (Figure 4). Cats with constipation had lower comfort ($P =$

0.0003), lower EWB ($P = 0.008$) and lower PWB scores ($P = 0.003$) than those with reportedly normal stools ([Figure 5](#)). Cats with moderate or severe muscle loss had lower EWB scores than those with normal to mild muscle loss ($P = 0.008$). Cats that were receiving five to seven medications daily had significant lower vitality, comfort and EWB and PWB scores than cats receiving zero to one medications daily ($P = 0.04$, $P = 0.01$, $P = 0.005$ and $P = 0.01$, respectively). The number of medications administered was negatively correlated with EWB ($P = 0.03$, $r = -0.25$). No cat was reported to be receiving more than seven medications.

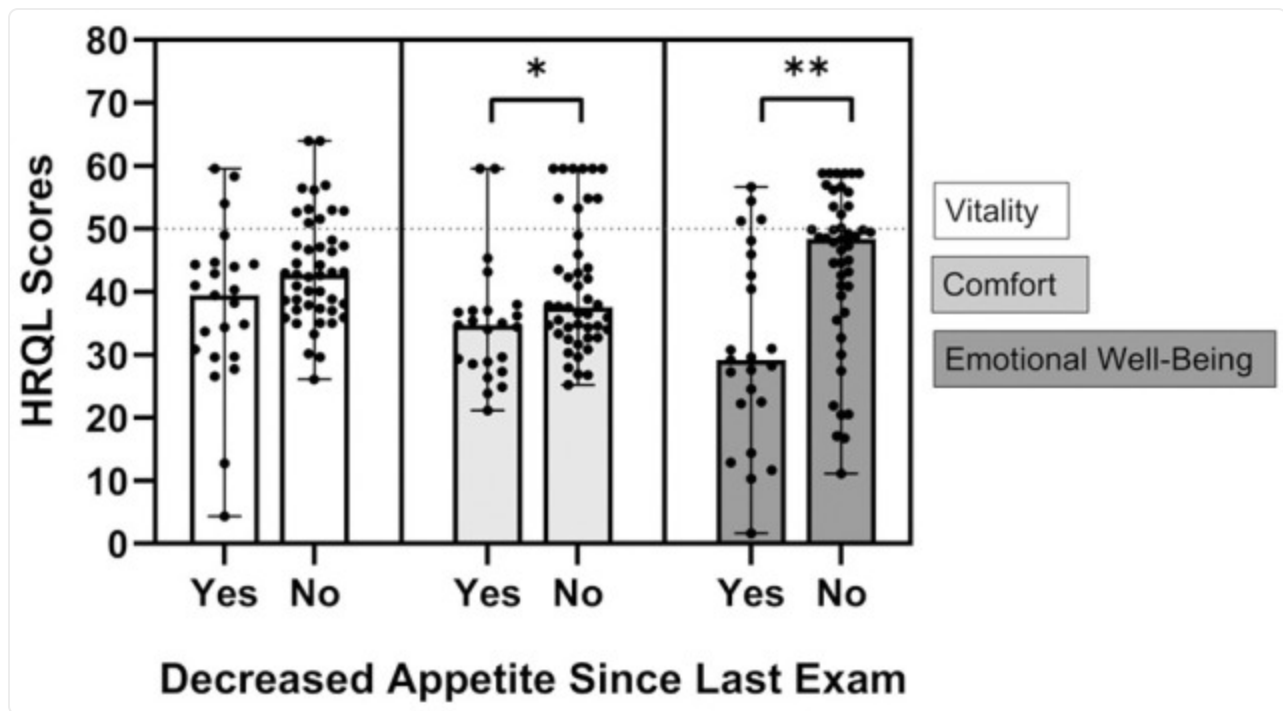
Figure 3.



[Open in a new tab](#)

Health-related quality of life (HRQoL) scores in cats with chronic kidney disease with and without anaemia (haematocrit <27%). Cats with anaemia had lower emotional wellbeing scores than those without (** $P = 0.005$). The dot and whisker plot portrays median and range, with dots representing individual cats for each domain. The dotted line represents the mean score for healthy cats

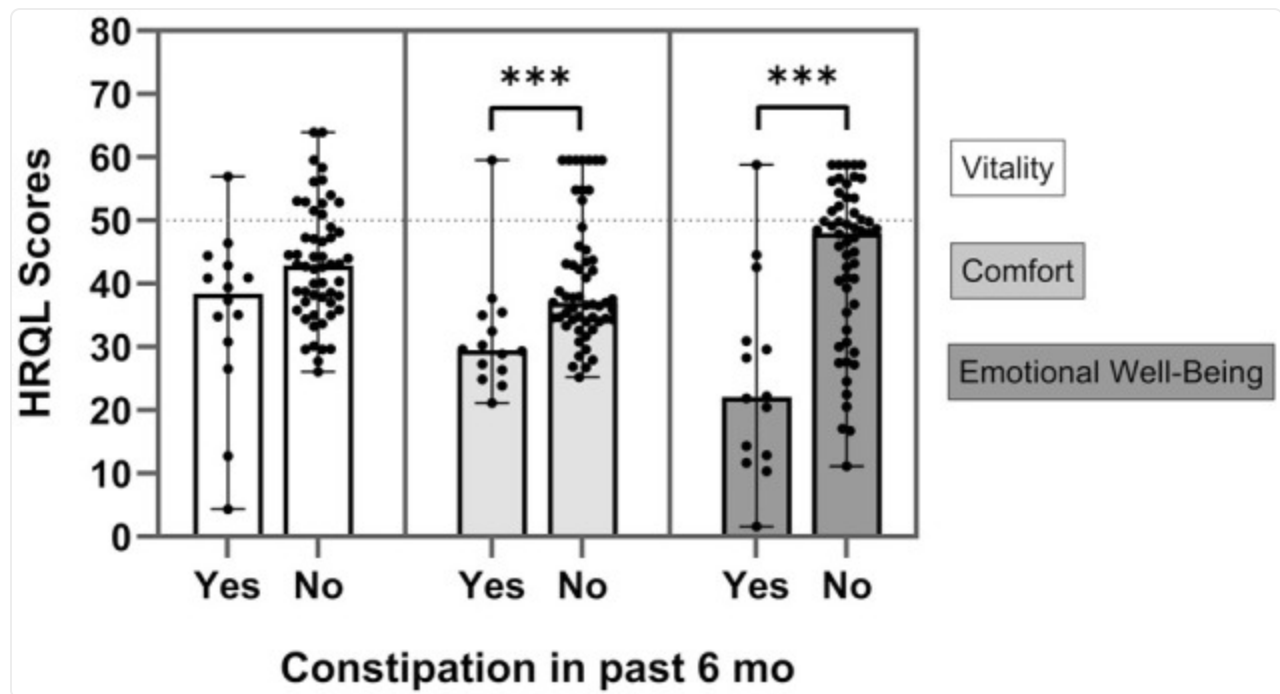
Figure 4.



[Open in a new tab](#)

Health-related quality of life (HRQoL) scores in cats with chronic kidney disease with normal vs decreased appetite. Cats with decreased appetite since the last examination had lower comfort and emotional wellbeing scores than cats with normal appetite (* $P = 0.04$ and ** $P = 0.001$, respectively). The dot and whisker plot portrays median and range, with dots representing individual cats for each domain. The dotted line represents the mean score for healthy cats

Figure 5.



[Open in a new tab](#)

Health-related quality of life (HRQoL) scores in cats with chronic kidney disease with and without constipation. Cats with signs of constipation in the past 6 months had lower comfort and emotional wellbeing scores than those with reportedly normal stools ($***P = 0.0003$ [comfort] and $***P = 0.008$ [emotional wellbeing], respectively). The dot and whisker plot portrays median and range, with dots representing individual cats for each domain. The dotted line represents the mean score for healthy cats

There were no significant differences in HRQoL scores for cats with or without hypertension, active UTI in the past year, hydration status or BCS categories. There was no significant correlation between any of the three HRQoL domains and systemic blood pressure, serum creatinine, potassium or phosphorus concentrations.

Discussion

This study assessed HRQoL in cats with CKD using a validated metric and compared these data to clinical and clinicopathologic parameters. Cats with CKD have significantly decreased HRQoL in

comparison with healthy cats, and cats with late-stage CKD have significantly decreased HRQoL scores in comparison with cats with earlier stage CKD. In addition, several key clinical parameters were associated with decreased HRQoL scores including anaemia, decreased appetite, decreased muscle mass and constipation. Although these results demonstrate association and not causation, they provide a foundation for clinical assessment of patients as well as a basis for interventional studies to further demonstrate the relationship between these factors and HRQoL.

In humans, multiple studies have found that CKD significantly affects HRQoL in both non-dialysis and dialysis-dependent patients, impacting primarily physical but also mental HRQoL domains.⁴⁻⁸ Although many studies describe a correlation between severity and progression of disease and decreasing HRQoL,^{4,7} these results are inconsistent.¹⁶ These findings are similar to those of our study, where no correlation between serum creatinine concentration and HRQoL domains was seen, but EWB scores were decreased in late-stage disease compared with early-stage disease. Similarly, in the study by Bijmans et al,¹¹ HRQoL scores were subjectively decreased in cats with IRIS stage 3 and 4 CKD, in comparison with IRIS stage 2, but failed to reach statistical significance. Alternatively, recent studies on human CKD hypothesise that symptom burden associated with CKD likely has a more impactful role on HRQoL than stage of disease.^{8,17-19} Symptoms such as fatigue, decreased appetite, anaemia and constipation are commonly mentioned specifically.^{6,8} Importantly, many of these CKD symptoms are considered modifiable factors, which, if targeted with management, have the potential to improve HRQoL.^{6,19}

Abnormalities in appetite are common in cats with CKD, and appetite dysregulation may play a role.^{20,21} Poor appetite and concomitantly low body weight are associated with a poorer prognosis.^{22,23} In our study, cats with decreased appetite had lower EWB scores than cats with normal appetite. In the study by Bijmans et al,¹¹ which assessed cats with CKD using a different instrument, QoL scores related to eating, including 'liking food' and 'appetite', were also decreased in comparison with young and old healthy cats. Very similar findings regarding the relationship between appetite and HRQoL are present in human medicine. Children with conservatively managed predialysis CKD with 'good' or 'very good' appetite scores have higher PedsQL (both self and parent-proxy scores) in comparison with children who have an appetite described as 'poor' or 'very poor'.²⁴ CKD patients receiving haemodialysis who self-reported 'poor' or 'very poor' appetite have lower kidney disease-specific HRQoL scores in a range of domains, including physical and mental/social, implying that a reduction in appetite has an effect on patients' lives that goes beyond the impact on food intake and nutritional status.²⁵ In addition, appetite management in elderly haemodialysis patients has been associated with improvements in both mental and physical domain kidney disease-specific HRQoL scores after treatment with an appetite stimulant.²⁶ These changes were accompanied by increased appetite, caloric intake,

weight and body mass index.²⁶ Addressing appetite therefore appears to be an important, modifiable HRQoL factor in CKD across species.⁶

Anaemia is a common clinicopathologic finding in cats with CKD. Its prevalence increases with disease severity, with up to 53–65% of cats affected in late-stage disease.^{23,27,28} It is associated with disease progression and may potentially contribute to progression.^{28–30} Clinical signs of anaemia, such as lethargy, weakness and poor appetite, may be vague and easily confused with other aspects of disease, and as such have a very plausible potential to effect HRQoL as evidenced by the significant decrease in EWB scores in our study. In humans, multiple studies have documented a relationship between anaemia and HRQoL.^{6,31} In a cross-sectional study of humans with CKD, patients with anaemia had lower HRQoL scores than those without anaemia.³¹ This effect was more apparent in non-dialysis patients with IRIS stage 3 and 4 CKD. In addition, tiredness and impaired activity were both greater in patients with anaemia. Findings were true regardless of the HRQoL scoring system used and were significant in both mental and physical domains.³¹ In a study evaluating daprodustat, a hypoxia-inducible factor prolyl hydroxylase inhibitor, treatment resulted in improvements in anaemia, associated fatigue and HRQoL compared with placebo.³² Importantly, a study assessing caregiver burden in partners of patients with CKD found that patient anaemia was associated with lower HRQoL and significantly higher caregiver burden.³³ Taken together, these findings identify anaemia as a key modifiable factor for improving HRQoL in human CKD patients.^{6,19} Given the prevalence of anaemia in feline CKD, this may represent an important target for tailored management.

Cats with CKD are more likely to experience constipation than healthy cats and are overrepresented in the population of feline patients presenting to the emergency room for constipation.^{34,35} The association between CKD and constipation is also recognised in humans, and is likely multifactorial in aetiology in both species, attributed to hydration status, electrolyte disturbances (specifically hypokalaemia), gastrointestinal physiologic alterations and pharmacologic management such as phosphate binders.^{35–40} When the effect on HRQoL was considered, cats with constipation in the past 6 months – defined as difficulty defecating, smaller or drier stools, or decreased frequency of bowel movements – had lower EWB and PWB HRQoL scores than those without constipation. These findings are similar to human data demonstrating that the presence and severity of constipation are associated with poorer HRQoL in both physical and mental domains.^{39,40} Constipation is therefore recognised as a modifiable factor for improving HRQoL in human CKD patients,⁶ and our results suggest a similar approach may be beneficial for feline patients.

Administering five or more medications to cats with CKD was associated with lowered HRQoL scores in all categories. Earlier literature demonstrates the difficulty many cat caregivers experience in administering medication to their pets,^{3,41,42} a phenomenon that over half of caregivers felt altered their relationship with their cat.⁴¹ Difficulty can be experienced with both short-term and long-term medication administration, with caregivers experiencing worry about the interaction and a percentage finding themselves unable to complete a course of medication.⁴¹ These factors are logically compounded with chronic administration of multiple medications. In human patients with CKD, polypharmacy is associated with decreased HRQoL.⁴³ When hyperpolypharmacy (≥ 10 classes of medications) and major polypharmacy (five to nine classes of medications) were compared with minor polypharmacy (four or fewer classes of medications), patients with hyper- and major polypharmacy had significantly lower HRQoL scores in both mental and physical domains.⁴³ Therefore, pill burden is likely an important part of developing a treatment plan for feline patients with CKD.

Given the propensity for disease in elderly cats, enrolling a population of healthy cats that were age-matched with CKD cats was not possible, and the CKD cat group that was enrolled was older in age than the healthy cat group. The impracticality of age-matching to CKD patients was also present in a previous study assessing HRQoL in cats with and without CKD.¹¹ In that study, HRQoL was assessed in young healthy (n = 105), old healthy (n = 36) and CKD cats (n = 70); it was concluded that young cats had significantly better HRQoL than older cats and those with CKD.¹¹ This was particularly true in domains including mobility, stools, playing and hunting.¹¹ This may reflect other age-related changes, such as osteoarthritis and increased incidence of concurrent disease. However, in that study, the health of the young cats was not confirmed by examination or laboratory work. In the present study, when only cats with CKD under the age of 10 years were considered in comparison with healthy control cats, there was still a statistically significant decrease in HRQoL scores in the cats affected by CKD. In addition, within the CKD cat group, there was a negative correlation with age for all three HRQoL domains. Overall, it can be concluded that age plays a role in HRQoL, but based on the data in the present study, changes in scores in CKD cats are not just attributable to ageing.

The data collected in this study demonstrate the validity of the VetMetrica instrument for evaluating cats with CKD. Validity provides evidence that a tool measures what it is intended to measure – in this case, HRQoL. However, validity is not an intrinsic property of a scale; evidence must be gathered for its use in different purposes, defined populations and specific contexts. Previous studies have established the validity of VetMetrica in healthy cats, cats with various chronic conditions and cats with osteoarthritis;^{13,15} however, this is the first study to validate its use in a feline CKD population. Instrument validity is established first through construction

methods, followed by testing for different types of construct validity. One such method, 'known-groups' testing, examines whether scores differ as hypothesised between groups – for example, healthy vs sick animals, or animals with increasing disease severity (mild < moderate < severe). In the present study, known-groups validity was supported by demonstrating different HRQoL profiles between healthy cats and those with CKD, between cats with IRIS stage 1/2 vs stage 3/4, anaemic vs non-anaemic cats, cats with abnormal vs normal appetite, cats with constipation vs normal stool and cats with varying degrees of muscle wasting.

Overlapping scores are still possible with known-group validation as complete separation is not expected and biological variation will be seen even in a 'normal' population. It is perhaps worth noting that there is much less variation in the healthy cats compared with the CKD cats. This is to be expected and that greater variability can be attributed to the heterogeneity of the sick population. In general, overlap could be due to the blurring of the distinction between IRIS stages. In particular, the overlap between healthy cats and those with stage 1/2 CKD is likely to be related to the fact that stage 1 cats may be asymptomatic and therefore present as healthy cats in terms of their PWB and EWB.

Limitations of this study include reliance on caretaker-reported data, which were based on observations rather than objective measurements and may have not been accurate (eg, assessment of constipation). In feline patients, a cat's individual experience cannot be truly assessed, making caregiver perception central to HRQoL evaluation, although such perception may be influenced by unrecognised factors. However, VetMetrica is an observer-based tool in which the caregiver reports behaviours that can then be interpreted to reflect the animal's emotional state, an approach favoured by the US Food and Drug Administration for non-verbal individuals, including infants, cognitively impaired individuals and animals.⁴⁴ In paediatric CKD patients, HRQoL assessments have been found to correlate moderately between patient self-reports and parental assessments, with stronger agreement for physical domains than mental domains.⁹ Another limitation was that two cats in the healthy group with high-normal serum creatinine were not imaged to differentiate IRIS stage 1 disease from a healthy status.

Conclusions

Cats with CKD have lower HRQoL scores than healthy cats. Several modifiable clinical signs, such as reduced appetite, anaemia and constipation, are associated with decreased HRQoL, suggesting that identifying and addressing these abnormalities may improve wellbeing in affected cats.

Supplemental Material

Supplemental Material

[sj-docx-1-jfm-10.1177_1098612X251367535.docx](#) (22.6KB, docx)

The VetMetrica HRQoL instrument consists of 20 questions. Additional questions for the caregiver about the clinical wellbeing and management of the cat were included as an addendum to the instrument. Veterinary staff who assessed the cat also provided information on their assessment and clinicopathologic data were collected.

Acknowledgments

The authors wish to thank the caregivers who participated in this study and their cats.

Footnotes

Accepted: 19 June 2025

Author note: The results of this study were presented, in part, as a research report at the American College of Veterinary Internal Medicine Annual Forum, 2022.

Supplementary material: The following file is available as supplementary material:

Appendix: The VetMetrica HRQoL instrument consists of 20 questions. Additional questions for the caregiver about the clinical wellbeing and management of the cat were included as an addendum to the instrument. Veterinary staff who assessed the cat also provided information on their assessment and clinicopathologic data were collected.

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article. Dr Jacqueline Reid was previously the director of NewMetrica.

Funding: The study was supported by Buttons Fund for Feline CKD Research at The Ohio State University.

Ethical approval: The work described in this manuscript involved the use of non-experimental (owned or unowned) animals and procedures that differed from established internationally recognised high standards ('best practice') of veterinary clinical care for the individual patient. The study therefore had prior ethical approval from an established (or ad hoc) committee as stated in the manuscript.

Informed consent: Informed consent (verbal or written) was obtained from the owner or legal custodian of all animal(s) described in this work (experimental or non-experimental animals, including cadavers, tissues and samples) for all procedure(s) undertaken (prospective or retrospective studies). No animals or humans are identifiable within this publication, and therefore additional informed consent for publication was not required.

ORCID iD: Jessica M Quimby  <https://orcid.org/0000-0002-1388-0452> 

Jacqueline Reid  <https://orcid.org/0000-0002-1673-9083> 

References

1. The World Health Organization quality of life assessment (WHOQOL): position paper from the World Health Organization. *Soc Sci Med* 1995; 41: 1403–1409. [[DOI](#) ] [[PubMed](#)] [[Google Scholar](#) ]
2. Hays RD, Reeve BB. **Measurement and modeling of health-related quality of life** . In: Killewo J, Heggenhougen HK, SR Q. (eds). *Epidemiology and demography in public health*. Academic Press, 2010, pp 195–220. [[Google Scholar](#) ]
3. Reynolds CA, Oyama MA, Rush JE, et al. Perceptions of quality of life and priorities of owners of cats with heart disease. *J Vet Intern Med* 2010; 24: 1421–1426. [[DOI](#) ] [[PubMed](#)] [[Google Scholar](#) ]
4. Mujais SK, Story K, Brouillette J, et al. Health-related quality of life in CKD patients: correlates and evolution over time. *Clin J Am Soc Nephrol* 2009; 4: 1293–1301. [[DOI](#) ] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#) ]

5. Porter A, Fischer MJ, Wang X, et al. Quality of life and outcomes in African Americans with CKD. *J Am Soc Nephrol* 2014; 25: 1849–1855. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
6. Fraser SDS, Phillips T. Quality of life in people with chronic kidney disease: focusing on modifiable risk factors. *Curr Opin Nephrol Hypertens* 2024; 33: 573–582. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
7. Wyld MLR, Morton RL, Clayton P, et al. The impact of progressive chronic kidney disease on health-related quality-of-life: a 12-year community cohort study. *Qual Life Res* 2019; 28: 2081–2090. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
8. Fletcher BR, Damery S, Aiyegbusi OL, et al. Symptom burden and health-related quality of life in chronic kidney disease: a global systematic review and meta-analysis. *PLoS Med* 2022; 19. DOI: 10.1371/journal.pmed.1003954. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
9. Buyan N, Turkmen MA, Bilge I, et al. Quality of life in children with chronic kidney disease (with child and parent assessments). *Pediatr Nephrol* 2010; 25: 1487–1496. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
10. Goldstein SL, Graham N, Warady BA, et al. Measuring health-related quality of life in children with ESRD: performance of the generic and ESRD-specific instrument of the Pediatric Quality of Life Inventory (PedsQL). *Am J Kidney Dis* 2008; 51: 285–297. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
11. Bijmans ES, Jepson RE, Syme HM, et al. Psychometric validation of a general health quality of life tool for cats used to compare healthy cats and cats with chronic kidney disease. *J Vet Intern Med* 2016; 30: 183–191. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
12. Sparkes AH, Caney S, Chalhoub S, et al. ISFM consensus guidelines on the diagnosis and management of feline chronic kidney disease. *J Feline Med Surg* 2016; 18: 219–239. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
13. Scott EM, Davies V, Nolan AM, et al. Validity and responsiveness of the generic health-related quality of life instrument (VetMetrica) in cats with osteoarthritis: comparison of vet and owner impressions of quality of life impact. *Front Vet Sci* 2021; 8. DOI: 10.3389/fvets.2021.733812. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

14. Davies V, Reid J, Scott EM. Optimisation of scores generated by an online feline health-related quality of life (HRQL) instrument to assist the veterinary user interpret its results. *Front Vet Sci* 2021; 7. DOI: 10.3389/fvets.2020.601304. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
15. Noble CE, Wiseman-Orr LM, Scott ME, et al. Development, initial validation and reliability testing of a web-based, generic feline health-related quality-of-life instrument. *J Feline Med Surg* 2019; 21: 84–94. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
16. Perlman RL, Finkelstein FO, Liu L, et al. Quality of life in chronic kidney disease (CKD): a cross-sectional analysis in the Renal Research Institute-CKD study. *Am J Kidney Dis* 2005; 45: 658–666. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
17. Speyer E, Tu C, Zee J, et al. Symptom burden and its impact on quality of life in patients with moderate to severe CKD: the international Chronic Kidney Disease Outcomes and Practice Patterns Study (CKDopps). *Am J Kidney Dis* 2024; 84: 696–707.e1. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
18. Yapa HE, Purtell L, Chambers S, et al. Factors influencing health-related quality of life in people with chronic kidney disease: a structural equation modelling approach. *J Clin Nurs* 2023; 32: 3445–3455. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
19. Phillips T, Harris S, Aiyegbusi OL, et al. Potentially modifiable factors associated with health-related quality of life among people with chronic kidney disease: baseline findings from the National Unified Renal Translational Research Enterprise CKD (NURTuRE-CKD) cohort. *Clin Kidney J* 2024; 17. DOI: 10.1093/ckj/sfae010. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
20. Markovich JE, Freeman LM, Labato MA, et al. Survey of dietary and medication practices of owners of cats with chronic kidney disease. *J Feline Med Surg* 2015; 17: 979–983. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
21. Brusach K, Lorbach S, Quimby J, et al. Measurement of ghrelin as a marker of appetite dysregulation in cats with and without chronic kidney disease. *Vet Sci* 2023; 10. DOI: 10.3390/vetsci10070464. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
22. Freeman LM, Lachaud MP, Matthews S, et al. Evaluation of weight loss over time in cats with chronic kidney disease. *J Vet Intern Med* 2016; 30: 1661–1666. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

23. King JN, Tasker S, Gunn-Moore DA, et al. Prognostic factors in cats with chronic kidney disease. *J Vet Intern Med* 2007; 21: 906–916. [[PubMed](#)] [[Google Scholar](#)]
24. Harmer M, Wootton S, Gilbert R, et al. Association of nutritional status and health-related quality of life in children with chronic kidney disease. *Qual Life Res* 2019; 28: 1565–1573. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
25. Zabel R, Ash S, King N, et al. Relationships between appetite and quality of life in hemodialysis patients. *Appetite* 2012; 59: 194–199. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
26. Zheng Z, Chen J, He D, et al. The effects of megestrol acetate on nutrition, inflammation and quality of life in elderly haemodialysis patients. *Int Urol Nephrol* 2019; 51: 1631–1638. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
27. Chalhoub S, Langston C, Eatroff A. Anemia of renal disease: what it is, what to do and what's new. *J Feline Med Surg* 2011; 13: 629–640. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
28. Elliott J. Therapeutics of managing reduced red cell mass associated with chronic kidney disease – is there a case for earlier intervention? *J Vet Pharmacol Ther* 2023; 46: 145–157. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
29. Chakrabarti S, Syme HM, Elliott J. Clinicopathological variables predicting progression of azotemia in cats with chronic kidney disease. *J Vet Intern Med* 2012; 26: 275–281. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
30. Geddes RF, Elliott J, Syme HM. Relationship between plasma fibroblast growth factor-23 concentration and survival time in cats with chronic kidney disease. *J Vet Intern Med* 2015; 29: 1494–1501. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
31. Eriksson D, Goldsmith D, Teitsson S, et al. Cross-sectional survey in CKD patients across Europe describing the association between quality of life and anaemia. *BMC Nephrol* 2016; 17. DOI: 10.1186/s12882-016-0312-9. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
32. Johansen KL, Cobitz AR, Singh AK, et al. The ASCEND-NHQ randomized trial found positive effects of daprodustat on hemoglobin and quality of life in patients with non-dialysis chronic kidney disease. *Kidney Int* 2023; 103: 1180–1192. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

33. Michalopoulos SN, Gauthier-Loiselle M, Aigbogun MS, et al. Patient and care partner burden in CKD patients with and without anemia: a US-based survey. *Kidney Med* 2022; 4. DOI: 0.1016/j.xkme.2022.100439. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
34. Jones SE, Quimby JM, Summers SC, et al. Survey of defecation habits in apparently healthy and chronic kidney disease cats. *J Feline Med Surg* 2022; 24: 131–141. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
35. Benjamin SE, Drobatz KJ. Retrospective evaluation of risk factors and treatment outcome predictors in cats presenting to the emergency room for constipation. *J Feline Med Surg* 2019; 22: 153–160. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
36. Sumida K, Molnar MZ, Potukuchi PK, et al. Constipation and incident CKD. *J Am Soc Nephrol* 2017; 28: 1248–1258. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
37. Sumida K, Yamagata K, Kovesdy CP. Constipation in CKD. *Kidney Int Rep* 2020; 5: 121–134. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
38. Lee A, Lambert K, Byrne P, et al. Prevalence of constipation in patients with advanced kidney disease. *J Ren Care* 2016; 42: 144–149. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
39. Ruszkowski J, Majkutewicz K, Heleniak Z, et al. Prevalence and severity of lower gastrointestinal symptoms amongst non-dialysis chronic kidney disease patients: a systematic review and meta-analysis. *J Clin Med* 2022; 11. DOI: 10.3390/jcm11216363. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
40. Ruszkowski J, Heleniak Z, Krol E, et al. Constipation and the quality of life in conservatively treated chronic kidney disease patients: a cross-sectional study. *Int J Med Sci* 2020; 17: 2954–2963. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
41. Taylor S, Caney S, Bessant C, et al. Online survey of owners' experiences of medicating their cats at home. *J Feline Med Surg* 2022; 24: 1283–1293. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
42. Siven M, Savolainen S, Rantila S, et al. Difficulties in administration of oral medication formulations to pet cats: an e-survey of cat owners. *Vet Rec* 2017; 180. DOI: 10.1136/vr.103991. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
43. Adjero L, Brothers T, Shawwa K, et al. The association between polypharmacy and health-related quality of life among non-dialysis chronic kidney disease patients. *PLoS One*

2023; 18. DOI: 10.1371/journal.pone.0293912. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

44. US Food and Drug Administration. Guidance report for industry. Patient-reported outcome measures: use in medical product development to support labeling claims. <https://www.fda.gov/media/77832/download> (2009, accessed 21 August 2025). [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

Supplemental Material

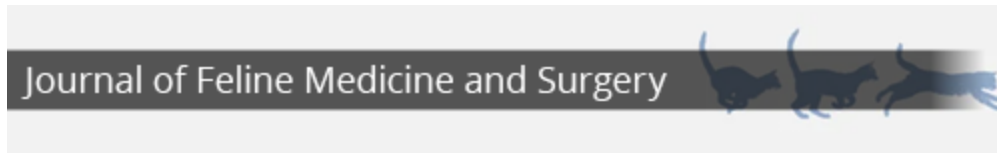
[sj-docx-1-jfm-10.1177_1098612X251367535.docx](#) (22.6KB, docx)

The VetMetrica HRQoL instrument consists of 20 questions. Additional questions for the caregiver about the clinical wellbeing and management of the cat were included as an addendum to the instrument. Veterinary staff who assessed the cat also provided information on their assessment and clinicopathologic data were collected.

Articles from Journal of Feline Medicine and Surgery are provided here courtesy of **SAGE Publications**

As a library, NLM provides access to scientific literature. Inclusion in an NLM database does not imply endorsement of, or agreement with, the contents by NLM or the National Institutes of Health.

Learn more: [PMC Disclaimer](#) | [PMC Copyright Notice](#)



J Feline Med Surg. 2025 Dec 15;27(12):1098612X251377486. doi: [10.1177/1098612X251377486](https://doi.org/10.1177/1098612X251377486) [↗](#)

Caregivers' perspectives on feline chronic kidney disease in Portugal: a questionnaire-based study

[Tomás Rodrigues Magalhães](#)^{1,2,3,✉}, [Ana Luísa Lourenço](#)^{2,3,4}, [Ronald Jan Corbee](#)⁵, [Inês Guerra](#)⁶, [Felisbina Luísa Queiroga](#)^{1,2,3}

[Author information](#) [Article notes](#) [Copyright and License information](#)

PMCID: PMC12709014 PMID: [41398709](#)

Abstract

Objectives

The study aimed to assess caregivers' perspectives on feline chronic kidney disease (CKD).

Methods

People living in Portugal caring for a cat with CKD were invited to participate in an online questionnaire.

Results

In total, 405 responses were considered, with most participants ($n = 309$, 76.3%) dealing with feline CKD for the first time. International Renal Interest Society (IRIS) CKD staging was explained to 260 (64.2%) respondents, but only 188 (46.4%) confirmed that blood pressure was assessed at diagnosis. The recommendation of a renal diet was received by 387 (95.6%) respondents, but only 341 (84.2%) gave it, and, of the latter, 139 (40.8%) were ultimately offering at least some non-renal food daily. Medications, supplements/nutraceuticals and subcutaneous fluid therapy were administered at home by 255 (63.0%), 204 (50.4%) and 205 (50.6%) respondents, respectively. Among the caregivers who used a phosphate binder ($n = 123$), at least 46 (37.4%) were giving it without food. After diagnosis, the caregiver–cat emotional bond remained the same, improved or worsened in 285 (70.4%), 106 (26.2%) and nine (2.2%) cases, respectively. Most respondents felt completely informed by their veterinarian ($n = 331$, 81.7%), complied with the recommended monitoring frequency ($n = 377$, 93.1%) and believed that their experience would not influence future decisions to get a new cat ($n = 221$, 54.6%). Clinical signs and proteinuria, the need for medication and nutraceuticals/supplements, the administration of erythrocyte-stimulating agents and subcutaneous fluids, and the monitoring frequency were higher ($P < 0.05$) in cats at late IRIS CKD stages.

Conclusions and relevance

Most caregivers surveyed felt informed; however, some recommendations were not completely followed. Veterinarian–caregiver communication must be improved to ensure greater adherence to medical recommendations, and an early diagnosis should be pursued to minimise the caregiver burden.

Keywords: Chronic kidney disease, questionnaire-based study, renal disease, veterinary communication, caregiver perspective

Introduction

Given the progressive and irreversible nature of feline chronic kidney disease (CKD), the therapeutic goal is to slow its progression, minimise the associated clinical signs, improve quality of life and increase survival.^{1,2} For that purpose, disease management should be tailored to each patient based on currently accepted international guidelines, and considering that each recommended intervention is supported by varying degrees of quality of evidence, according to the benefits described in the literature.^{2–4} Dietary phosphorus restriction, through the introduction of a renal diet, is one of the most evidence-based recommendations, particularly in cats at CKD stages 2–4.^{1,5} In line with this, the administration of phosphate binders has been

recommended when serum phosphate levels are not being controlled by the diet alone or when a transition to a low-phosphorus diet has not been successfully accomplished.^{2,6} Another common intervention applied to these patients is intravenous or subcutaneous fluid therapy to address dehydration and electrolyte and acid–base imbalances.^{7,8} Severe and sustained increases in the urine protein:creatinine ratio (UPC) and blood pressure (BP) are currently considered to be indications for the prescription of appropriate medications, such as angiotensin-converting enzyme inhibitors, angiotensin receptor blockers and/or calcium channel blockers.^{9,10} To date, it has not been demonstrated that these medications increase longevity; yet, the evidence supporting their impact on the improvement of quality of life was graded as ‘poor’ for proteinuria management and ‘good’ for hypertension management.³ Some additional treatments, namely those to correct anaemia and metabolic acidosis, are generally reserved for more advanced CKD stages (3 and 4), when these complications are more likely to occur and when ensuring quality of life is prioritised over delaying the disease’s progression.^{2,11}

Caregivers play a key role in ensuring that the aforementioned treatments are carried out properly, either by being responsible for implementing them at home or by taking their cats to receive further veterinary care.^{12–14} Their commitment to these treatment recommendations, as well as to follow-up appointments for clinical and laboratory monitoring, is crucial because of their clear impact on the disease’s course and prognosis.³ For example, an appropriate dietary intervention has been associated with significantly longer survival times, which can be achieved only if the caregiver ensures that the renal diet is properly consumed on a daily basis.^{15,16} Therefore, in order to enhance the care provided, it is important to evaluate the caregivers’ performance and identify opportunities for improvement, which has motivated several studies, although mostly limited to the UK^{14,17} and/or USA.^{12,13} As a result of cultural, socioeconomic and demographic differences, this sort of research should be expanded to other countries.

Clinical management of feline CKD was recently evaluated by the authors in a questionnaire-based study addressed to veterinarians working in Portugal, which identified some critical points that warrant further refinement to maximise cats’ prognosis.¹⁸ To complete this assessment and provide an overview regarding the current care of these patients in Portugal, the aim of the present study was to evaluate the caregivers’ perspective about several aspects related to their cats’ renal disease.

Materials and methods

To obtain information from cat caregivers in Portugal, an online questionnaire was designed, consisting mainly of multiple-choice and check-box questions, with a few short-answer inquiries

included (see [supplementary material](#) ). An optional open question was also incorporated at the end of the questionnaire, so the respondents could share additional information about their cat's condition, as well as comments regarding their own experience. The data obtained from this final question were not subject to formal analysis but rather used to provide the authors with more insights into the cat caregivers' perception and their common concerns. Before response collection, the questionnaire underwent an external validation process, involving a group of 10 cat caregivers of different ages, education levels and social backgrounds, to ensure that it would be easily understandable by the wider target population.

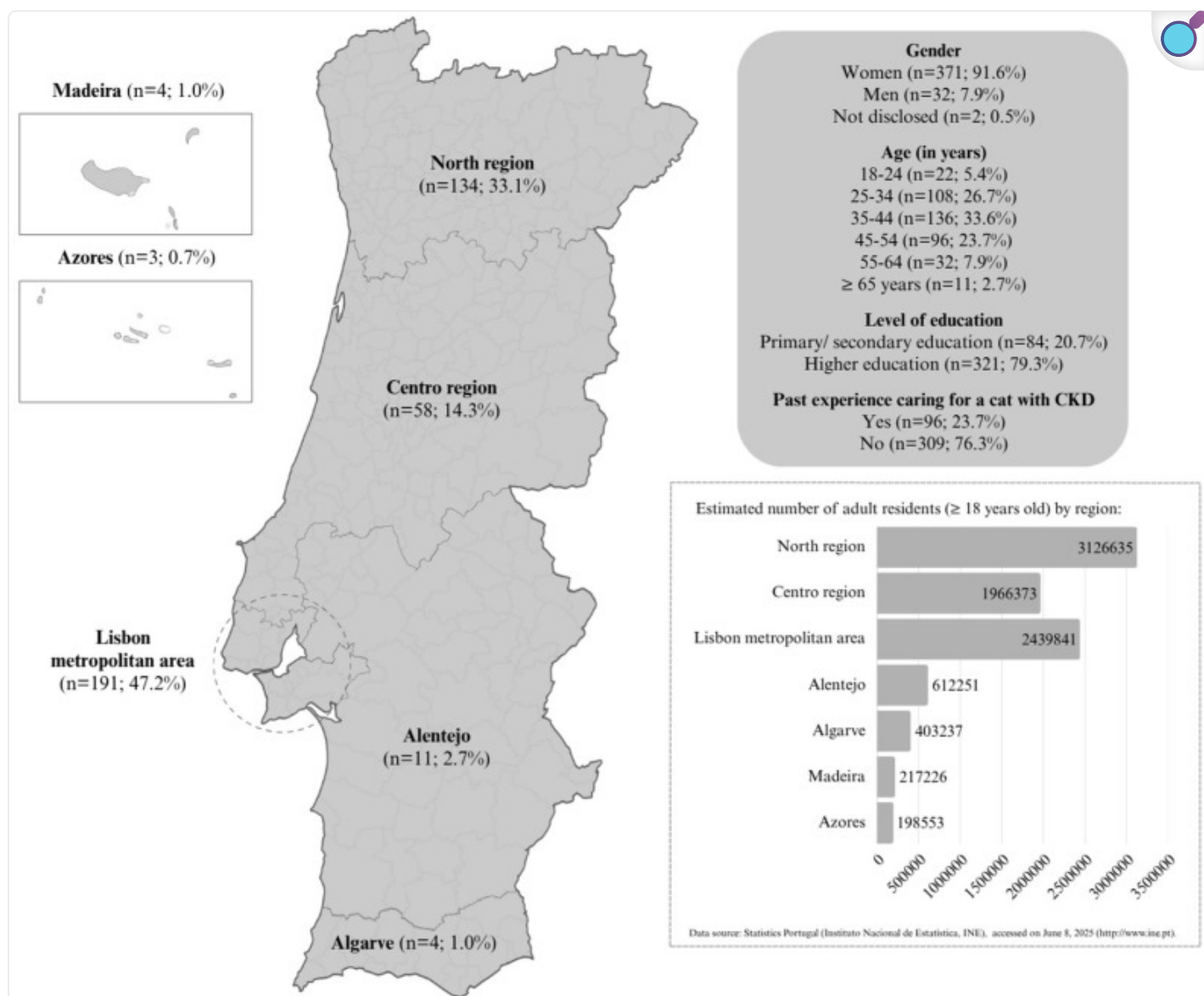
The final version of the questionnaire received a favourable ethical opinion from the Ethics Committee of the University of Trás-os-Montes and Alto Douro (Doc18-CE-UTAD-2024) and was made available through Microsoft Forms. Respondents were required to be aged over 18 years, live in Portugal and be caring for a cat with CKD at the time of participation. An image with a QR code and a short video were created to be shared on social media, and Portuguese veterinarians were also asked to collaborate by raising awareness among their clients. For the latter, all veterinary clinics and hospitals listed on the Directorate-General for Food and Veterinary Medicine (DGAV) website (<https://www.dgav.pt> , accessed on 24 January 2024) were contacted via email. In total, the period of active response collection lasted 13 months (January 2024 to February 2025).

All the data obtained were downloaded into an Excel spreadsheet (Excel version 2502; Microsoft Corp). Responses to short-answer questions, as well as those freely submitted when the 'other' option was selected, were grouped based on similarities in their content. Entries that failed to meet the inclusion criteria were rejected. The final database was then imported into SPSS Statistics software version 29.0.0.0 (IBM Corp) for descriptive analysis. χ^2 and Fisher's exact tests were also applied to investigate associations between the International Renal Interest Society (IRIS) CKD stage and some variables of clinical interest, with $P < 0.05$ considered statistically significant.

Results

In total, 451 caregivers participated, but 46 responses were ultimately excluded: 45 referring to cats that had already died at the time of participation and one due to the absence of a diagnosis. Therefore, 405 responses were considered valid and were included. A predominance of women, working-age adults, highly educated individuals and people living in the two largest urban areas (Porto and Lisbon) was found among respondents ([Figure 1](#)). As the target population was unknown and the questionnaire was freely accessible to all caregivers who met the criteria for participation, the response rate could not be determined.

Figure 1.



[Open in a new tab](#)

Sociodemographic characterisation of the 405 caregivers surveyed

Characteristics and environment of affected cats

Most cats were European (n = 228, 56.3%) or domestic shorthair/longhair (n = 134, 33.1%) and neutered (n = 391, 96.5%). Their median age at diagnosis was 10 years (range 4 months to 21

years). Concomitant diseases had been diagnosed in 175 (43.2%) cats, with periodontal disease (n = 41, 10.1%) and hyperthyroidism (n = 22, 5.4%) being the most common. The environment and resources provided to these cats are described in [Table 1](#).

Table 1.

Description of the environment and resources provided to the 405 cats

	Frequency
Environment	
Shared with other cats	263 (64.9)
Litter boxes	
More than the total number of cats	110 (27.2)
Distributed across different spaces of the household	119 (29.4)
Water supply	
Water bowls in multiple locations	249 (61.5)
Automatic fountains available	205 (50.6)
Water and food bowls placed next to each other	188 (46.4)
Diet (before diagnosis)	
Commercial dry food	396 (97.8)
Commercial wet food	241 (59.5)
Homemade food	13 (3.2)
Raw food	11 (2.7)

[Open in a new tab](#)

Data are n (%)

Clinical presentation, diagnosis and staging

At diagnosis, clinical signs were present in 298 (73.6%) cats, particularly weight loss (n = 172, 42.5%), polydipsia (n = 148, 36.5%), loss of appetite (n = 147, 36.3%), vomiting (n = 146, 36.0%), lethargy/weakness (n = 100, 24.7%) and polyuria (n = 94, 23.2%). In contrast, 107 (26.4%) cats showed no evidence of any abnormal clinical manifestations and were diagnosed through routine tests. The diagnosis was reportedly established mostly based on blood tests (n = 394, 97.3%), urinalysis (n = 288, 71.1%) and abdominal ultrasonography (n = 282, 69.6%).

Of the caregivers, 260 (64.2%) recalled having had the IRIS staging system explained, while 75 (18.5%) indicated they had not received that information and 70 (17.3%) were unsure. Of the 260 respondents who were informed, 69 (26.5%), 77 (29.6%), 41 (15.8%) and 15 (5.8%) stated their cat was diagnosed at stage 1, 2, 3 and 4, respectively. Substaging was also investigated, with 318 (78.5%) and 188 (46.4%) of all caregivers reporting that the UPC and BP had been measured at diagnosis, respectively. Among those who confirmed the UPC assessment, 201 (63.2%) and 71 (22.3%) replied that this urinary parameter was abnormal and normal, respectively. Of all cats whose BP was recorded, 95 (50.5%) had a normal value, 47 (25.0%) had hypertensive values and one (0.5%) was reported to be hypotensive. For the rest (n = 45, 23.9%), their caregivers were unable to specify the result.

Nutritional management

A therapeutic renal diet was recommended to 387 (95.6%) caregivers, mostly by a veterinarian (n = 383). Of all respondents, 345 (85.2%) confirmed that they were aware of the benefits of a renal diet ([Table 2](#)). In addition, some misconceptions were mentioned, namely that the renal diet had fat (n = 2) and potassium (n = 2) restriction, a higher phosphate (n = 1) and protein (n = 1) content, or no protein at all (n = 1).

Table 2.

The benefits of the therapeutic renal diet described by the 405 caregivers surveyed

Benefits reported by caregivers	Frequency
Protein restriction	130 (32.1)
Supports kidney function	84 (20.7)
Controls/slows disease progression	71 (17.5)
Phosphate restriction	57 (14.1)
Sodium intake control	22 (5.4)
Improved urinary tract health	17 (4.2)
Water supply	15 (3.7)
Reduction of clinical signs	13 (3.2)
Better protein quality	12 (3.0)
Increased survival time	10 (2.5)
Improved general condition/quality of life	8 (2.0)
Not aware of its benefits	60 (14.8)

[Open in a new tab](#)

Data are n (%). Only benefits described by five or more respondents were considered

Most caregivers had switched their cat to a renal diet (n = 341, 84.2%). Within this group, most dietary transitions were completed in 2 weeks or less (n = 190, 55.7%) or without a documented gradual protocol (n = 112, 32.8%), differing from the standard recommendations given for these cases. Of those who did transition their cat to a renal diet, 139 (40.8%) respondents were regularly offering other foods ([Table 3](#)).

Table 3.

Food transition period and percentage of renal diet in the cat's daily feeding, according to 341 caregivers

	Frequency
Food transition period carried out	
No transition/immediately	112 (32.8)
<1 week	107 (31.4)
1–2 weeks	83 (24.3)
3–4 weeks	32 (9.4)
5–6 weeks	1 (0.3)
7–8 weeks	2 (0.6)
>8 weeks	4 (1.2)
Percentage of the renal diet consumed	
100% (total) of the daily feeding	202 (59.2)
<100% but >75% of the daily feeding	69 (20.2)
≤75% but >50% of the daily feeding	34 (10.0)
≤50% but >25% of the daily feeding	16 (4.7)
≤25% but >0% of the daily feeding	8 (2.3)
0% (none) of the daily feeding	12 (3.5)
Type(s) of renal diet provided	
Dry commercial food	307 (90.0)
Wet commercial food	227 (66.6)
Homemade food	13 (3.8)
Raw food	3 (0.9)
Reasons not to offer a renal diet exclusively	
Partial or complete food refusal	93 (27.3)
Difficulty in controlling prescribed food intake (multi-cat household)	14 (4.1)

	Frequency
Cat's specific preference for a non-renal wet food	14 (4.1)
Cost/financial burden of the renal diet	11 (3.2)
Other nutritional management required (concomitant disease)	9 (2.6)
No noticeable improvement in the cat's overall condition	2 (0.6)
Other	4 (1.2)

[Open in a new tab](#)

Data are n (%)

Additional therapeutic approaches

Overall, 255 (63.0%), 204 (50.4%) and 205 (50.6%) cats were receiving medications, nutraceuticals/supplements and subcutaneous fluids, respectively ([Table 4](#)). In addition, 33 (8.1%) were on erythrocyte-stimulating agent (ESA) therapy.

Table 4.

Medications and nutraceuticals/supplements administered and main aspects of the subcutaneous fluid therapy, according to the 405 caregivers surveyed

	Frequency
Medications	
Medication to control proteinuria (eg, ACE inhibitor or ARB)	148 (36.5)
Antiemetic (eg, maropitant)	88 (21.7)
Appetite stimulant (eg, oral or transdermal mirtazapine)	81 (20.0)
Medication to control hypertension (eg, CCB or ARB)	64 (15.8)
Histamine type-2 receptor antagonists (eg, famotidine)	36 (8.9)
Proton pump inhibitors (eg, omeprazole)	21 (5.2)
Thiamazole	6 (1.5)
Gabapentin/pregabalin	5 (1.2)
None	150 (37.0)
Nutraceuticals/supplements	
Phosphate binders	123 (30.4)
Multivitamin concentrates	50 (12.3)
Iron and B vitamin products	30 (7.4)
Omega-3 fatty acid supplements	24 (5.9)
Potassium supplements	17 (4.2)
<i>Lespedeza</i> -based products	13 (3.2)
Probiotics	13 (3.2)
Glycosaminoglycans-rich products	6 (1.5)
Tryptophan	5 (1.2)
None	201 (49.6)
Fluid therapy administration	
Applied only at home	120 (29.6)

	Frequency
Applied only at the veterinary clinic/hospital	55 (13.6)
Applied in both locations	30 (7.4)
Not applied	199 (49.1)
Not sure	1 (0.2)
Reasons for not receiving subcutaneous fluids	
Not recommended by the attending veterinarian	162 (40.0)
No need at the time	28 (6.9)
Cat did not tolerate the procedure	15 (3.7)
No willingness to subject the cat to the procedure	6 (1.5)
No noticeable improvement in the cat's general condition	3 (0.7)

[Open in a new tab](#)

Data are n (%). Only medications and nutraceuticals/supplements reported by five or more respondents were considered

ACE = angiotensin-converting enzyme; ARB = angiotensin receptor blocker; CCB = calcium channel blocker

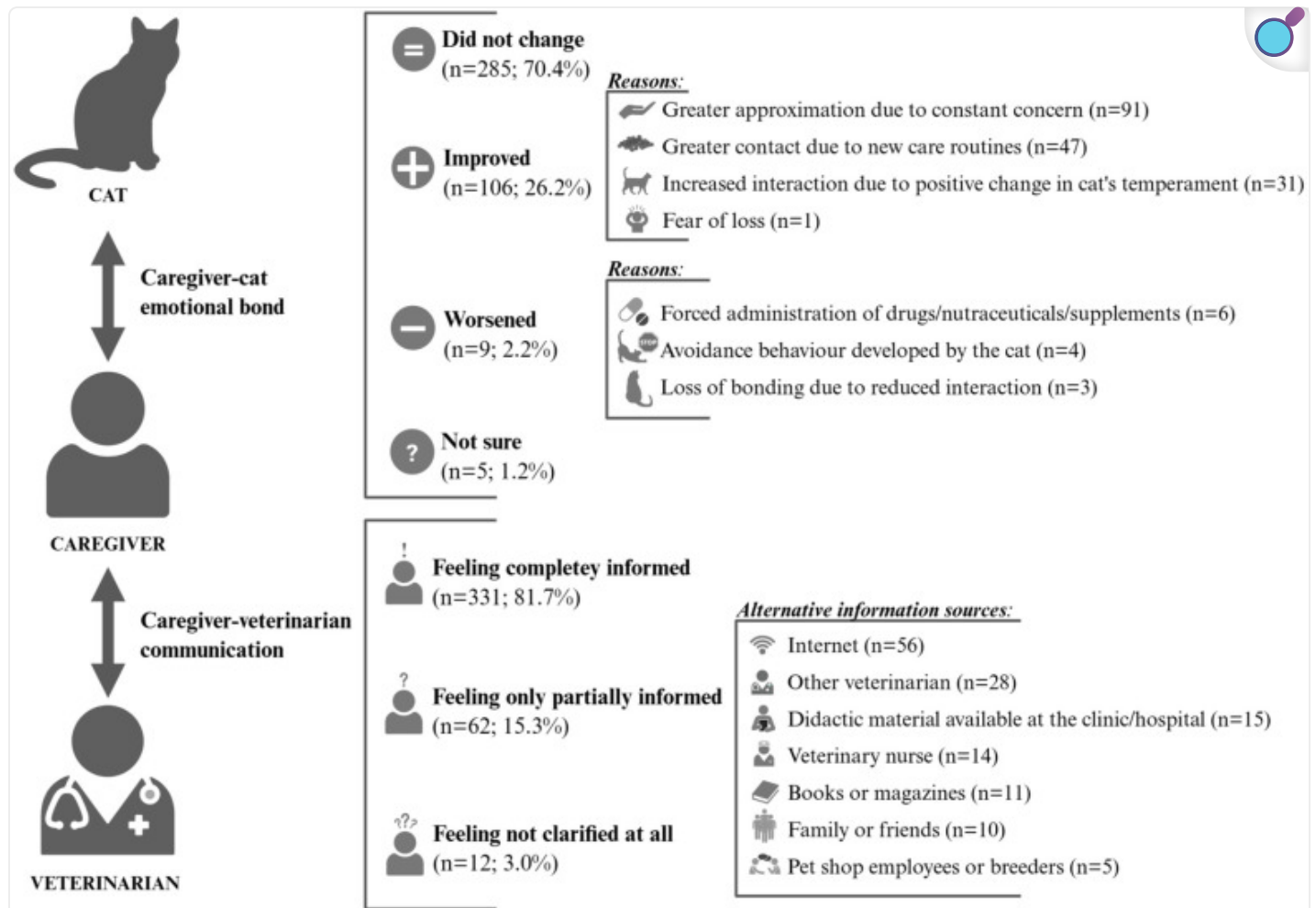
Of the 201 cats that had proteinuria at diagnosis, half (n = 100) were on antiproteinuric medication at the time the caregiver completed the questionnaire, whereas among the 47 cats that were hypertensive at presentation, 27 (57.4%) were being treated for that condition.

Of the 123 caregivers who administered a phosphate binder, the majority provided it with food, either in a single meal (n = 30, 24.4%) or across two or more meals (n = 32, 26.0%). A subset (n = 46, 37.4%) administered the binder directly into the cat's mouth, independent of mealtimes. Among the same 123 caregivers, 101 (82.1%) reported having previously attempted a dietary transition, and 64 (52.0%) administered the phosphate binder in addition to an exclusive renal diet.

Caregiver–cat emotional bond, caregiver–veterinarian communication and disease monitoring

For most respondents, the emotional bond with their cat remained the same after diagnosis (n = 285, 70.4%), and the information given by the veterinarian was considered sufficient to feel fully informed (n = 331, 81.7%) (Figure 2).

Figure 2.



[Open in a new tab](#)

Caregiver–cat emotional bond and caregiver–veterinarian communication after diagnosis, according to the 405 caregivers surveyed (Created in BioRender. Magalhães, T. (2025) <https://BioRender.com/rrltckw>)

Moreover, according to the majority (n = 235, 58.0%), when their cat was in a stable stage of the disease, the veterinarian recommended reassessment every 4–5 months or more often, and 377

(93.1%) reported complying with the medical recommendation ([Table 5](#)).

Table 5.

Monitoring frequency recommended by the veterinarian for cats in a stable stage of renal disease, according to the 405 caregivers surveyed

	Frequency
Monitoring frequency recommended by their veterinarian	
More than once a month	22 (5.4)
Monthly	56 (13.8)
Every 2–3 months	126 (31.1)
Every 4–5 months	31 (7.7)
Every 6 months	98 (24.2)
Every 7 or more months	27 (6.7)
Only when the cat became worse	45 (11.1)
Adherence to follow-up recommendation received	
Yes	377 (93.1)
No	28 (6.9)
Reasons for non-adherence	
Cat's stress during transportation and/or at the clinic/hospital	16 (4.0)
Cost/financial burden of the appointments and/or tests	13 (3.2)
Not thinking the cat needed to be monitored so regularly	6 (1.5)
Lack of time for veterinary visits	6 (1.5)
Forgetting to schedule an appointment	2 (0.5)

[Open in a new tab](#)

Data are n (%)

Most respondents (n = 388, 95.8%) indicated an understanding that feline CKD is a progressive condition managed through supportive care aimed at slowing progression and enhancing quality of life. A smaller proportion (n = 13, 3.2%) expressed the belief that a curative outcome might be possible. In addition, 121 (29.9%) respondents found the information provided about the disease sufficient to understand its general aspects, while 128 (31.6%) believed that more educational resources were needed. Overall, 221 (54.6%) respondents stated that the impact the disease had on their cat's life and their own would not influence future decisions regarding having a new cat, with 81 (20.0%) believing it would and the remaining 103 (25.4%) not committing to any answer.

Potential associations between IRIS CKD stage and variables of clinical interest

When the analysis was narrowed to cats with a known IRIS CKD stage at diagnosis (n = 202), some statistically significant associations were identified ([Table 6](#)).

Table 6.

Associations between disease stage (early vs late) and variables of clinical interest in the 202 cats with known International Renal Interest Society CKD stage at diagnosis

	Frequency		P value
	Early CKD stages (n = 146)	Late CKD stages (n = 56)	
Clinical signs at presentation			<0.001
Yes	97 (66.4)	55 (98.2)	
No	49 (33.6)	1 (1.8)	
Proteinuria at diagnosis			<0.001
Proteinuric	67 (45.9)	40 (71.4)	
Non-proteinuric	44 (30.1)	3 (5.4)	
Not measured	10 (6.8)	4 (7.1)	
Don't know/don't remember	25 (17.1)	9 (16.1)	
Transition to a renal diet			0.378
Yes	127 (87.0)	46 (82.1)	
No	19 (13.0)	10 (17.9)	
Medication administration			<0.001
Yes	73 (50.0)	48 (85.7)	
No	73 (50.0)	8 (14.3)	
Nutraceutical/supplement administration			0.036
Yes	70 (47.9)	34 (60.7)	
No	76 (52.1)	21 (37.5)	
Not specified	0 (0.0)	1 (1.8)	
ESA therapy			0.026
Yes	11 (7.5)	8 (14.3)	

	Frequency		<i>P</i> value
	Early CKD stages (n = 146)	Late CKD stages (n = 56)	
No	133 (91.1)	44 (78.6)	<0.001
Don't know/don't remember	2 (1.4)	4 (7.1)	
Fluid therapy			
Yes	56 (38.4)	41 (73.2)	0.004
No	90 (61.6)	15 (26.8)	
Recommended monitoring frequency			
More than once a month	4 (2.7)	9 (16.1)	
Monthly	17 (11.6)	14 (25.0)	
Every 2–3 months	47 (32.2)	12 (21.4)	
Every 4–5 months	13 (8.9)	4 (7.1)	
Every 6 months	50 (34.2)	13 (23.2)	
Every 7 or more months	10 (6.8)	2 (3.6)	
Only when the cat became worse	5 (3.4)	2 (3.6)	

[Open in a new tab](#)

Data are n (%). *P* value represents the level of significance

Other variables were evaluated, such as breed, sex/reproductive status, age (≤ 10 years vs > 10 years) and blood pressure at diagnosis, but were not significantly associated with disease stage

CKD = chronic kidney disease; ESA = erythrocyte-stimulating agent

Discussion

The present findings showed that most cats already had clinical signs at diagnosis of CKD, with these being more frequent among late-stage patients.¹⁹ However, over one-quarter of the cats were identified solely through routine tests, which is a higher relative percentage than those

reported in other studies.^{14,20} This difference may be caused by a greater awareness in recent years about the importance of check-up screening tests for early diagnosis.²¹ Moreover, the IRIS staging system was explained to a larger percentage of caregivers compared with a previous survey-based study involving UK-based owners (64.2% vs 20.5%),¹⁴ reflecting that IRIS staging has been increasingly adopted by veterinarians over time. This approach may benefit from further reinforcement to support caregiver understanding of the progressive nature of feline CKD and its associated prognosis. Enhanced communication and education efforts could potentially increase the proportion of caregivers who are fully informed and who recognise the primary goals of CKD management. For substaging, UPC was determined in most cats, but BP measurement was seemingly often overlooked, which is consistent with the very limited percentage of Portuguese veterinarians (19.1%) who reported performing BP measurement on all their newly diagnosed cats.¹⁸ Some of the reasons behind this include equipment unavailability, difficulty in performing and/or interpreting BP measurement, stress/lack of patient compliance, and caregivers' financial and/or time constraints. Particular obstacles must be identified to find solutions to systematise this procedure, allowing early detection of systemic hypertension, which is a highly prevalent condition in these patients.²² Still, it should be noted that since the diagnosis and staging were reported by the caregiver and were not cross-checked with the attending veterinarian's records (to maintain full anonymity), there may be some discrepancies due to misunderstanding/miscommunication.

The treatment recommendations published by IRIS² and in the 'ISFM consensus guidelines on the diagnosis and management of feline chronic kidney disease'³ have been the most widely applied in the therapeutic management of feline CKD. Although both were developed by a panel of experts and reflect current evidence, these guidelines should be interpreted as general recommendations rather than strict rules, as each case has its specific circumstances. Either way, dietary intervention is consistently considered the foundation of this approach, often relying on the prescription of a renal diet, which is typically characterised by restricted phosphorus content, moderate to restricted high-quality protein, elevated concentrations of omega-3 fatty acids, and added potassium and B vitamins.^{1,5} Beyond adequate nutritional supply tailored to the patient's needs, it is crucial that the diet ensures the required calorie intake, as cats with significant weight loss tend to have poorer outcomes.^{23,24} According to the findings of the present study, a renal diet was recommended by the veterinarian for most cats, but in 20.5% of those who transitioned, it represented 75% or less of their daily feeding, which is in line with the perception of suboptimal food acceptance shared by the Portuguese veterinarians.¹⁸ The cat's refusal to eat the renal diet was indicated as the main limiting factor, which, based on the remaining findings, could be explained by the apparent inadequate diet introduction by a large proportion of caregivers who did not carry out a gradual transition period or did it for a duration that may have been insufficient. Rather than being the result of poor adherence to veterinary recommendations, whether due to

hastiness or misunderstanding, this is probably a consequence of many professionals recommending switching diets immediately or too quickly in this country, as also previously identified.¹⁸ However, it should be noted that although the guidelines typically recommend a gradual transition of 4 weeks or more,³ the most appropriate period depends on each patient and may be shorter in cats that have a good appetite and are more receptive to change, or slower in more selective/'fussy' eaters and/or unwell cats. Regarding the latter scenario, most did indeed show clinical signs at diagnosis, particularly weight loss, so their appetite and willingness to accept new foods were probably already compromised. Elliot et al¹⁷ recently reported that veterinary recommendation was the factor that most frequently influenced UK owners to feed their cat a renal diet. It is expected that the same would have been verified with the Portuguese caregivers, suggesting that if they had allegedly received more effective counselling at the time of the dietary recommendation, namely by informing them of all the potential benefits, their adherence might have been higher. For example, although it depends on several factors such as disease stage and associated comorbidities, previous studies have suggested that a cat on a renal diet may potentially live 2–3 times longer than one fed a non-renal diet,^{15,16} which, if communicated, could make caregivers more prone to keep their cat on a renal diet long term. Yet, only 2.5% of caregivers mentioned being aware of this benefit, revealing an opportunity for improvement for the veterinary team. Therefore, to increase the success rate of its implementation, this therapeutic diet should be initiated as early as clinically recommended, when appetite is not impaired and the cat is apparently stable, and then gradually introduced, according to each cat's individual response.^{3,5}

Most caregivers were administering at least one medication, with antiproteinuric drugs being the most frequently listed, which is aligned with the significant percentage of cats that were proteinuric at diagnosis. Intervention to address proteinuria is advised, as it has been considered a negative prognostic factor.^{25,26} Systemic hypertension has also been associated with the development and aggravation of proteinuria, even though their correlation has not yet been well established and the reverse also seems to occur.^{22,27,28} Therefore, drugs for treating hypertension were also reported among the medications most frequently administered, albeit to a lesser extent. In fact, as CKD progresses, the risk of developing hypertension tends to increase,²⁹ which may explain the higher number of cats on treatment for this condition compared with those identified as hypertensive at diagnosis. Even so, this discrepancy may also be the result of more cats actually having hypertension at presentation than were reported (particularly among those whose caregivers were unsure or could not remember) and/or due to the fact that Semintra (Boehringer Ingelheim) was listed as an option for the treatment of both hypertension and proteinuria, which may have caused misleading selection. Another noteworthy finding was that less than 60% of cats diagnosed with these two conditions were reportedly on appropriate medication when the caregiver responded to the questionnaire, suggesting that the remainder were either not receiving

the recommended treatment or no longer required it because of disease stabilisation. Future studies should explore in greater depth this potential inconsistency and the criteria being used to support therapeutic decisions regarding these drugs.

Phosphate binders were commonly used by respondents; however, in some cases, their administration may not have aligned with best practice. For instance, over one-third of caregivers reported giving phosphate binders without regard for feeding times, which can reduce their effectiveness.^{2,3} In addition, 17.9% of those using phosphate binders had not attempted a transition to a renal diet, which has been considered the recommended first-line approach for managing hyperphosphataemia.^{3,6}

Approximately half of the cats were receiving subcutaneous fluids, a therapy frequently prescribed by veterinarians in this country.¹⁸ In most of those cases, the procedure was carried out at home, which, according to the findings described by Cooley et al,¹³ was considered an 'easy' and 'no stress' experience by the majority of the owners surveyed. However, evidence of its long-term benefits, as well as the ideal patient selection criteria, the target fluid volume to be administered and the most suitable treatment regimen, are still lacking.^{4,7} This, combined with the fact that it may potentially be associated with adverse effects, particularly in cats with underlying cardiovascular disease and/or hypoproteinaemia,^{7,30} and that some owners have been reported to administer fluids without assessing their cat's hydration status,¹³ suggests that this therapy should be carefully considered on a case-by-case basis. Thus, the authors suggest that other strategies to increase water intake, such as feeding wet food and providing several separated water bowls and water fountains, should be prioritised before prescribing in-home fluid therapy, as was already done by many (but not all) of the caregivers surveyed. Regarding ESA therapy, only a minority of the cats were undergoing it, as expected, since this type of treatment has generally been reserved only for symptomatic anaemia or when the packed cell volume is persistently below 20%.^{3,11} Finally, it should be noted that cats were more subjected to medication, nutraceuticals/supplementation, subcutaneous fluids and ESAs when they were diagnosed at late IRIS CKD stages, emphasising the greater frequency of secondary complications at a more advanced disease state.^{19,31} Therefore, less care was required to be provided at home by caregivers with cats diagnosed in early stages, reducing their burden.

Despite all the precautions and procedures normally recommended for managing the disease, a majority of the caregivers considered that the relationship with their cat did not change after diagnosis, which shows that a normal interaction can be preserved without any significant detriment to the existing emotional bond. Interestingly, in approximately one-quarter of the respondents, this connection even seemed to have improved because of greater contact and

consequent interaction, albeit it cannot be ruled out that this may have resulted from the cat being more tolerant and cooperative due to unwellness. In contrast, the impact was viewed as negative by a few caregivers because of the forced medication administration, an issue that has been subject to research. Providing caregivers with a demonstration of how to administer the medication in the most effective and cat-friendly way, and choosing products ideally in liquid form, registered for cats and with greater palatability, are some of the recommendations to overcome this difficulty and minimise the associated stress.³²⁻³⁴

Although a high percentage of the caregivers surveyed felt fully informed by their veterinarian, 18.3% still had questions, which may have led to some of the misconceptions about the renal diet and the inappropriate in-home management practices reported. When they were unclear, respondents often used the internet for guidance, which could include not only websites with professionally reviewed information but also unsupervised sources with inaccurate advice. Considering the importance of evidence-based veterinary medicine,³⁵ a collective effort must be made by the veterinary community to ensure that caregivers are fully informed, reducing the need to seek information through alternative channels. To facilitate this, more veterinary educational resources with easy-to-understand and easy-to-apply content should be developed, filling the gap highlighted by 31.6% of respondents.

One of the main relevant findings of the survey-based study of Portuguese veterinarians was the concerning discrepancy found between the recommended monitoring frequency and the periodicity truly achievable in most cases,¹⁸ which was further explored here: 70.9% of veterinarians in the preceding study stated they recommended clinical monitoring every 2–3 months or more often in their stable patients, whereas only 50.4% of the caregivers in the present study indicated having received this same instruction. Furthermore, given that the majority of caregivers believed they were adhering to the reassessment periodicity advised, a possible miscommunication between veterinary professionals and their clients was suggested. Still, at least for the late stages, a significantly higher recommended frequency was reported, which is consistent with the current guidelines, since uraemic crises are more frequent and quality of life can deteriorate faster.^{2,31} Apart from raising awareness of the importance of more periodic check-ups, an attempt should be made to minimise the associated stress, as concerns about this were pointed out by non-compliant caregivers. Careful preparation of veterinary visits and cat-friendly handling techniques should be ensured to overcome this limiting factor.^{36,37} Pre-appointment gabapentin is also a commonly used option; however, in these patients, a dose reduction is advisable given the renal impairment.^{38,39}

An increased caregiver burden would be expected among this population, as has been described for this and other chronic feline diseases.⁴⁰⁻⁴³ Nevertheless, only 20% of caregivers clearly stated that their experience would influence their decision to get a new cat in the future, suggesting that for most of them, the challenges and difficulties reported did not outweigh the positive impact of the human–animal bond on their lives.^{44,45} However, as entries relating to already deceased cats were not accepted, it is possible that caregivers who experienced greater distress due to the disease may be underrepresented, namely those who immediately chose euthanasia upon diagnosis. In any case, strategies to ensure continuous caregiver support should be promoted in all cases to minimise this potentially negative impact of the disease.

This study has some limitations that should be acknowledged. Given the absence of a national registry for this feline disease (unlike for cancer⁴⁶), it was not possible to estimate the target population size or characterise it, thereby precluding a proper assessment of the risk of sampling bias. Beyond that, the questionnaire was time-consuming and could only be completed online, which may have led to a greater representation of more dedicated caregivers, while limiting participation to those with internet access. Since respondents had to recall past information, particularly about the diagnosis, some difficulty may have been also encountered, occasionally resulting in more ‘don’t know/don’t remember’ responses or potential loss of data accuracy. Conformity bias cannot be ruled out either, although the assurance of anonymity and the use of neutral wording are expected to have minimised this potential issue. Finally, considering the overrepresentation of female respondents, it is conceivable that the present findings may not adequately reflect the views of male cat caregivers. This is a frequent issue in research, with other recent questionnaire-based studies on cat owners reporting participant pools in which women accounted for more than 90% of those surveyed.^{17,47,48} Greater willingness to respond to online surveys,^{49,50} more use of social media,⁵¹ and increased bonding with and concern for their cat^{52,53} may explain this female predominance. However, it is also likely that more women than men are taking on the role of primary caregiver for domestic cats, explaining this higher proportion, which may in fact be consistent with the target population.

Despite this, to the best of the authors’ knowledge, this is one of the most comprehensive caregiver survey-based studies on feline CKD published to date, and the first to be conducted in a southern European country. Another distinguishing feature was the decision not to include entries regarding cats that were already deceased at the time of completion, as this loss could have negatively influenced some of the answers and would not reflect current veterinary care. Furthermore, the final response rate was considered satisfactory, especially if compared with other similar survey-based studies with longer collection periods¹⁴ and/or larger target populations,^{12-14,17,54} adequately representing the population group under investigation.

Conclusions

This study showed that most caregivers in Portugal felt supported by their veterinarian in managing feline CKD, although several inadequate practices were being carried out, such as lack of BP measurement at diagnosis, potentially improper transition to a renal diet and incorrect administration of phosphate binders. The importance of early diagnosis was also highlighted, as it was associated with a lower prevalence of clinical signs and less in-home care. Overall, study findings showed that veterinarian–caregiver communication should be refined to ensure better adherence, maximising therapeutic success.

Supplemental Material

Online questionnaire provided to Portuguese caregivers of cats diagnosed with chronic kidney disease

[sj-docx-1-jfm-10.1177_1098612X251377486.docx](#) ^(42.9KB, docx)

'Feline chronic kidney disease from the caregivers' perspective'.

Acknowledgments

The authors would like to thank all the cat caregivers who responded to the questionnaire, as well as all the veterinarians who encouraged their eligible clients to participate.

Footnotes

Accepted: 14 August 2025

Author note: Preliminary results were presented at the ISFM Congress 2024.

Supplementary material: The following file is available as supplementary material:

Online questionnaire provided to Portuguese caregivers of cats diagnosed with chronic kidney disease: ‘Feline chronic kidney disease from the caregivers’ perspective’.

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding: This work was supported by the PhD Studentship number 2021.07272.BD (DOI: 10.54499/2021.07272.BD) provided by Fundação para a Ciência e a Tecnologia (FCT) and financed by National Funds (FCT/Ministério da Ciência, Tecnologia e Ensino Superior, Portugal) under the project number UID/00772.

Ethical approval: This work did not involve the use of animals and therefore ethical approval was not specifically required for publication in *JFMS*.

Informed consent: This work did not involve the use of animals (including cadavers) and therefore informed consent was not required. No animals or people are identifiable within this publication, and therefore additional informed consent for publication was not required.

ORCID iD: Tomás Rodrigues Magalhães  <https://orcid.org/0000-0003-3478-2840> 

Felisbina Luísa Queiroga  <https://orcid.org/0000-0001-6130-8381> 

References

1. Parker VJ. Nutritional management for dogs and cats with chronic kidney disease. *Vet Clin North Am Small Anim Pract* 2021; 51: 685–710. [[DOI](#) ] [[PubMed](#)] [[Google Scholar](#) - 2. International Renal Interest Society. Treatment recommendations for CKD in cats. <https://www.iris-kidney.com/iris-guidelines-1>  (2023, accessed March 2025).
- 3. Sparkes AH, Caney S, Chalhoub S, et al. ISFM Consensus guidelines on the diagnosis and management of feline chronic kidney disease. *J Feline Med Surg* 2016; 18: 219–239. [[DOI](#) ] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#) - 4. Korman RM, White JD. Feline CKD: current therapies—what is achievable? *J Feline Med Surg* 2013; 15 Suppl 1: 29–44. [[DOI](#) ] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#) 

5. Polzin DJ, Churchill JA. Controversies in veterinary nephrology: renal diets are indicated for cats with International Renal Interest Society chronic kidney disease stages 2 to 4: the pro view. *Vet Clin North Am Small Anim Pract* 2016; 46: 1049–1065. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
6. Kidder AC, Chew D. Treatment options for hyperphosphatemia in feline CKD: what's out there? *J Feline Med Surg* 2009; 11: 913–924. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
7. Pardo M, Spencer E, Odunayo A, et al. 2024 AAHA fluid therapy guidelines for dogs and cats. *J Am Anim Hosp Assoc* 2024; 60: 131–163. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
8. Langston C, Gordon D. Effects of IV fluids in dogs and cats with kidney failure. *Front Vet Sci* 2021; 8: 659960. DOI: 10.3389/fvets.2021.659960. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
9. Vaden SL, Elliott J. Management of proteinuria in dogs and cats with chronic kidney disease. *Vet Clin North Am Small Anim Pract* 2016; 46: 1115–1130. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
10. Taylor SS, Sparkes AH, Briscoe K, et al. ISFM consensus guidelines on the diagnosis and management of hypertension in cats. *J Feline Med Surg* 2017; 19: 288–303. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
11. Chalhoub S, Langston C, Eatroff A. Anemia of renal disease: what it is, what to do and what's new. *J Feline Med Surg* 2011; 13: 629–640. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
12. Markovich JE, Freeman LM, Labato MA, et al. Survey of dietary and medication practices of owners of cats with chronic kidney disease. *J Feline Med Surg* 2015; 17: 979–983. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
13. Cooley CM, Quimby JM, Caney SM, et al. Survey of owner subcutaneous fluid practices in cats with chronic kidney disease. *J Feline Med Surg* 2018; 20: 884–890. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
14. Caney SM. An online survey of dietary and phosphate binder practices of owners of cats with chronic kidney disease. *J Feline Med Surg* 2017; 19: 1040–1047. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

15. Elliott J, Rawlings JM, Markwell PJ, et al. Survival of cats with naturally occurring chronic renal failure: effect of dietary management. *J Small Anim Pract* 2000; 41: 235–242. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
16. Plantinga EA, Everts H, Kastelein AM, et al. Retrospective study of the survival of cats with acquired chronic renal insufficiency offered different commercial diets. *Vet Rec* 2005; 157: 185–187. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
17. Elliott J, Reyes-Hughes H, Hibbert A, et al. Owners' experiences of caring for cats with chronic kidney disease in the UK. *J Feline Med Surg* 2025; 27. DOI: 10.1177/1098612X251314769. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
18. Magalhães TR, Lourenço AL, Corbee RJ, et al. Clinical management of feline chronic kidney disease in Portugal: a questionnaire-based study. *J Feline Med Surg* 2023; 25. DOI: 10.1177/1098612X231206125. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
19. International Renal Interest Society. IRIS staging of CKD. <https://www.iris-kidney.com/iris-guidelines-1> (2023, accessed March 2025).
20. Elliott J, Barber PJ. Feline chronic renal failure: clinical findings in 80 cases diagnosed between 1992 and 1995. *J Small Anim Pract* 1998; 39: 78–85. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
21. Quimby J, Gowland S, Carney HC, et al. 2021 AAHA/AAFP feline life stage guidelines. *J Feline Med Surg* 2021; 23: 211–233. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
22. Lawson JS, Jepson RE. Feline comorbidities: the intermingled relationship between chronic kidney disease and hypertension. *J Feline Med Surg* 2021; 23: 812–822. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
23. Freeman LM, Lachaud MP, Matthews S, et al. Evaluation of weight loss over time in cats with chronic kidney disease. *J Vet Intern Med* 2016; 30: 1661–1666. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
24. Hall JA, Fritsch DA, Jewell DE, et al. Cats with IRIS stage 1 and 2 chronic kidney disease maintain body weight and lean muscle mass when fed food having increased caloric density, and enhanced concentrations of carnitine and essential amino acids. *Vet Rec* 2019; 184: 190. DOI: 10.1136/vr.104865. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
25. Syme HM, Markwell PJ, Pfeiffer D, et al. Survival of cats with naturally occurring chronic renal failure is related to severity of proteinuria. *J Vet Intern Med* 2006; 20: 528–535.

[\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

26. King JN, Tasker S, Gunn-Moore DA, et al. Prognostic factors in cats with chronic kidney disease. *J Vet Intern Med* 2007; 21: 906–916. [\[PubMed\]](#) [\[Google Scholar\]](#)

27. de Freitas MN, de Freitas MS, da Silva TF, et al. Blood pressure and proteinuria in older cats and cats with chronic kidney disease. *Vet World* 2025; 18: 527–533. [\[DOI\]](#) [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

28. Reynolds BS, Lefebvre HP. Feline CKD: pathophysiology and risk factors – what do we know? *J Feline Med Surg* 2013; 15 Suppl 1: 3–14. [\[DOI\]](#) [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

29. Hori Y, Heishima Y, Yamashita Y, et al. Relationship between indirect blood pressure and various stages of chronic kidney disease in cats. *J Vet Med Sci* 2018; 80: 447–452. [\[DOI\]](#) [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

30. Pouchelon JL, Atkins CE, Bussadori C, et al. Cardiovascular–renal axis disorders in the domestic dog and cat: a veterinary consensus statement. *J Small Anim Pract* 2015; 56: 537–552. [\[DOI\]](#) [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

31. Brown CA, Elliott J, Schmiedt CW, et al. Chronic kidney disease in aged cats: clinical features, morphology, and proposed pathogeneses. *Vet Pathol* 2016; 53: 309–326. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

32. Odom TF, Riley CB, Benschop J, et al. Medication compliance by cat owners prescribed treatment for home administration. *J Vet Intern Med* 2025; 39: e17298. DOI: 10.1111/jvim.17298. [\[DOI\]](#) [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

33. Taylor S, Caney S, Bessant C, et al. Online survey of owners’ experiences of medicating their cats at home. *J Feline Med Surg* 2022; 24: 1283–1293. [\[DOI\]](#) [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

34. Sivén M, Savolainen S, Rantilä S, et al. Difficulties in administration of oral medication formulations to pet cats: an e-survey of cat owners. *Vet Rec* 2017; 180: 250. DOI: 10.1136/vr.103991. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

35. Block G. Evidence-based veterinary medicine—potential, practice, and pitfalls. *J Vet Intern Med* 2024; 38: 3261–3271. [\[DOI\]](#) [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

36. Rodan I, Dowgray N, Carney HC, et al. 2022 AAFP/ISFM cat friendly veterinary interaction guidelines: approach and handling techniques. *J Feline Med Surg* 2022; 24: 1093–1132. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
37. Taylor S, St Denis K, Collins S, et al. 2022 ISFM/AAFP cat friendly veterinary environment guidelines. *J Feline Med Surg* 2022; 24: 1133–1163. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
38. Erickson A, Harbin K, MacPherson J, et al. A review of pre-appointment medications to reduce fear and anxiety in dogs and cats at veterinary visits. *Can Vet J* 2021; 62: 952–960. [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
39. Quimby JM, Lorbach SK, Saffire A, et al. Serum concentrations of gabapentin in cats with chronic kidney disease. *J Feline Med Surg* 2022; 24: 1260–1266. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
40. Nakano Y, Matsushima M, Nakamori A, et al. Depression and anxiety in pet owners after a diagnosis of cancer in their pets: a cross-sectional study in Japan. *BMJ Open* 2019; 9. DOI: 10.1136/bmjopen-2018-024512. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
41. Shaevitz MH, Tullius JA, Callahan RT, et al. Early caregiver burden in owners of pets with suspected cancer: owner psychosocial outcomes, communication behavior, and treatment factors. *J Vet Intern Med* 2020; 34: 2636–2644. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
42. Spitznagel MB, Gober MW, Patrick K. Caregiver burden in cat owners: a cross-sectional observational study. *J Feline Med Surg* 2023; 25. DOI: 10.1177/1098612X221145835. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
43. Henning J, Nielson T, Nettifee J, et al. Understanding the impacts of feline epilepsy on cats and their owners. *Vet Rec* 2021; 189: e836. DOI: 10.1002/vetr.836. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
44. Friedmann E, Son H. The human–companion animal bond: how humans benefit. *Vet Clin North Am Small Anim Pract* 2009; 39: 293–326. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
45. Hodgson K, Darling M. Zooeyia: an essential component of ‘One Health’. *Can Vet J* 2011; 52: 189–191. [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
46. Pinello K, Pires I, Castro AF, et al. Vet-OncoNet: developing a network of veterinary oncology and reporting a pioneering Portuguese experience. *Vet Sci* 2022; 9: 72. DOI:

10.3390/vetsci9020072. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

47. Filipe JFS, Lauzi S, Pina L, et al. A survey of Italian cat owners' attitudes towards cat vaccination through a web-based questionnaire. BMC Vet Res 2021; 17: 267. DOI: 10.1186/s12917-021-02981-z. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

48. Stallwood J, Shirlow A, Hibbert A. A UK-based survey of cat owners' perceptions and experiences of antibiotic usage. J Feline Med Surg 2020; 22: 69–76. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

49. Mulder J, de Bruijne M. Willingness of online respondents to participate in alternative modes of data collection. Surv Pract 2019; 12. DOI: 10.29115/SP-2019-0001. [[DOI](#)] [[Google Scholar](#)]

50. Smith WG. Does gender influence online survey participation? A record-linkage analysis of university faculty online survey response behavior. <https://eric.ed.gov/?id=ED501717> (2008, accessed June 2025).

51. Laor T. My social network: group differences in frequency of use, active use, and interactive use on Facebook, Instagram and Twitter. Technol Soc 2022; 68: 101922. DOI: 10.1016/j.techsoc.2022.101922. [[Google Scholar](#)]

52. Bouma EMC, Reijgwart ML, Dijkstra A. Family member, best friend, child or 'just' a pet, owners' relationship perceptions and consequences for their cats. Int J Environ Res Public Health 2021; 19: 193. DOI: 10.3390/ijerph19010193. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

53. Adamelli S, Marinelli L, Normando S, et al. Owner and cat features influence the quality of life of the cat. Appl Anim Behav Sci 2005; 94: 89–98. [[Google Scholar](#)]

54. Jones SE, Quimby JM, Summers SC, et al. Survey of defecation habits in apparently healthy and chronic kidney disease cats. J Feline Med Surg 2022; 24: 131–141. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

Online questionnaire provided to Portuguese caregivers of cats diagnosed with chronic kidney disease

[sj-docx-1-jfm-10.1177_1098612X251377486.docx](#) (42.9KB, docx)

'Feline chronic kidney disease from the caregivers' perspective'.

Articles from Journal of Feline Medicine and Surgery are provided here courtesy of **SAGE Publications**

As a library, NLM provides access to scientific literature. Inclusion in an NLM database does not imply endorsement of, or agreement with, the contents by NLM or the National Institutes of Health.

Learn more: [PMC Disclaimer](#) | [PMC Copyright Notice](#)



Animals (Basel). 2026 Apr 25;16(9):1319. doi: [10.3390/ani16091319](https://doi.org/10.3390/ani16091319) [↗](#)

Enhancing Detection of Feline Chronic Kidney Disease Through Smart Litter Box Monitoring

[Natalie Langenfeld-McCoy](#)^{1,*}, [LeAnn Snow](#)¹, [Heidi Gordon](#)¹, [Zachary George](#)², [Jessica Quimby](#)², [Olivia Arndt](#)¹, [Sarah Thomas](#)¹, [Nicholas Schoeneck](#)¹, [Ragen T S McGowan](#)¹

Editor: Joseph Bartges

[Author information](#) [Article notes](#) [Copyright and License information](#)

PMCID: PMC13162653 PMID: [42121739](#)

Simple Summary

Chronic kidney disease (CKD) is a common and serious health issue in older cats that is irreversible and often challenging to detect, especially in the early stages of the disease. This study investigated how a new technology, the smart litter box monitor, could help detect behavioral changes related to CKD by tracking how cats use the litter box. We compared cats with CKD (Renal group) to a group of similar age and sex matched cats with no known conditions (Non-Renal group), analyzing their elimination patterns. The findings showed that the Renal group urinated more often, spent a longer time eliminating, and covered their waste with less force and for a shorter duration than the Non-Renal group. The smart litter box monitor was able to detect these behaviors and identify them as unique to this group of cats. This could help caregivers and veterinarians identify CKD and manage the condition better because the monitor offers an in-home, ongoing 24/7 passive data collection of elimination behavior trends and changes.

Keywords: chronic kidney disease (CKD), feline elimination behavior, litter box monitor, supervised machine learning, smart pet lot devices, health monitoring, CKD detection

Abstract

Chronic kidney disease (CKD) is a prevalent condition in cats and recognized as a leading cause of mortality among cats generally, yet it is difficult to detect by cat caregivers, especially in its early stages. This retrospective study aimed to enhance detection of CKD by cat caregivers through a smart litter box monitor technology that tracks feline elimination behaviors. We established cohorts of cats with CKD and cats with no known conditions and analyzed behavioral features captured by the monitor. A mixed-effects model identified significant differences, which were then integrated into a machine learning framework for predictive modeling. The model achieved a weighted F1-score of 92.7% in training using cross-validation and 89.9% in validation, demonstrating high precision for CKD predictions. Key findings included a behavioral profile unique to cats with CKD characterized by increased urination frequency, longer elimination durations, and reduced post-elimination covering behavior. These results suggest that the smart litter box monitor can provide valuable, continuous, and non-invasive data for CKD detection and management.

1. Introduction

Chronic kidney disease (CKD) is one of the most common irreversible, and often progressive, syndromes experienced by pet cats [1,2,3,4]. Estimates range from 31% in cats > 15 years old [1] to 30–40% of cats > 10 years old [5] and it is considered a primary cause of mortality in cats overall [6,7,8]. There is also evidence of higher prevalence: one study found CKD in 50% of randomly selected cats aged 6 months to 20 years [9]. Despite the high prevalence, signs of CKD are often missed by pet caregivers because they may not know what changes to look for, or because even known behaviors are difficult to reliably observe and track. Moreover, clinical diagnosis presents a substantial challenge for veterinarians who have limited visibility into a cat's behavior at home and typically only encounter patients at annual visits or once overt signs have advanced sufficiently to raise caregiver concern.

Some of the literature suggests that CKD, in its early stages, is characterized by clinical changes that are present but typically too subtle for caregivers to readily detect [10]. As such, there is general interest in studying risk factors, biomarkers, and detection mechanisms to better understand and treat CKD as early as possible [1,2,5,6,11,12]. Due to the complexity of the syndrome, timely detection and diagnosis of CKD can be difficult [11,13]. Diagnosis of CKD is

further complicated by the presence of non-specific clinical signs (i.e., weight loss, dehydration, vomiting, altered appetite) and comorbidities (i.e., hyperthyroidism, lower urinary tract disease) [6,10]. In one cohort case study [14], the most frequent clinical signs preceding diagnosis were reported to be weight loss (49.9%; 311), excessive thirst (polydipsia) (38.6%; 241) and excessive urination (polyuria) (24.5%; 153). At diagnosis, over 56.6% of the cohort cats (354) had two or more compatible clinical signs recorded, while nearly 25% (155) had no clinical signs recorded [14].

Biomarkers hold promise as possible sources of early detection; however, they also hold inherent limitations like lacking a single salient marker for assessment, non-renal factors, poor sensitivity required for reliable early detection, misinterpretations due to wide reference ranges, and lacking robust validation [5,6,7,8,11,15]. Overall, routine blood and urine testing, in combination with medical histories and clinical findings, are considered the most reliable means for screening kidney disease [3,8,10,16]. And still, up to 75% of functional renal mass may be lost before azotemia (persistently increased creatinine and BUN concentration) is detected by testing [17].

Algorithms and machine learning have started to be leveraged with promising results as a possible tool for the detection of CKD, in some cases combining biomarkers and risk factor features in a recurrent neural network (RNN) [15], and in others evaluating predictive confidences of multivariate biomarkers using random forest (RF) and support vector machine (SVM) modeling to detect up to 6 months earlier than typical diagnosis [8]. Digital tools may contribute enhanced detection and may also facilitate managing the chronic, progressive course of CKD in which regular monitoring is widely acknowledged as essential for optimizing the efficacy of interventions [3]. This is one of the emerging benefits of the smart litter box monitor, a device that passively collects continuous longitudinal data of a cat's elimination patterns [18]. The load cell sensors imbedded in the smart litter box monitor detect signals related to cat behaviors in and around the litter box. AI models were rigorously trained by a supervised machine learning methodology in which hundreds of thousands of events time-synced with video footage were labeled. The system predicts cat or human interaction, the type of cat event that is occurring (i.e., urination, defecation, non-elimination), specific cat IDs within a multi-cat household, and weight measurements, among other features related to cat elimination behaviors [18]. The smart litter box monitor also alerts cat caregivers to changes in cats' weight and elimination behaviors. Due to the signal detection capabilities of the smart litter box monitor and its alerting capabilities, this device may support caregivers by providing health- and behavior-related insights.

For example, a group of researchers have identified elimination differences (both urination and defecation) among cats with CKD [1] using the same smart litter box monitor technology [18]. This

2025 study found that cats with CKD had a higher mean number of urination events per day compared to the healthy cat group. This same study was primarily focused on defecation patterns and found that CKD cats defecate less frequently than healthy cats. More specifically, cats with CKD had a lower mean number of defecation events per day over the 30-day study period, and CKD cats defecated less frequently per day and aggregated across the observation periods. This study provided the first published evidence for the capabilities of the smart litter box monitor to assess meaningful changes as related to CKD.

The present retrospective study aims to demonstrate that the smart litter box monitor may be a useful tool empowering cat caregivers and veterinarians with AI-driven data collected passively, continuously, and noninvasively for CKD detection. The aim of this study was to synthesize complex elimination behavior features from a smart litter box monitor and develop a predictive model for the detection of CKD. We hypothesized that we could detect behavioral differences between cats with CKD (Renal group) and cats with no known conditions (Non-Renal group) using features from sensors within the smart litter box monitor technology, and that these features would support the development of a predictive model to identify cats exhibiting a unique behavioral profile indicative of CKD.

2. Materials and Methods

This retrospective study analyzed data collected from January 2023 to September 2025. The dataset utilized in this study was compiled from two primary sources: the Nestlé Purina PetCare Center (NPPC) in Missouri, USA, and various in-home (IH) environments. Cats were categorized into two groups, termed Renal and Non-Renal. The NPPC cats in the Renal group were all diagnosed with CKD by a veterinarian and receiving veterinary standard of care for CKD. Diagnosis of CKD was based on creatinine greater than 1.6, blood urea nitrogen greater than reference range, and USG < 1.035, in combination with patient history and physical exam findings. The sample of IH cats comprised two groups: (1) clinical study cats that had confirmed veterinary diagnoses of CKD and receiving CKD standard of care [\[1\]](#); and (2) general population cats, categorized based on caregiver-reported CKD diagnoses with unknown treatment status.

The data from this retrospective study had been collected under Institutional Animal Care and Use (IACUC) approval or informed consent of data usage. The Nestlé Purina IACUC and Ohio State University IACUC numbers were NT9913 and IACUC-2023A00000032, respectively. For all IH data collection, signed informed consent waivers were gathered. Additionally, caregiver-reported health information was obtained from a voluntary in-app health survey from cat caregivers utilizing the

smart litter box monitors and contributed a group of cats with reported “no known health conditions” which were included in the Non-Renal group.

Cats from this data pool were split into two groups for modeling purposes: a Leave-One-Out Cross-Validation (LOOCV) set was used for training the model, and a second group, called the validation set, was used to evaluate how the model would perform on unseen cats. The training cats included the 13 CKD cats (4 from NPPC and 9 from IH) with confirmed International Renal Interest Society (IRIS) stage (stage 1 ($n = 2$), stage 2 ($n = 4$), stage 3 ($n = 5$), stage 4 ($n = 2$)) and 72 cats (22 from NPPC and 50 from IH) with no known conditions. For each cat in the Renal group there was a cat of similar age in the Non-Renal group, but the Non-Renal group had a wider range of ages and a lower average age (see [Table S1](#)). The validation Renal group comprised 7 cats (1 from NPPC and 6 from IH) with reported CKD and 44 cats (3 from NPPC and 41 from IH) with no known conditions. IRIS stage was not recorded for all cats in the Renal group validation set, thereby simulating a real-world use case of the smart litter box monitor technology. However, the inability to confirm self-reported diagnoses of the IH cats in the validation set presents a limitation and added uncertainty. Each cat in both the training and the validation group contributed multiple weeks of data from the smart litter box monitor technology. Data ranged from a minimum of 2 weeks to a maximum of 104 weeks (median: 6 weeks) for each cat. To prevent data leakage from the multiple weeks of data, LOOCV was employed at the cat level, grouping all weekly observations from each cat so that individuals appeared exclusively in either the training or test set within a fold. The dataset was partitioned into 13 folds, each containing one renal cat and a subset of non-renal cats, ensuring that no observations from the same cat were shared between training and testing sets.

To begin the data preparation, data filtering criteria were established to include weeks with a minimum of four days of recorded data to ensure inclusion of households with cats actively using the smart litter box monitor. Outlier measurements of the urination and/or defecation weight of output were identified and removed, specifically those below -150 g or above 350 g. Combo events, defined as urination and defecation occurrences within a single elimination event, were aggregated into respective counts. Exploratory Data Analysis (EDA) was conducted by aggregating features at the weekly level for each cat, with modeling features computed using a 7-day rolling window. The EDA identified a wide initial pool of 120 potential statistical features (e.g., mean, medians, totals) from the available smart litter box monitor data so a feature preselection process was implemented to eliminate redundancy and retain only distinct features to use for the mixed model analysis step in the process, as shown in [Figure 1](#).

Figure 1.

Data Preparation



Feature Preselection



Mixed Model Analysis



Scaling of Features



Model Development
and Selection



Model Evaluation



Precision

Recall

F1-Score

[Open in a new tab](#)

Model development process.

Ultimately, 24 explainable features that could be measured by the smart litter box monitor were selected for further analysis to both validate known behavioral differences and uncover new differences between the two groups, including the following:

- Mean Daily Visits in a Week;
- Mean Transition/Covering Duration in a Week;
- Mean Transition/Covering Duration of Urination Events in a Week;
- Mean Elimination Duration of Defecation Events in a Week;
- Mean Event Duration in a Week;
- Sum of Weight of Output of Urination Events in a Week;
- Mean Elimination Duration of Urination Events in a Week;
- Mean Event Duration of Defecation Events in a Week;
- Mean Elimination Duration in a Week;
- Mean Covering Up Intensity of Urination Events in a Week;
- Mean Event Duration of Urination Events in a Week;
- Mean Entry and Digging Duration in a Week;
- Mean of Weight of Output of Defecation Events in a Week;
- Mean Transition/Covering Duration of Defecation Events in a Week;
- Mean Event Duration of Non-Elimination Events in a Week;
- Sum of Weight of Output of Defecation Events in a Week;
- Mean of Weight of Output of Urination Events in a Week;
- Mean Entry and Digging Duration of Urination Events in a Week;
- Standard Deviation of Weight in a Week;
- Mean Digging Up Intensity of Urination Events in a Week;
- Mean Daily Non-Elimination Events in a Week;
- Mean Entry and Digging Duration of Defecation Events in a Week;
- Mean Daily Defecation Events in a Week.

A mixed-effects model was then employed to identify features significantly associated with the two groups (Renal vs. Non-Renal), accounting for individual variability across cats. Mixed-effects models were conducted using R Software (v4.4.1) [19] and glmmTMB (v 1.1.12) packages [20]. The model was first structured as Model v1: $\text{Feature} \sim \text{Condition} + (1|\text{pet_id})$.

However, the litter box maintenance practices at NPPC involved daily scooping, whereas in-home maintenance revealed that 25% of boxes were scooped daily, another 25% once a week, and 50% between 2 and 6 times per week. Previous research indicates that litter box maintenance can significantly influence behavior [21,22], necessitating the inclusion of Source as a control variable in the mixed models. To identify features impacted by Source, a second model was structured as Model v2: $\text{Feature} \sim \text{Source} + (1|\text{pet_id})$.

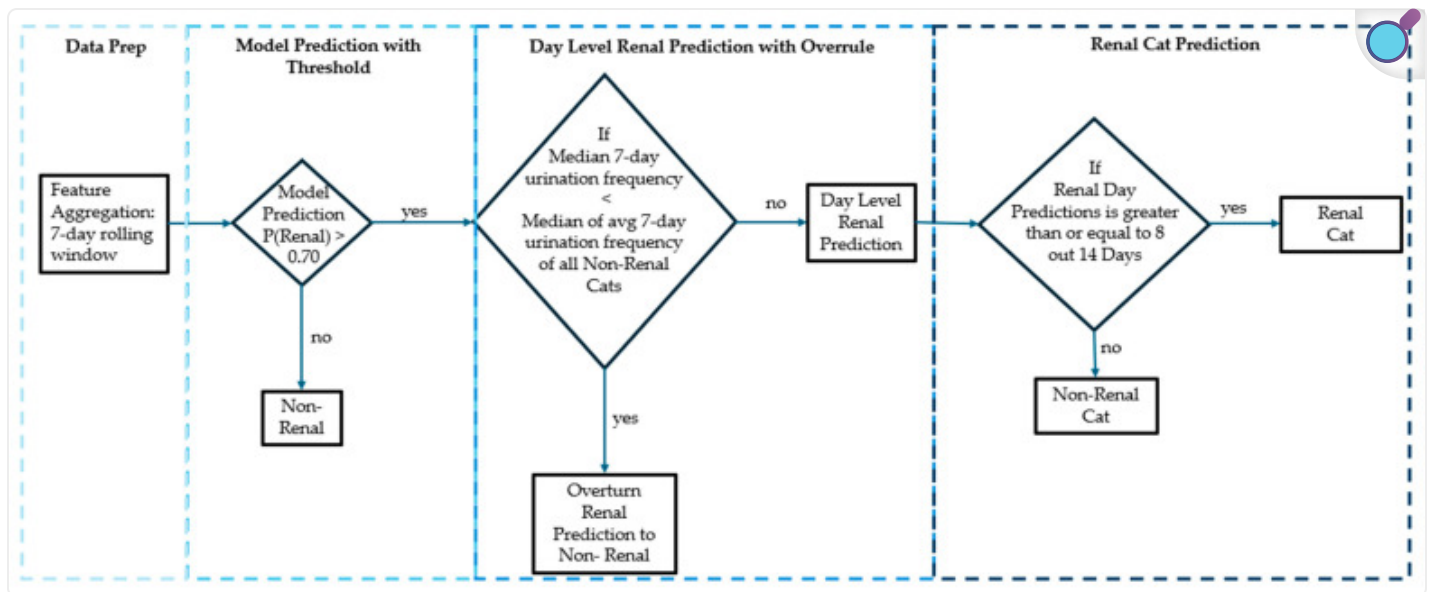
Finally, to ascertain which features exhibited a statistically significant relationship with the health condition while controlling for environmental influence ($\text{Feature} \sim \text{Source}$), the mixed-effects model that included Condition, Source, and the interaction fixed effects was deployed Model v3: $\text{Feature} \sim \text{Condition} + \text{Source} + \text{Condition}:\text{Source} + (1|\text{pet_id})$.

To mitigate environmental impacts on key features that were significantly related to Condition, standardization was performed on those features that were also significantly related to the environment, scaling feature values within the NPPC and IH groups independently.

Significant features identified through mixed modeling analyses were inputted into a predictive modeling framework for further refinement. Modeling was conducted in Python (v.3.12.9) [23] using common machine learning packages (numpy 2.1.3; pandas 2.3.0; catboost 1.2.10; scikit-learn 1.5.1) [24,25]. A CatBoost model was constructed, chosen for its effectiveness in handling categorical variables and robustness against overfitting [26] and performed better (28% lift in Renal precision) than more simple models like logistic regression. Given the limited number of cats with CKD in the dataset, LOOCV was utilized for model training and testing. This approach maximizes the use of limited data, providing a more accurate estimate of model performance while reducing bias and detecting overfitting.

In predictive modeling, a common threshold for positive predictions is 0.5; however, a threshold of 0.7 was employed in this analysis to enhance precision (Figure 2). We systematically evaluated thresholds of 0.5, 0.6, 0.7, and 0.8 and found that a threshold of 0.7 provided the most favorable balance between precision and F1-score for the renal class. This elevated threshold reduces the likelihood of false positives, particularly critical in medical contexts.

Figure 2.



[Open in a new tab](#)

Flow for predicting Renal.

Additionally, a Renal prediction override mechanism was established, requiring an elevated number of urination events over a 7-day rolling window to further refine predictions. The trained model generates daily predictions based on data collected over the preceding 7 days. A cat is classified as having CKD if it is predicted in the Renal group for 8 out of 14 consecutive days, allowing for variability in the data and reducing the risk of misclassification due to transient conditions.

To optimize model performance, Recursive Feature Elimination was employed, systematically removing the least significant features and re-evaluating the model's performance iteratively to select for the most salient features ([Table 1](#)).

Table 1.

Selected features.

Feature Name	Feature Importance
Mean Daily Urination Events in a Week	23.01
Mean Covering Up Intensity of Urination Events in a Week	16.66
Mean Elimination Duration of Urination Events in a Week	15.14
Mean Transition/Covering Duration in a Week	12.56
Mean Transition/Covering Duration of Urination Events in a Week	11.25
Mean Elimination Duration in a Week (Scaled)	10.99
Sum of Weight of Output of Urination Events in a Week	10.40

[Open in a new tab](#)

The performance metrics assessed included precision (true positive predictions), recall (sensitivity), and F1-score, with a focus on optimizing precision for the Renal group and recall for the Non-Renal group. Misclassification analyses were performed for incorrect predictions.

3. Results

Several significant elimination behavior differences among cats with CKD were observed, particularly regarding their urination events and related in-box activities ([Table 2](#)).

Table 2.

Results.

Feature Name	Condition	Count	Mean	Std.	Model v1	Model v2	Condition <i>p</i> Value	Source <i>p</i> Value	Condition <i>p</i> Value
Mean Daily Non-Elimination Events in a Week	Non-Renal	816	1.2	1.8	0.552	<0.001	0.008		
	Renal	117	1.2	2.1					
Mean Daily Urination Events in a Week	Non-Renal	816	2.3	1.0	< 0.001	0.641	< 0.001		
	Renal	117	4.6	2.1					
Mean Daily Defecation Events in a Week	Non-Renal	816	0.8	0.5	0.968	0.146	0.010		
	Renal	117	0.7	0.6					
Mean Daily Visits in a Week	Non-Renal	816	4.1	2.5	< 0.001	0.001	< 0.001		
	Renal	117	6.3	4.0					
Mean Event Duration of Non-Elimination Events in a Week	Non-Renal	629	38.2	32.1	0.260	0.002	-		
	Renal	71	54.3	43.5					
Mean Event Duration of Urination Events in a Week	Non-Renal	808	70.4	20.5	0.037	0.297	0.305		
	Renal	117	58.9	18.1					
Mean Entry and Digging Duration of Urination Event in a Week	Non-Renal	726	18.7	8.9	0.321	<0.001	0.361		
	Renal	117	13.6	6.1					

					Model v1	Model v2	
Feature Name	Condition	Count	Mean	Std.	Condition <i>p</i> Value	Source <i>p</i> Value	Condition <i>p</i> Value
Mean Elimination Duration of Urination Events in a Week	Non-Renal	709	17.8	8.3	0.015	0.128	0.250
	Renal	117	25.2	9.1			
Mean Transition/Covering Duration of Urination Events in a Week	Non-Renal	726	22.9	11.3	0.001	0.962	0.015
	Renal	117	14.0	6.7			
Mean Digging Up Intensity of Urination Events in a Week	Non-Renal	646	0.1	0.2	0.396	0.593	0.964
	Renal	108	0.1	0.1			
Mean Covering Up Intensity of Urination Events in a Week	Non-Renal	651	0.2	0.2	0.036	0.931	0.400
	Renal	112	0.1	0.1			
Mean Event Duration in a Week	Non-Renal	816	78.5	25.4	0.009	<0.001	0.122
	Renal	117	64.0	21.1			
Mean Entry and Digging Duration in a Week	Non-Renal	730	19.8	8.2	0.068	0.013	0.133
	Renal	117	14.3	5.9			
Mean Elimination Duration in a Week	Non-Renal	713	17.4	7.9	0.030	0.026	0.877
	Renal	117	24.9	8.2			
Mean Transition/Covering Duration in a Week	Non-Renal	730	24.0	11.7	<0.001	0.190	0.009
	Renal	117	14.7	6.9			

					Model v1	Model v2	
Feature Name	Condition	Count	Mean	Std.	Condition <i>p</i> Value	Source <i>p</i> Value	Condition <i>p</i> Value
Sum of Weight of Output of Urination Events in a Week	Non-Renal	808	469.8	476.8	0.015	0.603	0.028
	Renal	117	1007.4	634.5			
Mean of Weight of Output of Urination Events in a Week	Non-Renal	808	36.0	29.2	0.315	0.142	0.715
	Renal	117	37.9	22.6			
Standard Deviation of Weight in a Week	Non-Renal	598	40.1	47.5	0.374	0.196	0.848
	Renal	113	61.8	56.3			
Mean Event Duration of Defecation Events in a Week	Non-Renal	724	135.2	38.7	0.025	0.253	0.032
	Renal	78	114.4	34.1			
Mean Entry and Digging Duration of Defecation Events in a Week	Non-Renal	477	33.9	20.4	0.872	0.961	0.344
	Renal	78	28.7	16.4			
Mean Elimination Duration of Defecation Events in a Week	Non-Renal	455	20.9	11.7	0.003	0.483	0.061
	Renal	78	32.5	12.9			
Mean Transition/Covering Duration of Defecation Events in a Week	Non-Renal	477	39.2	22.3	0.187	0.259	0.629
	Renal	78	33.0	31.4			
Sum of Weight of Output of Defecation Events in a Week	Non-Renal	724	129.4	156.9	0.312	0.901	0.232
	Renal	97	89.7	140.5			

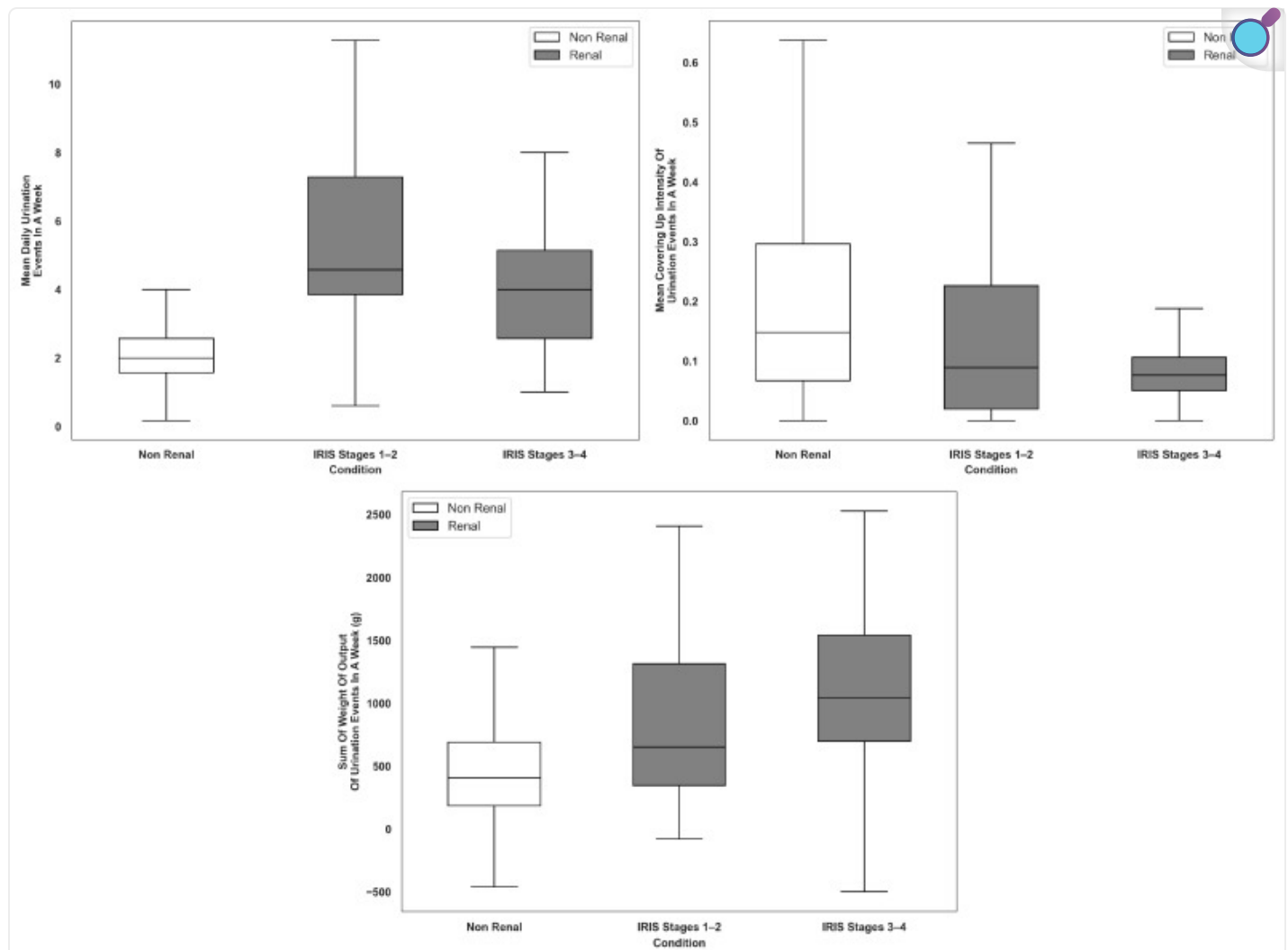
					Model v1	Model v2	
Feature Name	Condition	Count	Mean	Std.	Condition <i>p</i> Value	Source <i>p</i> Value	Condition <i>p</i> Value
Mean of Weight of Output of Defecation Events in a Week	Non-Renal	721	31.0	33.5	0.172	0.992	0.842
	Renal	95	21.8	25.5			

[Open in a new tab](#)

Bolded numbers correspond to statistically significant values related to condition.

Cats in the Renal group demonstrated an increased frequency of daily urination events compared to cats in the Non-Renal group with a mean of 4.6 events per week for cats with CKD compared to 2.3 in cats with no known conditions ($p < 0.001$). This increase was one of the most prominent indicators of CKD ([Table 1](#)). Throughout various elimination events, cats with CKD spent more time eliminating (urination, defecation, or combo). The duration of voiding was also notably longer for these cats during urination events (25.2 s vs. 17.8 s; $p = 0.015$). Interestingly, while the overall event duration for urination was shorter in cats with CKD (58.9 s vs. 70.4 s; $p = 0.037$), this was attributed to significantly reduced post-elimination behaviors, such as covering (14.0 s vs. 22.9 s; $p = 0.001$) and covering intensity or the force that cats are using to move the litter ($p = 0.036$). The cats in the Renal group exhibited a tendency to cover their waste with less intensity. Moreover, cats in the Renal group spent less time covering their waste after urination. The overall time spent covering, regardless of the event, was reduced compared to the Non-Renal group, indicating a broader behavioral shift among the cats in the Renal group. Finally, the cats in the Renal group recorded an increased total weight of output during urination over a 7-day period (1007.4 g vs. 469.8 g; $p = 0.015$), suggesting a rise in urine volume. The respective behavior findings among the seven most important features as selected by RFE from the set of statistically significant features across the Non-Renal and Renal groups stratified by IRIS stages are shown in [Figure 3](#) and [Figure 4](#).

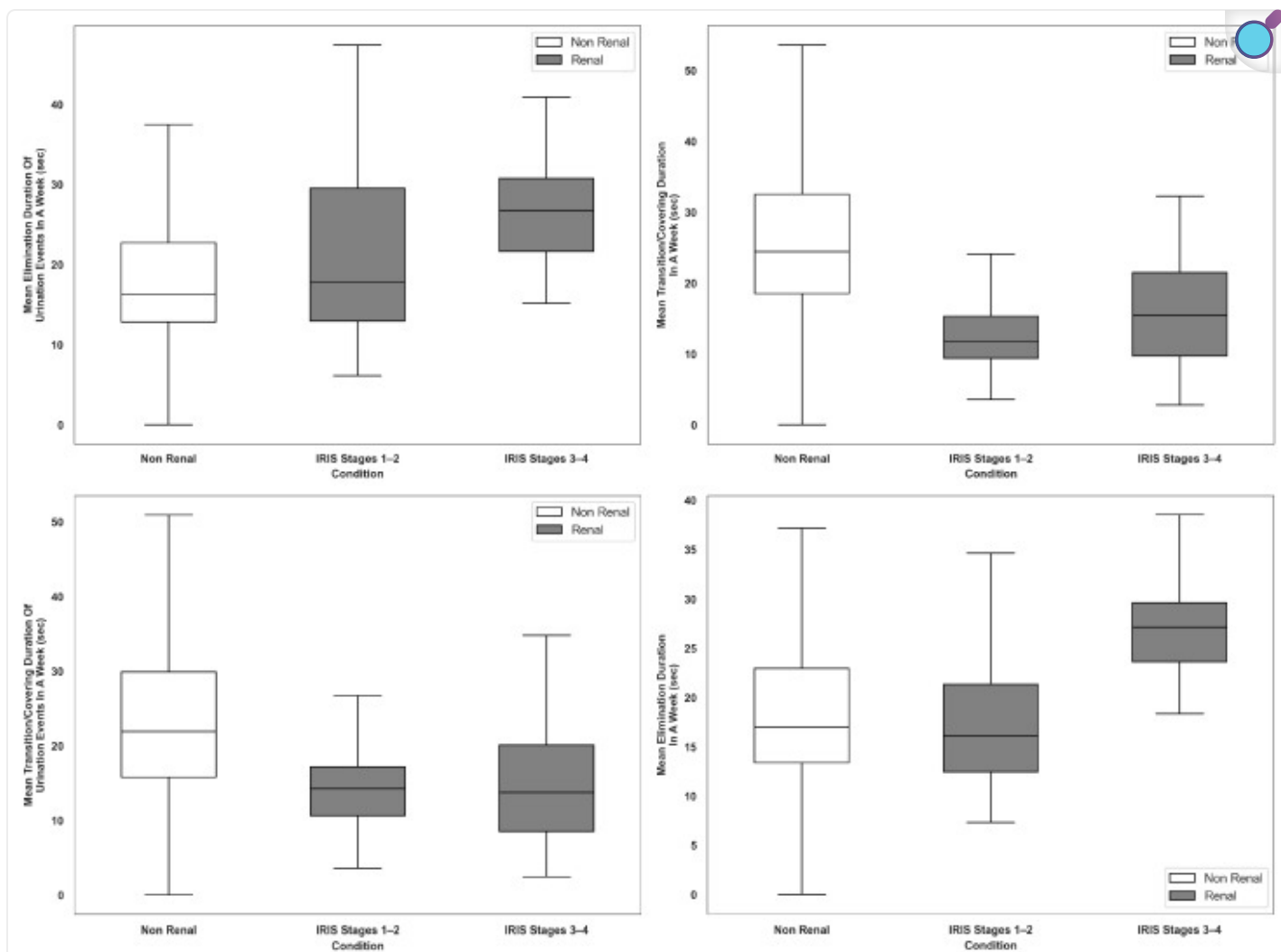
Figure 3.



[Open in a new tab](#)

Boxplots of non-duration-based features for Non-Renal and Renal by IRIS stages for significant features.

Figure 4.



[Open in a new tab](#)

Boxplots of duration-based features for Non-Renal and Renal by IRIS stages for significant features.

These features collectively highlight a unique behavioral profile in cats with CKD characterized by more frequent urination, longer voiding durations, reduced covering behavior, and increased urine output as compared to cats with no known conditions. While some of these features have been anecdotally reported, this study objectively confirmed differences in behavioral profiles. Some of these behaviors (i.e., reduced covering behavior) are consistent across IRIS stages 1-2 and IRIS

stages 3–4 while others are more pronounced in IRIS stages 3–4 compared to IRIS stages 1–2 (i.e., longer voiding durations).

In terms of defecation behavior, cats in the Renal group showed a slightly lower frequency of daily defecation events (0.7 vs. 0.8 per week; $p = 0.010$) and shorter event durations (114.4 s vs. 135.2 s; $p = 0.025$). However, they exhibited longer elimination durations during defecation (32.5 s vs. 20.9 s; $p = 0.003$), again suggesting prolonged posturing or straining. Additional features such as mean total daily visits and transition/covering durations across all visits showed significant differences: the Renal group visits were higher, and the transition/covering durations were briefer compared to the Non-Renal group.

As [Table 3](#) indicates, the model achieved 100% precision for the Renal group in the LOOCV set and 66.7% precision for the Renal group in the validation set. Here, precision represents the ratio of accurate Renal predictions (i.e., the predicted CKD cat was in the Renal group) over total Renal predictions.

Table 3.

Model results.

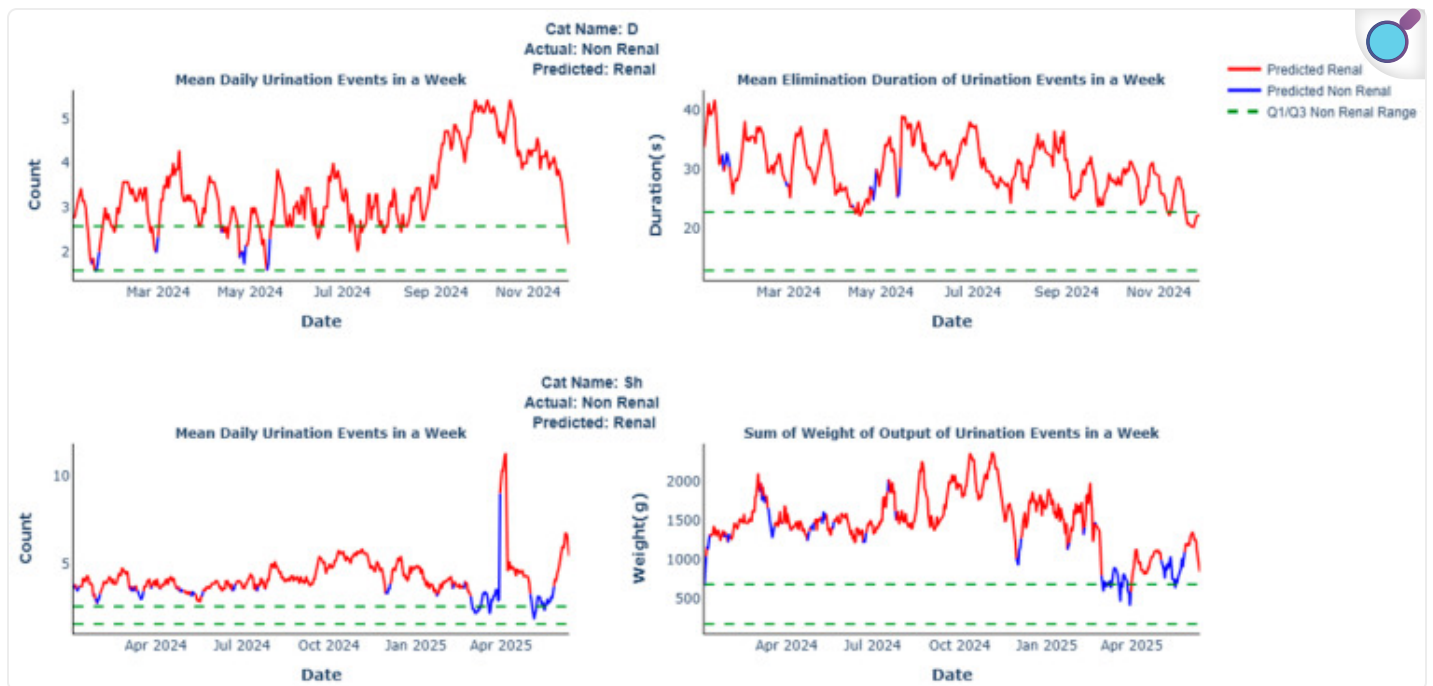
		Total	Predicted Non-Renal	Predicted Renal	Precision	Recall	F1-Score
LOOCV	Non-Renal	72	72	0	92.3%	100.0%	96.0%
	Renal	13	6	7	100.0%	53.8%	70.0%
	Weighted				93.5%	92.9%	92.0%
Validation	Non-Renal	44	42	2	93.3%	95.5%	94.4%
	Renal	7	3	4	66.7%	57.1%	61.5%
	Weighted				89.7%	90.2%	89.9%

[Open in a new tab](#)

In the LOOCV set, all cats predicted to have CKD were indeed in the Renal group. In the validation set, two of the cats who were predicted to have CKD were reported by their caregivers as having “no known conditions.” Furthermore, a 100% recall rate and 95.5% recall rate for the Non-Renal group in the LOOCV set and validation set respectively demonstrate that the majority of cats with no known conditions were accurately identified as such. Here, recall represents the ratio of true Non-Renal predictions (i.e., the cat predicted to have no known conditions was in the Non-Renal group) over the total number of cats in the Non-Renal group.

The misclassification analyses revealed that six of the cats in the Renal group who did not receive a CKD prediction had a behavioral profile that mirrored cats with no known conditions (IRIS stage 1 ($n = 1$), stage 2 ($n = 2$), stage 3 ($n = 2$), and stage 4 ($n = 1$)). In the validation set, the recall for cats with no known conditions was 95.5%, while the precision of Renal predictions was 66.7%. This lower precision was due to the two cats predicted as having CKD, despite their caregivers self-reporting “no known conditions.” Specifically, [Figure 5](#) shows that cat D exhibited high daily urination and prolonged voiding durations, while cat Sh had a high frequency of daily urinations and an increased weight of output as compared to the Non-Renal group ranges. The behavioral data for these two cats aligns with profiles unique to cats with CKD. It is unknown whether these cats were diagnosed with renal disease after the initial caregiver self-report.

Figure 5.

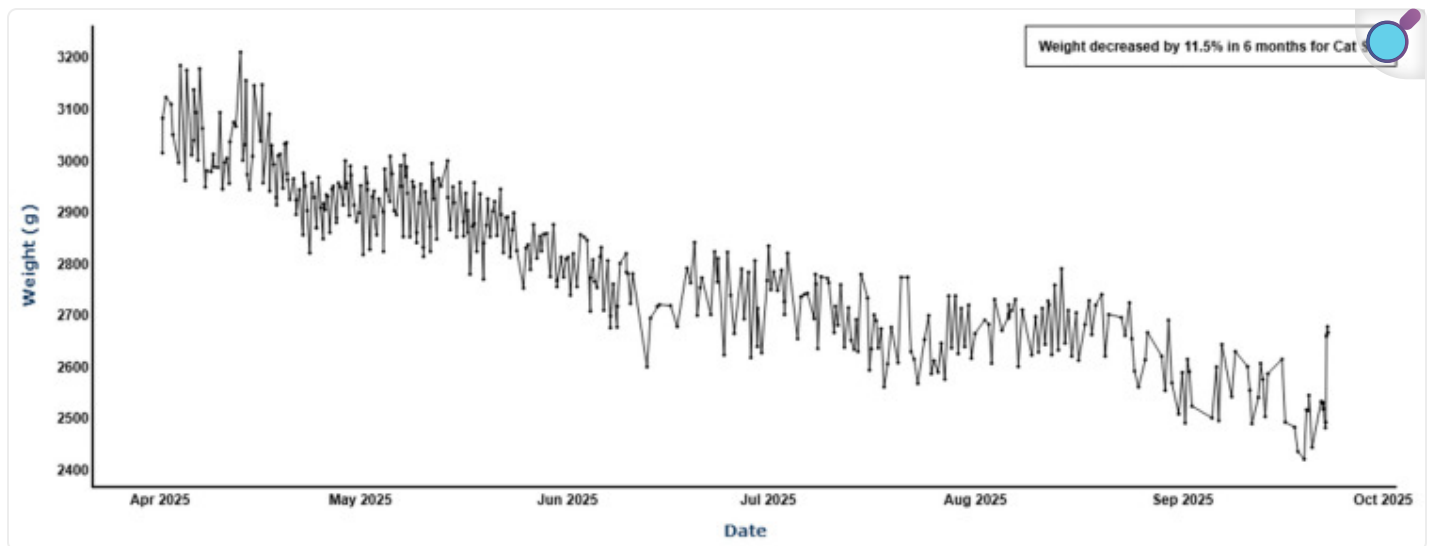


[Open in a new tab](#)

Renal misprediction examples.

Further, we provide an illustrative example of how this model can be combined with the weight tracking technology of the smart litter box monitor. One of the validation set cats Sp was correctly predicted as having CKD and also showed steady weight decline that was detected by the smart litter box monitor ([Figure 6](#)):

Figure 6.

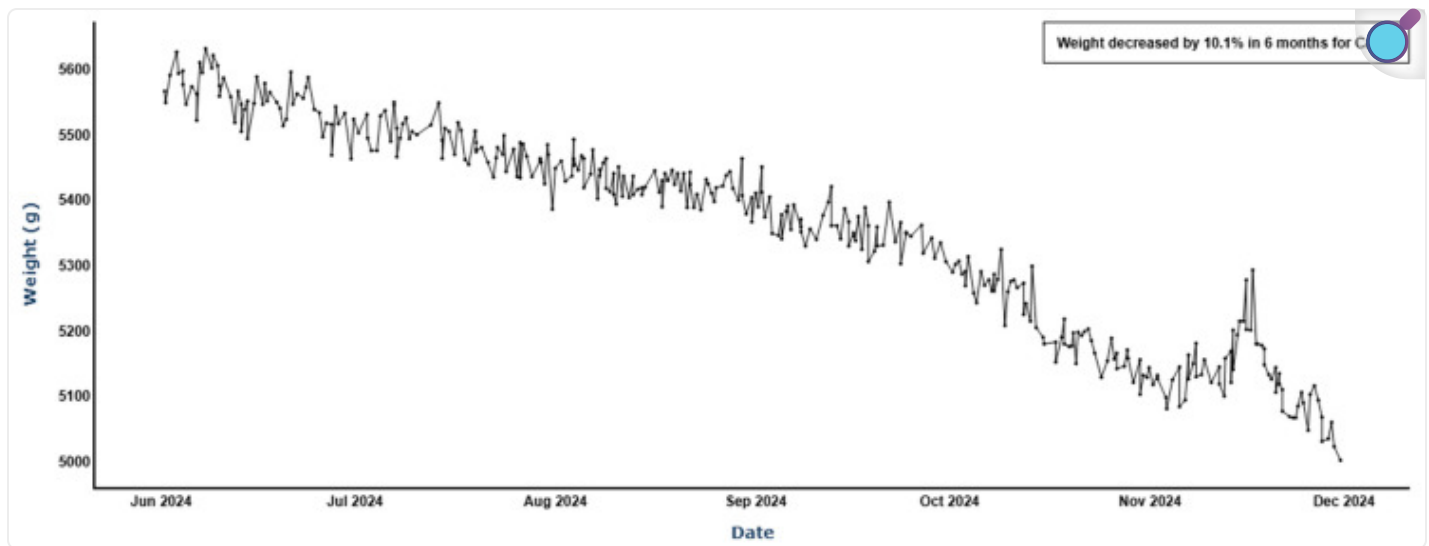


[Open in a new tab](#)

Weight loss trend for cat Sp (11.5% decrease over 6 months), * weight alert triggered.

Comparatively, cat Z in the validation group was mispredicted as not having CKD despite the caregiver reporting a veterinary CKD diagnosis. Regardless of an absence of the behavioral profile of a CKD cat, this cat had prominent weight loss trends that would have triggered a concerning weight loss alert by the smart litter box monitor. So, while Z may have been “asymptomatic” from the perspective of the machine learning model, a clue into his health state may be more evident by his weight trend as shown in [Figure 7](#):

Figure 7.



[Open in a new tab](#)

Weight loss trend for cat Z (10.1% decrease over 6 months), * weight alert triggered.

4. Discussion

Due to the model's optimized precision in predictions for cats with CKD and recall for predictions for cats with no known conditions, the results indicate a reliable, statistically linked behavioral profile for cats with CKD defined as the feature constellation of increased urination frequency, increased urinary output, longer elimination durations, and briefer, less intense post-elimination covering behavior. The model achieved weighted F1-scores of 92.7% (LOOCV) and 89.9% (validation), indicating robust predictive performance and minimization of false positives. The model performance was also consistent across cats representing different IRIS stages, suggesting that the results were not driven by a single disease stage. Key features like increased urination, reduced covering behaviors, and increased urine output were already observable in IRIS stages 1–2 as can be seen in [Figure 3](#) and [Figure 4](#). Overall, the results mirror findings from other studies and general knowledge concerning CKD.

Polyuria (excessive urination) is a common clinical sign preceding CKD diagnosis [14]. In particular, the specific finding of statistically relevant increased weight of urinary output corresponds to the observed symptom of increased urine production which can indicate onset of

renal issues [27]. Patterns of higher urination frequency were also found in another study [1]; interestingly, while the most important feature in our dataset was the increased mean daily urination events in a week, the previous study observed that cats with CKD had a higher mean number of urination events per day [1].

Our results also show that while the cats in the Renal group spent more time voiding, the total duration of their elimination events was shorter compared to the cats in the Non-Renal group. This relates to the briefer durations that the cats in the Renal group spent in the post-elimination phase of covering their waste. Further, while it was not statistically significant, the results indicated that the pre-elimination phase was also shorter among the cats in the Renal group. In other words, they also did not spend as much time sniffing or digging prior to eliminating compared to cats in the Non-Renal group. This observation aligns with general understanding of cats living with CKD, especially in later stages, where weight loss and frailty are common symptoms [28,29]. As such, less intense digging/covering behavior among cats in the Renal group is intuitively sound. Musculoskeletal diseases such as osteoarthritis are common comorbidities; however, since the cats in the Non-Renal group were age matched, the differences in covering behavior between the Renal and Non-Renal groups can be attributed to CKD.

Despite the overall success and focus on optimizing precision for CKD predictions to increase the confidence in positive CKD predictions, this design choice did come at the cost of an increased false-negative rate for CKD cats. It was noted that some cats in the Renal group for both the LOOCV and validation sets would not receive a CKD prediction due to a less pronounced behavioral profile. This finding highlights an area for improvement in the predictive model, suggesting that while the accuracy for identified cats with CKD is high, there are some cats with CKD that are not recognized by the model. It is possible that this may indicate they lack overt symptomology or that they are receiving treatment (e.g., medication, supplement, and/or therapeutic diet) whereby expected behaviors are masked. In other cases, treatments such as fluid therapy and prednisone may exacerbate behaviors such as increased urination frequency and output, and treatments for osteoarthritis may improve mobility and alter covering behaviors. Accordingly, model performance should be interpreted within the context of a conservative screening strategy that prioritizes confidence in positive predictions over a comprehensive detector of all renal cases. These findings are expected to generalize primarily to exclusively indoor cats that reliably utilize monitored litter boxes for elimination, where continuous passive data can be consistently captured.

The study was limited by both the small sample size, the limited range of ages of CKD cats, and by the general categorization of “CKD diagnosis” to define the Renal group instead of specific disease

stage groupings. A further limitation of this study was that we could not validate the precise health states of the in-home cats beyond confirming a general CKD diagnosis as reported by cat caregivers. However, this limitation does mimic the real-world scenario of the smart litter box monitor technology. Cats listed with “no known condition” may have underlying health issues such as early-stage CKD that are otherwise undetected. Due to the retrospective design constraints, we could not effectively establish a “healthy” control group or validate all health states of the cats with a veterinarian.

General consensus is that CKD is typically only detected very late when the kidney dysfunction is irreversible, and is likely an underdiagnosed health condition [14,30]. Lab diagnostics may not detect azotemia until <75% of renal mass deteriorates [17]. The smart litter box monitor technology, and specifically the machine learning model developed in this study may be a tool for cat caregivers to identify cats demonstrating behavioral profiles consistent with those of CKD cats. The machine learning model combined with the weight tracking component of the smart litter box monitor [18] offers a means for an at-home tool providing continuous data that cat caregivers can bring to their veterinarians to supplement the periodic opportunities to screen for renal health.

While the results of the machine learning model are promising, the limited sample size impacts population-level generalizability. Future work will investigate larger, more balanced cohorts with expanded age ranges for CKD cats across IRIS stages to determine the feasibility of detecting the unique stages and defining the profiles unique to each stage. Further, assessing longitudinal CKD data will help to better track, and potentially predict, the progression of CKD via elimination-related behavioral changes detected by the smart litter box monitor. Larger datasets would also enable us to split the stages and do more careful evaluations within and across CKD groups. Also, while weight trends are incredibly relevant as one of several variables for detecting CKD onset and progression, the machine learning model uses features based on the last 7 days to make a prediction for a single day. For this reason, weight was not a salient feature for the model, but as its own longer-term metric combined with the more precise flagging mechanism of the model, general weight trends from the smart litter box monitor can become more meaningful in the context of CKD detection. Perhaps a future model iteration can combine a behavioral feature set with weight trends. Future work should also investigate cats with other health conditions that may exhibit similar litter box behavior changes to CKD, such as diabetes mellitus, hyperthyroidism, mobility, or musculoskeletal diseases. Successful identification of other disease signatures using similar model development could also further validate this model.

Taken together, the technology of the smart litter box monitor holds promise for effective flagging of behaviors associated with health conditions like CKD while also offering a source for post-

diagnostic monitoring via continuous in-home data that leverages a baseline reference to alert caregivers of significant changes to a cat's typical behaviors. Caregivers of cats with CKD have reported being negatively impacted by necessary changes to daily routines or restrictions on their lives after receiving a diagnosis [31,32]. At-home, non-invasive, passive continuous data collection may also provide an ease of burden or peace of mind around disease management for caregivers.

5. Conclusions

CKD is a leading cause of mortality among aging cats rife with challenges associated with early detection and diagnosis due to non-specific symptoms [6,8,9,10]. Despite numerous methodologies employed for timely detection and management of the disease, there remains a critical need for improved detection techniques. This study's findings highlight the potential of innovative technologies, such as the smart litter box monitor's machine learning model, to facilitate the identification of behavioral changes indicative of chronic kidney disease. This device, combined with machine learning algorithms, demonstrates promise in providing passively collected longitudinal data to support caregiver and veterinarian decision-making alongside standard veterinary clinical diagnostics.

Moreover, while the model achieved high precision and recall rates, it also revealed limitations, particularly in identifying all CKD cases. This emphasizes the importance of continuous refinement of detection methodologies, including further exploration of specific IRIS stages and the integration of additional health metrics. Future research should focus on expanding datasets and enhancing the model's capabilities, ensuring that it can effectively support cat caregivers and veterinarians in the detection and management of CKD, thereby contributing to better health outcomes for feline patients.

Acknowledgments

We thank Tiger Analytics for their partnership and support in statistics and data science execution.

Supplementary Materials

The following supporting information can be downloaded at:

<https://www.mdpi.com/article/10.3390/ani16091319/s1> , Table S1: signalment data by group for training cats. Table S2: medication list for 13 training cats.

[animals-16-01319-s001.zip](#) (86.8KB, zip)

Author Contributions

Conceptualization, R.T.S.M. and N.L.-M.; methodology, N.L.-M.; software, N.L.-M.; validation, N.L.-M., H.G. and J.Q.; formal analysis, N.L.-M.; investigation, N.L.-M.; resources, R.T.S.M.; data curation, N.L.-M., L.S., O.A., S.T., N.S., Z.G. and J.Q.; writing—original draft preparation, L.S. and N.L.-M.; writing—review and editing, L.S., N.L.-M., H.G., R.T.S.M., O.A., S.T., N.S., Z.G. and J.Q.; visualization, N.L.-M.; supervision, R.T.S.M. and N.L.-M.; project administration, R.T.S.M. and N.L.-M.; funding acquisition, R.T.S.M. and N.L.-M. All authors have read and agreed to the published version of the manuscript.

Institutional Review Board Statement

The data from this retrospective study had been collected under Institutional Animal Care and Use (IACUC) approval or informed consent of data usage. The Nestlé Purina IACUC and Ohio State University IACUC numbers were NT9913 and IACUC-125 2023A00000032, respectively.

Informed Consent Statement

Informed consent was obtained from all subjects involved in the study.

Data Availability Statement

The datasets presented in this article are not readily available due to the proprietary nature of the research and technology development. Requests to access the datasets should be directed to natalie.langenfeld-mccoy@rd.nestle.com.

Conflicts of Interest

Authors N. Langenfeld-McCoy, L. Snow, H. Gordon, O. Arndt, N. Schoeneck, S. Thomas, and R.T.S. McGowan are employed by the company Nestle Purina Research. Author J. Quimby serves on an advisory board for Nestle Purina. All the authors declare that the research was conducted in the

absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Funding Statement

This research received no external funding.

Footnotes

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.

References

1. George Z.M., Quimby J.M., Jones S., Brusach K.B., Rudinsky A.J. Quantification of defecation frequency in cats with and without chronic kidney disease. *J. Feline Med. Surg.* 2025;27:1098612X251348011. doi: 10.1177/1098612X251348011. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
2. McLeland S., Quimby J., Lappin M.R. Alpha-enolase staining patterns in the renal tissues of cats with and without chronic kidney disease. *Veter. Immunol. Immunopathol.* 2019;212:23–26. doi: 10.1016/j.vetimm.2019.04.009. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
3. Sparkes A.H., Caney S., Chalhoub S., Elliott J., Finch N., Gajankayake I., Langston C., Lefebvre H.P., White J., Quimby J. ISFM consensus guidelines on the diagnosis and management of feline chronic kidney disease. *J. Feline Med. Surg.* 2016;18:219–239. doi: 10.1177/1098612X16631234. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
4. Chakrabarti S., Syme H.M., Elliott J. Clinicopathological variables predicting progression of azotemia in cats with chronic kidney disease. *J. Veter. Intern. Med.* 2012;26:275–281. doi: 10.1111/j.1939-1676.2011.00874.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
5. Kovarikova S. Urinary biomarkers of renal function in dogs and cats: A review. *Veter. Med.* 2015;60:589–602. doi: 10.17221/8527-VETMED. [[DOI](#)] [[Google Scholar](#)]

6. Kongtasai T, Paepe D, Meyer E, Mortier F, Marynissen S, Stammeleer L, Defauw P, Daminet S. Renal biomarkers in cats: A review of the current status in chronic kidney disease. *J. Veter. Intern. Med.* 2022;36:379–396. doi: 10.1111/jvim.16377. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
7. Finch N.C., Syme H.M., Elliott J. Risk factors for development of chronic kidney disease in cats. *J. Veter. Intern. Med.* 2016;30:602–610. doi: 10.1111/jvim.13917. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
8. Vanden Broecke E, Van Mulders L, De Paepe E, Paepe D, Daminet S, Vanhaecke L. Early detection of feline chronic kidney disease via 3-hydroxykynurenine and machine learning. *Sci. Rep.* 2025;15:6875. doi: 10.1038/s41598-025-90019-x. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
9. Marino C.L., Lascelles B.D.X., Vaden S.L., Gruen M.E., Marks S.L. Prevalence and classification of chronic kidney disease in cats randomly selected from four age groups and in cats recruited for degenerative joint disease studies. *J. Feline Med. Surg.* 2014;16:465–472. doi: 10.1177/1098612X13511446. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
10. Nassar G., Mahmoud A.R., Mohamed S., El-Sheikh A.K., Bayoumi Y. Chronic kidney disease in cats. *Egypt. J. Veter. Sci.* 2025;56:109–119. doi: 10.21608/ejvs.2025.366943.2688. [[DOI](#)] [[Google Scholar](#)]
11. Cobrin A.R., Blois S.L., Kruth S.A., Abrams-Ogg A.C.G., Dewey C. Biomarkers in the assessment of acute and chronic kidney diseases in the dog and cat. *J. Small Anim. Pract.* 2013;54:647–655. doi: 10.1111/jsap.12150. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
12. Li Q., Cominetti O., Holzwarth J.A., Summers S., Wang X., Dayon L. Integrated multi-omics analysis of renal metabolism in domestic cats with spontaneous chronic kidney disease. *Commun. Biol.* 2025;8:1794. doi: 10.1038/s42003-025-09164-8. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
13. Paepe D., Daminet S. Feline CKD: Diagnosis, staging and screening—What is recommended? *J. Feline Med. Surg.* 2013;15:15–27. doi: 10.1177/1098612X13495235. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
14. Conroy M., Brodbelt D.C., Neill D., Chang Y.-M., Elliott J. Chronic kidney disease in cats attending primary care practice in the UK: A VetCompass™ study. *Veter. Rec.* 2019;184:526. doi: 10.1136/vr.105100. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

15. Bradley R., Tagkopoulos I., Kim M., Kokkinos Y., Panagiotakos T., Kennedy J., De Meyer G., Watson P., Elliott J. Predicting early risk of chronic kidney disease in cats using routine clinical laboratory tests and machine learning. *J. Veer. Intern. Med.* 2019;33:2644–2656. doi: 10.1111/jvim.15623. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
16. Nickel M.R., Sweet H.M., Lee A., Bohaychuk-Preuss K., Varnhagen C., Olson M. A saliva urea test strip for use in feline and canine patients: A pilot study. *J. Veter. Diagn. Investig.* 2022;34:496–503. doi: 10.1177/10406387221086917. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
17. Jepson R.E., Brodbelt D., Vallance C., Syme H.M., Elliott J. Evaluation of predictors of the development of azotemia in cats. *J. Veter. Intern. Med.* 2009;23:806–813. doi: 10.1111/j.1939-1676.2009.0339.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
18. Snow L., Langenfeld-McCoy N., Dussan H., Arndt O., Schoeneck N., Thomas S., McGowan R.T.S. Enhancing cat care: Unveiling the technology of intelligent litter box monitoring. *Appl. Anim. Behav. Sci.* 2025;292:106811. doi: 10.1016/j.applanim.2025.106811. [[DOI](#)] [[Google Scholar](#)]
19. R Core Team . R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing; Vienna, Austria: 2023. [(accessed on 16 April 2026)]. Available online: <https://www.R-project.org/> [[Google Scholar](#)]
20. Brooks M.E., Kristensen K., van Benthem K.J., Magnusson A., Berg C.W., Nielsen A., Højsgaard S. glmmTMB balances speed and flexibility among packages for zero-inflated generalized linear mixed modeling. *R J.* 2017;9:378–400. doi: 10.32614/RJ-2017-066. [[DOI](#)] [[Google Scholar](#)]
21. Ellis J.J., McGowan R.T.S., Martin F. Does previous use affect litter box appeal in multi-cat households? *Behav. Process.* 2017;141:284–290. doi: 10.1016/j.beproc.2017.02.008. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
22. McGowan R.T.S., Ellis J.J., Bensky M.K., Martin F. The ins and outs of the litter box: A detailed ethogram of cat elimination behavior in two contrasting environments. *Appl. Anim. Behav. Sci.* 2017;194:67–78. doi: 10.1016/j.applanim.2017.05.009. [[DOI](#)] [[Google Scholar](#)]
23. Python Software Foundation Python 3.12. 2023. [(accessed on 21 April 2026)]. Available online: <https://www.python.org/downloads/release/python-3120/> [[Google Scholar](#)]

24. Oliphant T.E. A Guide to NumPy. CreateSpace Independent Publishing Platform; Scotts Valley, CA, USA: 2015. [[Google Scholar](#)]
25. Pedregosa F., Varoquaux G., Gramfort A., Michel V., Thirion B., Grisel O., Duchesnay É. Scikit-learn: Machine learning in Python. J. Mach. Learn. Res. 2011;12:2825–2830. [[Google Scholar](#)]
26. Dorogush A.V., Ershov V., Gulin A. CatBoost: Gradient boosting with categorical features support. arXiv. 2018 doi: 10.48550/arXiv.1810.11363.1810.11363 [[DOI](#)] [[Google Scholar](#)]
27. Grauer G.F. Early Detection of renal damage and disease in dogs and cats. Veter. Clin. N. Am. Small Anim. Pract. 2005;35:581–596. doi: 10.1016/j.cvsm.2004.12.013. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
28. Greene J.P., Lefebvre S.L., Wang M., Yang M., Lund E.M., Polzin D.J. Risk factors associated with the development of chronic kidney disease in cats evaluated at primary care veterinary hospitals. J. Am. Veter. Med. Assoc. 2014;244:320–327. doi: 10.2460/javma.244.3.320. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
29. Pye C.R., Dowgray N.J., Eyre K., Pinchbeck G., Biourge V., Moniot D., Comerford E., German A.J. Longitudinal changes in bodyweight, body condition, and muscle condition in ageing pet cats: Findings from the Cat Prospective Ageing and Welfare Study. Front. Veter. Sci. 2025;12:1654002. doi: 10.3389/fvets.2025.1654002. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
30. Chen H., Dunaevich A., Apfelbaum N., Kuzi S., Mazaki-Tovi M., Aroch I., Segev G. Acute on chronic kidney disease in cats: Etiology, clinical and clinicopathologic findings, prognostic markers, and outcome. J. Veter. Intern. Med. 2020;34:1496–1506. doi: 10.1111/jvim.15808. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
31. Elliott J., Reyes-Hughes H., Hibbert A., Blackwell E., Finch N.C. Owners' experiences of caring for cats with chronic kidney disease in the UK. J. Feline Med. Surg. 2025;27:1098612X251314769. doi: 10.1177/1098612X251314769. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
32. Boyd L.M., Langston C., Thompson K., Zivin K., Imanishi M. Survival in Cats with Naturally Occurring Chronic Kidney Disease (2000–2002) J. Vet. Intern. Med. 2008;22:1111–1117. doi: 10.1111/j.1939-1676.2008.0163.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

[animals-16-01319-s001.zip](#) (86.8KB, zip)

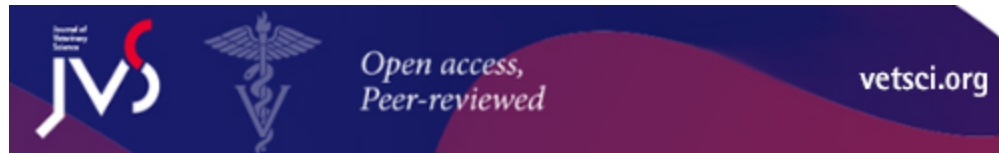
Data Availability Statement

The datasets presented in this article are not readily available due to the proprietary nature of the research and technology development. Requests to access the datasets should be directed to natalie.langenfeld-mccoy@rd.nestle.com.

Articles from Animals : an Open Access Journal from MDPI are provided here courtesy of
Multidisciplinary Digital Publishing Institute (MDPI)

As a library, NLM provides access to scientific literature. Inclusion in an NLM database does not imply endorsement of, or agreement with, the contents by NLM or the National Institutes of Health.

Learn more: [PMC Disclaimer](#) | [PMC Copyright Notice](#)



J Vet Sci. 2026 Mar 26;27(2):e30. doi: [10.4142/jvs.25273](https://doi.org/10.4142/jvs.25273) [↗](#)

Association between chronic kidney disease progression and serum feline pancreatic lipase concentrations in cats

[Jihyun Kim](#)^{1,2}, [Youngmin Yun](#)^{1,✉}

[Author information](#) [Article notes](#) [Copyright and License information](#)

PMCID: PMC13062426 PMID: [41947688](#)

Abstract

Importance

Serum feline pancreatic lipase (fPL) is used to diagnose pancreatitis, but interpretation is challenging because fPL may be affected by non-pancreatic disease, including reduced renal function.

Objective

To evaluate the association between chronic kidney disease (CKD) stage and serum fPL in cats and characterize how renal biomarkers relate to fPL.

Methods

This retrospective study evaluated medical records of cats presented to a secondary referral veterinary hospital between February 2024 and March 2025. Inclusion required same-day testing of complete blood count, serum biochemistry, blood gas analysis, urinalysis, serum fPL, and abdominal ultrasonography. CKD staging followed International Renal Interest Society guidelines using serum creatinine, urine specific gravity (USG), and ultrasonography. Nonparametric group comparisons were performed using the Kruskal–Wallis test, and associations between fPL and renal biomarkers were assessed using Spearman’s correlation.

Results

fPL increased significantly with advancing CKD stage ($p < 0.001$). Furthermore, fPL was positively correlated with blood urea nitrogen ($\rho = 0.516, p < 0.001$), creatinine ($\rho = 0.459, p < 0.001$), and inorganic phosphorus ($\rho = 0.312, p < 0.001$). Additionally, fPL and USG exhibited an inverse correlation ($\rho = -0.502, p < 0.001$). Ionized calcium showed no correlation with fPL ($\rho = 0.012, p = 0.898$).

Conclusions and Relevance

In this clinical cohort, higher CKD stage was associated with higher serum fPL. Because fPL alone is insufficient to diagnose pancreatitis and may increase with reduced renal function, mild-to-moderate fPL elevations in cats with CKD should be interpreted in the context of clinical signs and pancreatic imaging findings.

Keywords: Feline medicine, renal insufficiency, azotemia, pancreatitis, lipase

INTRODUCTION

Chronic kidney disease (CKD) is one of the most prevalent conditions affecting geriatric cats. Its reported prevalence rates in cats older than 10 years exceed 30%–40% [[1,2,3,4,5](#)]. Common historical and clinical symptoms of CKD include weight loss, polyuria, polydipsia, and persistently low urine specific gravity (USG; < 1.035) [[6,7,8,9](#)]. According to the consensus guidelines of the International Society of Feline Medicine, a clinical diagnosis of feline CKD is established based on the presence of azotemia in conjunction with inadequate concentrated urine [[10](#)].

Feline pancreatitis, once considered an uncommon clinical entity, is now recognized as a frequently occurring condition with a prevalence rate comparable to that observed in humans and

dogs. A 2007 histopathological study involving 115 feline necropsy cases demonstrated evidence of pancreatic inflammation in 67% of the participants; chronic pancreatitis, acute pancreatitis (AP), and acute-on-chronic pancreatitis accounted for 50.4%, 6.1%, and 9.6%, respectively. Furthermore, 45% of clinically healthy cats had histopathological signs of pancreatitis [11]. Serum feline pancreatic lipase (fPL) concentration has since emerged as a sensitive and specific biomarker for the non-invasive diagnosis of feline pancreatitis [12,13,14,15].

In human medicine, a substantial body of evidence indicates that the pancreas is adversely affected by uremia [16,17,18]. A strong association has been documented between end-stage renal disease (ESRD) and AP [19,20]. A nationwide cohort study conducted in Germany that involved more than 70,000 patients with ESRD undergoing hemodialysis reported an annual incidence of AP at 67 cases per 100,000 individuals. This was markedly higher than the 19.7 cases per 100,000 observed in the general population [21]. Similarly, a prospective cohort study involving 2,603 ESRD patients in Taiwan found that the incidence of AP was more than threefold more prevalent among patients with ESRD than in the general population. Several pathophysiological mechanisms have been proposed to explain this relationship. One hypothesis suggests that calcium in the peritoneal dialysate may diffuse through the peritoneum, leading to localized hypercalcemia within the pancreas and further exacerbation of pancreatic dysfunction [22]. Additionally, increased concentrations of gastrointestinal hormones such as cholecystokinin, gastric inhibitory polypeptide, and glucagon that occur in renal insufficiency may stimulate pancreatic enzyme hypersecretion and contribute to pancreatic injury [23].

Despite the recognized association between renal dysfunction and pancreatitis in humans, the relationship between CKD and pancreatitis in cats remains difficult to define. Importantly, diagnosis of feline pancreatitis is challenging and typically relies on integration of clinical signs, laboratory abnormalities, and imaging findings rather than a single biomarker. In addition, pancreatic lipase is cleared in part by the kidneys; therefore, reduced glomerular filtration rate may contribute to higher measured lipase concentrations even in the absence of primary pancreatic inflammation.

Therefore, this study aimed to evaluate how serum fPL concentrations vary across CKD stages and how fPL relates to commonly measured renal biomarkers in cats. We hypothesized that fPL concentrations would show an association with markers of reduced renal function, emphasizing the need for cautious interpretation of fPL in cats with CKD.

METHODS

Ethical approval

This retrospective study involved a review of existing clinical data and did not require experimental interventions; therefore, ethical approval was waived. Written informed consent was obtained from all cat owners prior to diagnostic evaluation and use of clinical data. All clinical procedures were performed under the supervision of licensed veterinarians in accordance with institutional and national guidelines. Patient data were anonymized before analysis to ensure confidentiality.

Study design

This retrospective, observational study examined the electronic medical records of cats presented to the secondary referral veterinary hospital between February 2024 and March 2025.

Inclusion criteria

Cats were eligible for inclusion if they underwent a comprehensive diagnostic evaluation comprising complete blood count (CBC), blood gas analysis, serum biochemistry, measurement of serum fPL concentration, urinalysis, and abdominal ultrasonography on the same day as their clinical examination.

CBC was performed using an automated hematology analyzer (ProCyte Dx; IDEXX Laboratories, USA). Serum biochemical analysis was conducted using an automated dry chemistry analyzer (FUJI DRY-CHEM NX700; Fujifilm, Japan). Urinalysis, including USG measurement, was performed using a refractometer on urine samples obtained by cystocentesis at the time of initial diagnostic evaluation during abdominal ultrasonography, prior to therapeutic interventions. Abdominal ultrasonographic examinations were performed using an ultrasound system (ProSound α 7; Hitachi Aloka Medical, Japan) equipped with a 12-MHz linear transducer. Examinations were conducted as part of routine clinical care by a board-certified veterinary radiologist and typically included evaluation of multiple abdominal organs, including the kidneys and pancreas. However, pancreatic ultrasonographic findings and standardized clinical signs compatible with pancreatitis were not consistently recorded across cases and therefore, were not used to define pancreatitis for the purposes of this analysis. A total of 169 cats met the initial inclusion criteria during the study period.

Exclusion criteria

Fifty-seven cats were excluded based on the following criteria: acute kidney injury secondary to nephrotoxic agents such as non-steroidal anti-inflammatory drugs; infectious etiologies including pyelonephritis, neoplastic diseases, or urinary tract obstruction; administration of immunosuppressive therapy (e.g., prednisolone) owing to potential effects on fPL concentrations and USG; and use of diuretics for the management of hypertrophic cardiomyopathy. A total of 112 cats were included in the final analysis after applying these criteria.

Clinical assessment

The 112 cats were categorized into five groups based on serum creatinine concentration, USG, and renal ultrasonographic findings. The normal group had serum creatinine concentrations of < 1.6 mg/dL, USG of ≥ 1.035 , and no detectable abnormalities on renal ultrasonography. CKD staging was performed according to the 2019 International Renal Interest Society (IRIS) guidelines: stage 1 was characterized by serum creatinine concentrations of < 1.6 mg/dL and USG of < 1.035 ; stage 2 by serum creatinine concentrations between 1.6 and 2.8 mg/dL; stage 3 by serum creatinine concentrations between 2.9 and 5.0 mg/dL; and stage 4 by serum creatinine concentrations of > 5.0 mg/dL. This operational definition was used to identify early CKD (Stage 1) characterized by inadequate urine concentrating ability without azotemia and represents a study-specific modification of the IRIS staging system, in which USG is used for CKD diagnosis rather than formal staging [10]. Renal ultrasonographic abnormalities considered consistent with CKD included increased cortical echogenicity, loss of corticomedullary distinction, renal contour irregularities, and reduced renal size [24].

Serum fPL concentrations were measured using the Vcheck fPL (Bionote) quantitative immunoassay, which is a fluorescence immunochromatographic assay designed for fPL detection. The assay provides quantitative results within 15 min and has a dynamic measurement range of 1–30 ng/mL. According to the manufacturer's recommendations, values of ≤ 3.5 , 3.6–5.3, and ≥ 5.4 ng/mL were categorized as normal, equivocal, and above the assay cutoff, respectively.

Statistical analysis

Comparisons of serum fPL concentrations and USG across CKD stages were performed using the Kruskal–Wallis test due to the non-parametric distribution of the data. Associations between fPL concentrations and renal biomarkers, including blood urea nitrogen (BUN), creatinine, inorganic phosphorus (IP), ionized calcium (iCa), and USG, were assessed using Spearman's rank correlation coefficients. All p values were two-tailed, and p values of < 0.05 were considered statistically

significant. Statistical analyses were conducted using IBM SPSS Statistics, version 22 (IBM, USA), with the nonparametric tests and correlation modules applied.

RESULTS

Patient characteristics

A total of 112 cats were included in this study and categorized according to their CKD stage based on IRIS guidelines. The distribution of CKD stages was as follows: normal: 42 cats (37.5%); stage 1, 13 cats (11.6%); stage 2, 34 cats (30.4%); stage 3, 9 cats (8.0%); and stage 4, 14 cats (12.5%).

Regarding sex distribution, neutered males were predominant (n = 61, 54.5%), followed by spayed females (n = 45, 40.2%). Intact males and females comprised 2.7% and 2.7% of the population, respectively (n = 3 for both).

Domestic Shorthair cats were the most common (n = 53, 47.3%), followed by Persian (n = 15, 13.4%), Russian Blue (n = 8, 7.1%), and Siamese (n = 7, 6.3%) cats. Other breeds included Scottish Fold, Abyssinian, Turkish Angora, Munchkin, Ragdoll, Bengal, British Shorthair, Norwegian Forest, Maine Coon, and American Shorthair cats, each representing a smaller proportion of the population ([Table 1](#)).

Table 1. Demographic characteristics of 112 cats stratified by CKD stage based on the IRIS guidelines.

CKD IRIS stage	Normal	1	2	3	4	Total (%)
Number of cats (%)	42 (37.5)	13 (11.6)	34 (30.4)	9 (8.0)	14 (12.5)	112 (100.0)
Sex						
Intact male	1	0	0	1	1	3 (2.7)
Neutered male	17	9	20	6	9	61 (54.5)
Intact female	1	0	0	2	0	3 (2.7)
Spayed female	23	4	14	0	4	45 (40.2)
Breed						
Domestic Shorthair	19	7	18	3	6	53 (47.3)
Persian	4	1	8	1	1	15 (13.4)
Russian Blue	1	1	2	2	2	8 (7.1)
Siamese	1	0	3	1	2	7 (6.3)
Scottish Fold	3	1	1	0	1	6 (5.4)
Abyssinian	2	1	0	1	0	4 (3.6)
Turkish Angora	1	1	1	0	1	4 (3.6)
Munchkin	4	0	0	0	0	4 (3.6)
Ragdoll	2	0	1	0	0	3 (2.7)
Bengal	2	0	0	1	0	3 (2.7)
British Shorthair	2	0	0	0	0	2 (1.8)
Norwegian Forest	0	1	0	0	0	1 (0.9)

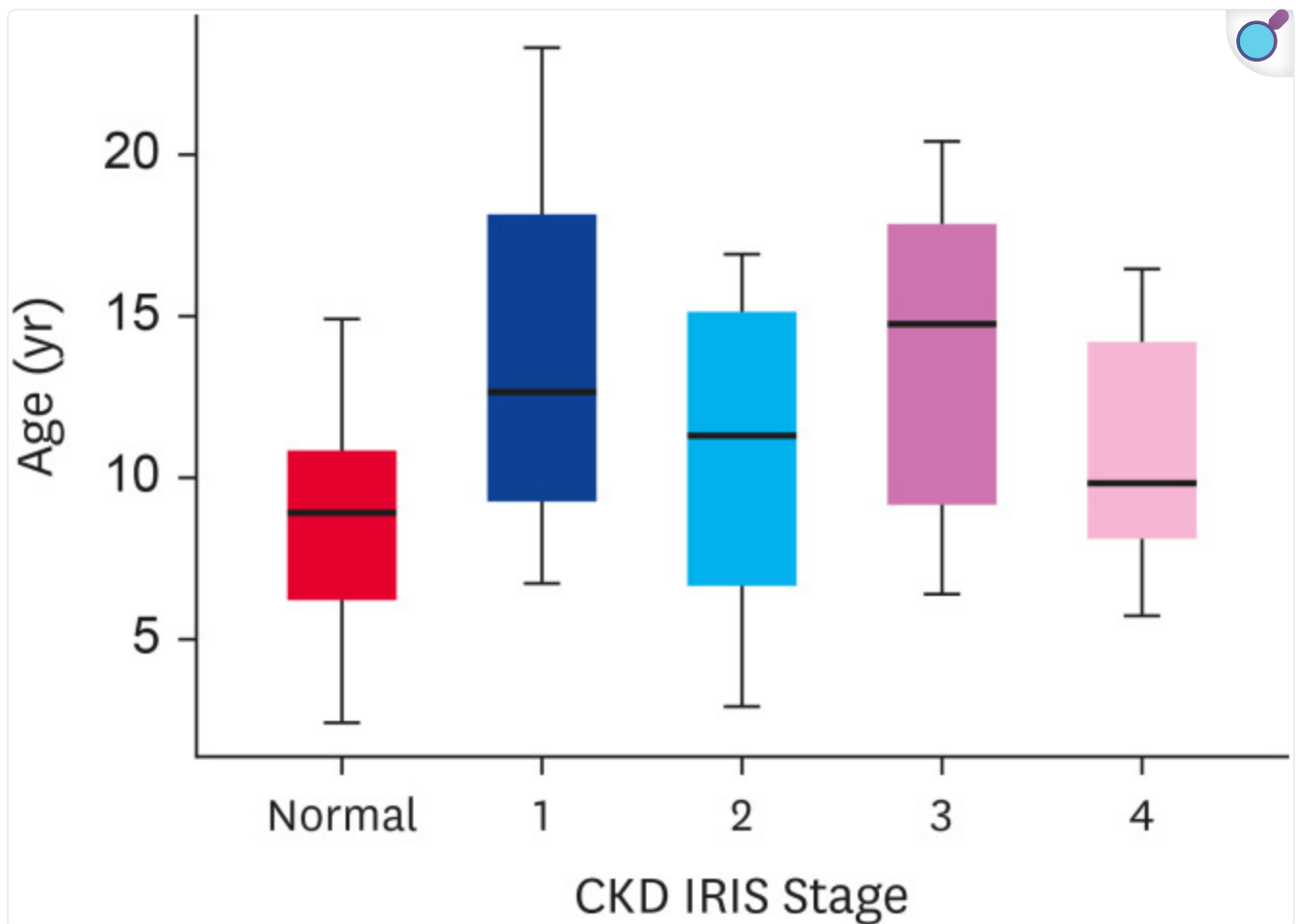
CKD IRIS stage	Normal	1	2	3	4	Total (%)
Maine Coon	1	0	0	0	0	1 (0.9)
American Shorthair	0	0	0	0	1	1 (0.9)
Age						
Median (min–max)	9.0 (2.0–15.1)	13.0 (6.4–23.7)	11.6 (2.6–17.1)	14.8 (5.7–21.7)	10.0 (5.4–18.4)	10.2 (2.0–23.7)

[Open in a new tab](#)

CKD, chronic kidney disease; IRIS, International Renal Interest Society.

The median age of the overall population was 10.2 years (range, 2.0–23.7 years). Median age tended to be higher in cats with CKD, although this trend was not strictly progressive with increasing stage: 9.0 years (2.0–15.1 years) for the normal group, 13.0 years (6.4–23.7 years) for stage 1, 11.6 years (2.6–17.1 years) for stage 2, 14.8 years (5.7–21.7 years) for stage 3, and 10.0 years (5.4–18.4 years) for stage 4 ([Fig. 1](#)).

Fig. 1. Age distribution according to CKD IRIS stage among 112 cats. Box-and-whisker plot showing the distribution of age across different CKD IRIS stages in cats. Each box represents the IQR, the line within each box indicates the median, and the whiskers extend to 1.5 times the IQR. Median age tends to be higher for cats with CKD, but this trend was not strictly progressive with increasing stage: 9.0 years (2.0–15.1 years) for the normal group; 13.0 years (6.4–23.7 years) for stage 1; 11.6 years (2.6–17.1 years) for stage 2; 14.8 years (5.7–21.7 years) for stage 3; and 10.0 years (5.4–18.4 years) for stage 4.



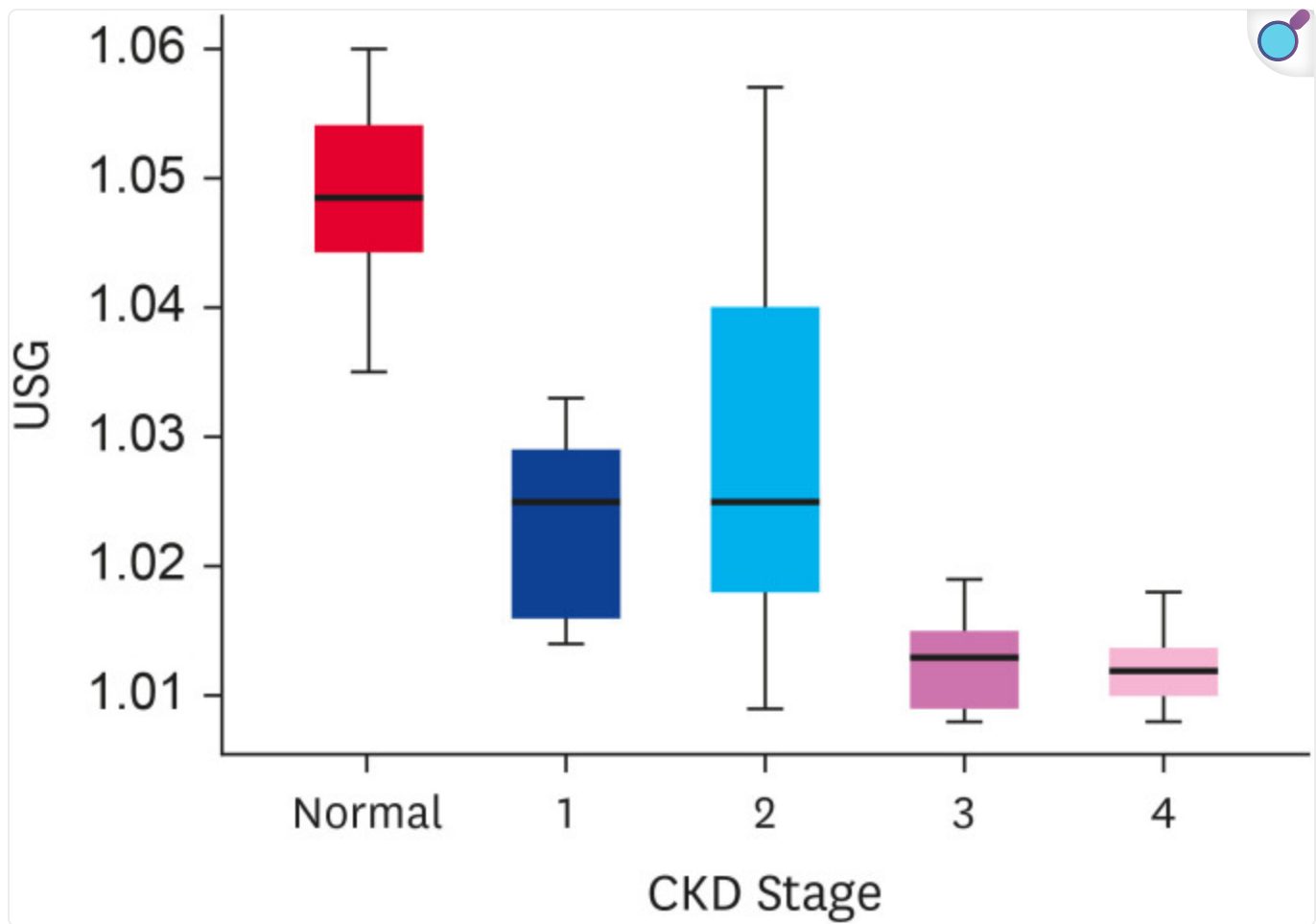
[Open in a new tab](#)

CKD, chronic kidney disease; IRIS, International Renal Interest Society; IQR, interquartile range.

USG was analyzed across CKD IRIS stages for 112 cats. For the normal group, USG values were appropriately high with a median of 1.049 (interquartile range [IQR], 1.044–1.054) and a range of 1.035–1.060, reflecting normal renal concentrating ability.

The median USG values for the CKD stages were as follows: stage 1: 1.025 (IQR, 1.016–1.029; 95% confidence interval [CI], 1.019–1.028); stage 2: 1.025 (IQR, 1.018–1.040; 95% CI, 1.023–1.033); stage 3: 1.013 (IQR, 1.009–1.015; 95% CI, 1.010–1.016); and stage 4: 1.012 (IQR, 1.010–1.014; 95% CI, 1.011–1.014) ([Fig. 2](#)).

Fig. 2. Distribution of USG across CKD stages in cats. Box-and-whisker plot illustrating the distribution of USG values among cats at different CKD IRIS stages.



[Open in a new tab](#)

USG, urine specific gravity; CKD, chronic kidney disease; IRIS, International Renal Interest Society.

A significant decline in USG was observed with increasing CKD severity, although the trend was not strictly progressive. The USG values for cats with stage 1 disease showed a modest decrease relative to those for normal cats, while those with stage 2 disease had a wider distribution with some overlap. In contrast, marked reductions in USG were evident for stages 3 and 4, where values clustered tightly around 1.009–1.015, indicating impaired renal concentrating ability. A Kruskal–Wallis H test revealed that the differences in USG values across CKD stages were statistically significant ($H = 80.53, p < 0.001$).

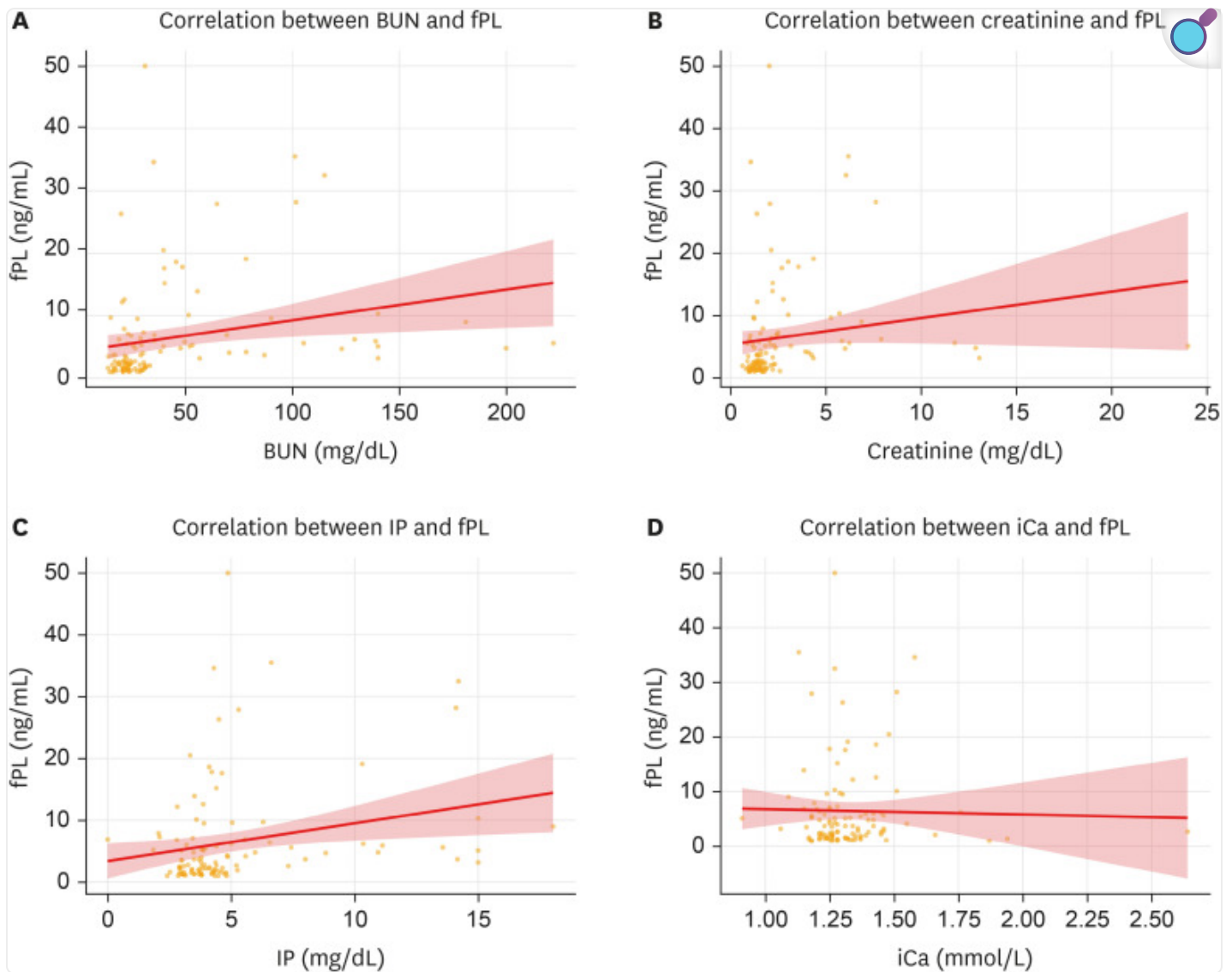
Post-hoc pairwise comparisons revealed that USG values in CKD stages 1 to 4 were significantly lower than those in the normal group (all $p < 0.001$). USG was also significantly lower for stages 3 and 4 than for stages 1 and 2 (all $p < 0.001$). No significant difference in USG was observed for CKD stages 3 and 4 ($p = 0.944$).

Relationship between renal biomarkers and fPL concentration

BUN

Serum fPL concentration showed a moderate positive association with BUN (Spearman's $\rho = 0.516$, $p < 0.001$), indicating that cats with higher BUN tended to have higher fPL concentrations ([Fig. 3](#)).

Fig. 3. Scatter plots depicting correlations between fPL concentrations and renal biomarkers in 112 cats: (A) BUN, (B) Creatinine, (C) IP, and (D) iCa.



[Open in a new tab](#)

Each point represents an individual cat. The red line indicates the line of best fit obtained from simple linear regression, and the shaded area represents the 95% confidence interval of the regression line. Positive correlations are observed between fPL and BUN, Creatinine, and IP concentrations. No significant correlation is found between iCa and fPL concentrations.

fPL, feline pancreatic lipase; BUN, blood urea nitrogen; IP, inorganic phosphorus; iCa, ionized calcium.

Creatinine

Serum fPL concentration showed a moderate positive association with serum creatinine (Spearman's $\rho = 0.459$, $p < 0.001$), indicating that cats with higher creatinine tended to have higher fPL concentrations.

IP

Serum fPL concentration showed a weak-to-moderate positive association with IP (Spearman's $\rho = 0.312$, $p < 0.001$).

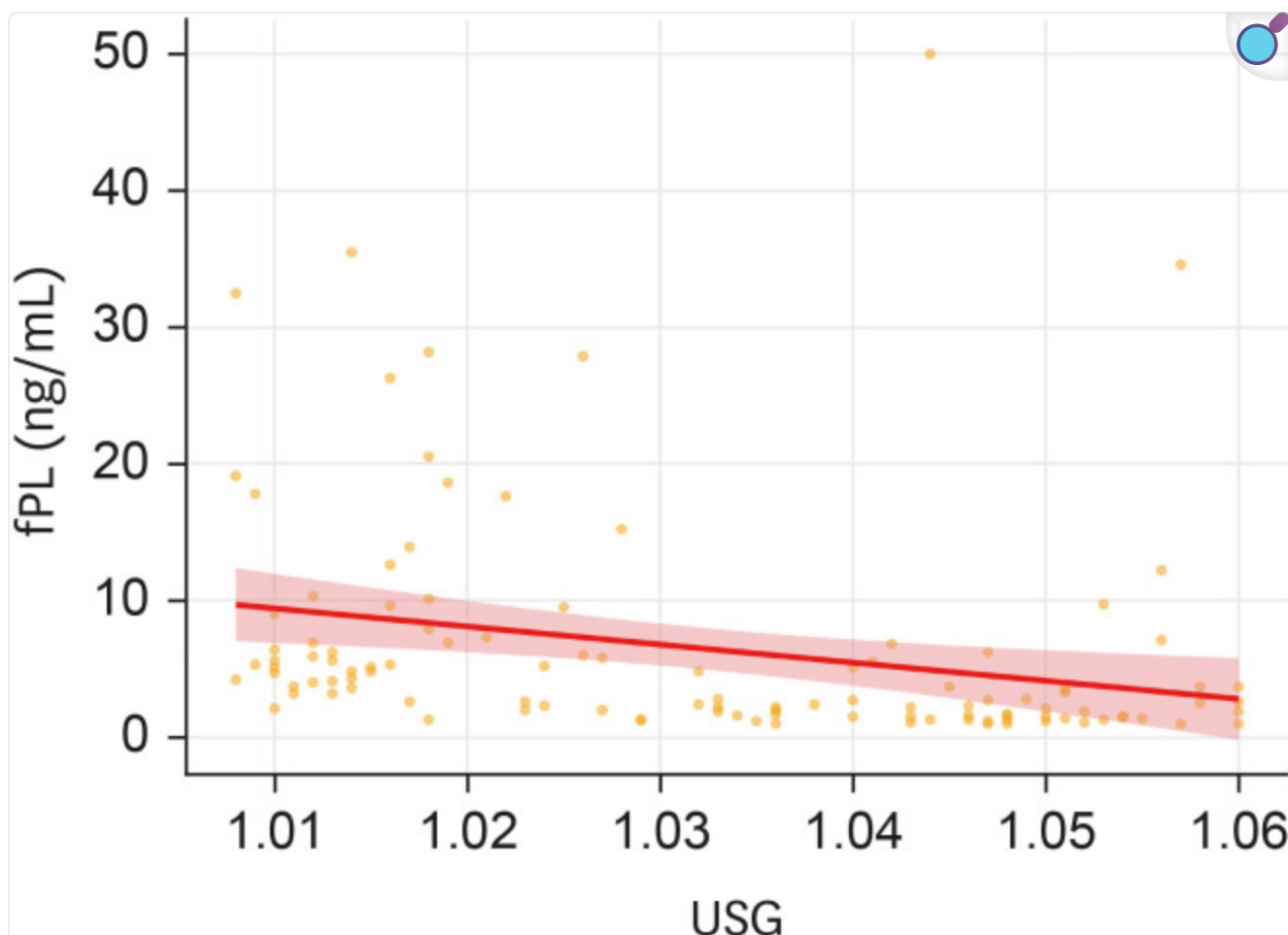
iCa

No meaningful association was identified between serum iCa and fPL concentration (Spearman's $\rho = 0.012$, $p = 0.898$).

USG

Serum fPL concentration showed a moderate negative association with USG (Spearman's $\rho = -0.502$, $p < 0.001$), indicating that cats with more dilute urine tended to have higher fPL concentrations ([Fig. 4](#)).

Fig. 4. Scatter plot depicting the correlation between USG and fPL concentrations in cats. A significant negative correlation is observed between USG and fPL concentrations, suggesting that lower USG values, indicative of impaired renal concentrating ability, are associated with higher fPL concentrations.



[Open in a new tab](#)

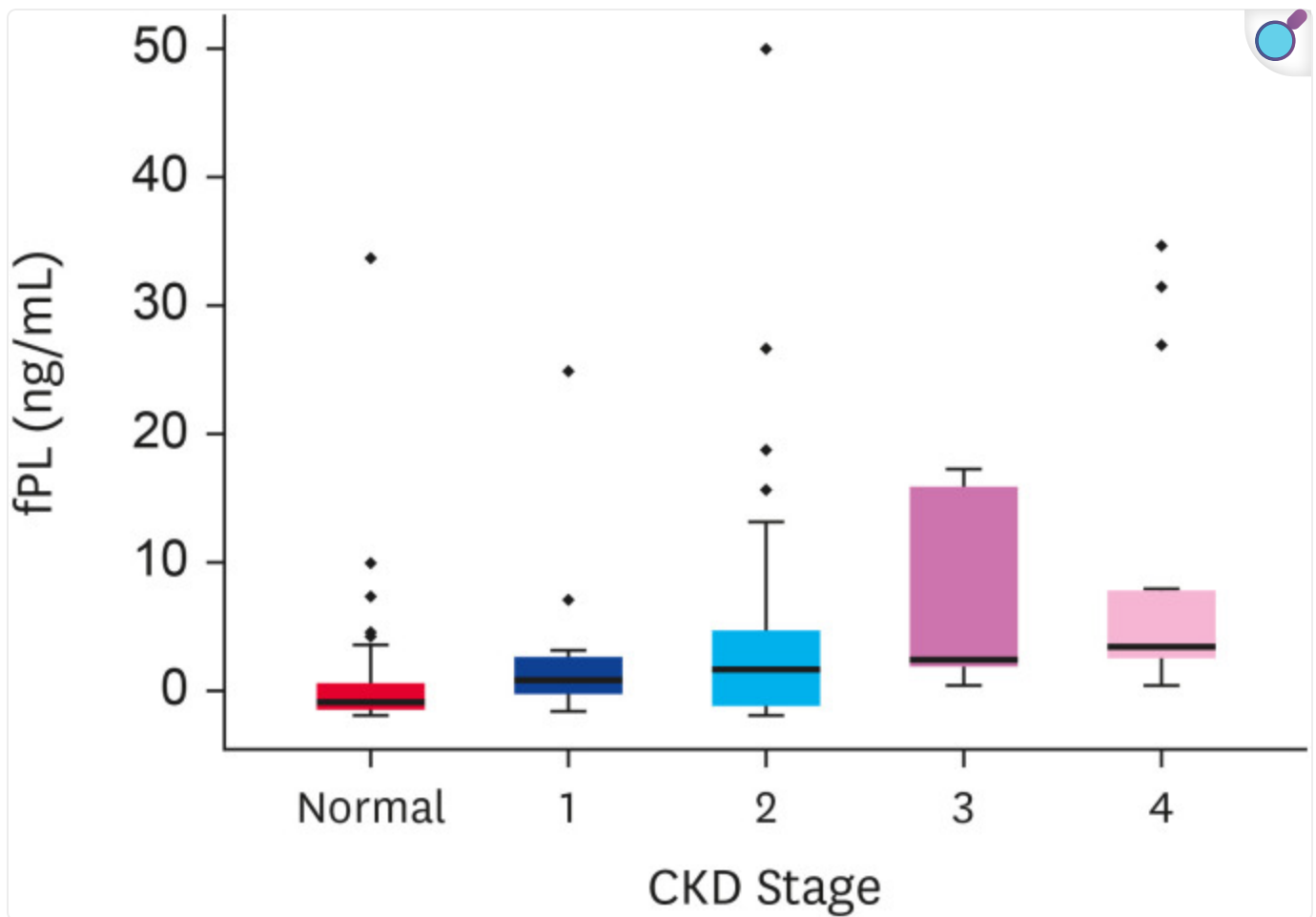
USG, urine specific gravity; fPL, feline pancreatic lipase.

Association between CKD stage and fPL

Serum fPL concentrations differed across CKD stages, with progressively higher values observed in cats with more advanced CKD ([Fig. 5](#)). Median fPL concentrations showed a progressive increase with advancing CKD stage: normal: 1.95 ng/mL (IQR, 1.4–3.325; 95% CI, 1.87–5.27); stage 1: 3.60

ng/mL (IQR, 2.6–5.3; 95% CI, 1.62–9.58); stage 2: 4.4 ng/mL (IQR, 1.7–7.2; 95% CI, 3.98–10.84); stage 3: 5.1 ng/mL (IQR, 4.1–17.8; 95% CI, 4.15–14.94); and stage 4: 6.05 ng/mL (IQR, 5.225–10.125; 95% CI, 5.40–18.35).

Fig. 5. Distribution of fPL concentrations across CKD stages in cats. Box-and-whisker plot depicting the distribution of fPL concentrations according to CKD IRIS stages in 112 cats. A progressive increase in fPL concentration is observed with advancing CKD stage. Markedly higher fPL concentrations are observed in cats with stages 3 and 4 of CKD.



[Open in a new tab](#)

fPL, feline pancreatic lipase; CKD, chronic kidney disease; IRIS, International Renal Interest Society.

The Kruskal–Wallis test revealed a significant difference in fPL concentrations across CKD stages ($H = 26.80$, $p < 0.001$). Post-hoc pairwise comparisons demonstrated that fPL concentrations were significantly higher in all CKD stages compared to the normal group (stage 1: $p = 0.04$; stage 2: $p = 0.04$; stage 3: $p < 0.01$; stage 4: $p < 0.01$). However, no significant differences were observed between adjacent CKD stages (stage 1 vs stage 2, $p = 0.95$; stage 2 vs stage 3, $p = 0.17$; stage 3 vs stage 4, $p = 0.48$). A significant difference was identified between early and advanced CKD, with stage 4 showing higher fPL concentrations than stage 1 ($p = 0.02$) and a borderline difference compared to stage 2 ($p = 0.05$).

DISCUSSION

This study assessed the association between serum fPL concentrations and CKD severity in cats, as well as their relationship with commonly used renal biomarkers. Serum fPL increased with worsening CKD stage and was associated with azotemia and reduced urine concentrating ability.

A progressive decline in USG was observed with advancing CKD stages, which were consistent with previous findings that associate CKD progression with impaired renal concentrating ability due to nephron loss [25,26]. However, no significant difference in USG was detected between CKD stages 3 and 4 ($p = 0.944$). This observation suggests that a critical threshold of nephron loss may have already been reached by stage 3, resulting in diminished concentrating capacity with minimal further decline as CKD advances to stage 4.

Serum fPL concentrations showed positive correlations with traditional renal biomarkers, including BUN, creatinine, and IP, all of which reflect impaired glomerular filtration and systemic azotemia. These associations suggest that declining renal function may influence measured fPL concentrations. The diagnosis of pancreatitis in cats remains inherently challenging, and an elevated fPL concentration alone is generally insufficient to establish clinically significant pancreatitis. Appropriate interpretation therefore requires integration of clinical signs and pancreatic ultrasonographic findings. Furthermore, pancreatic enzymes are known to increase when glomerular filtration is reduced, likely due to decreased renal clearance, which further complicates the interpretation of fPL results in cats with CKD.

A relevant experimental study in cats with induced CKD reported that serum feline trypsin-like immunoreactivity increased significantly with CKD, whereas serum fPL did not differ significantly from that in healthy controls and remained within reference intervals. This suggests that some pancreatic biomarkers may be variably influenced by renal dysfunction and highlights the need to interpret fPL results cautiously in the setting of azotemia [27].

In human medicine, the association between ESRD and AP is well-documented and attributed to multiple mechanisms. First, hypercalcemia is recognized as a leading cause of AP in advanced CKD, irrespective of dialysis status. Epidemiological data indicate that patients with CKD stage 5 who are not on dialysis and those undergoing dialysis are 4.59- and 1.65-fold, respectively, more likely to develop hypercalcemia-induced AP ($p < 0.01$) [19,20]. Second, the kidneys play a critical role in the degradation of gastrointestinal hormones such as cholecystokinin; impaired clearance can lead to persistent pancreatic stimulation and increased risk of AP. Third, intradialytic hypotension has been implicated in mesenteric ischemia, a known trigger for AP, with studies linking hypotension, high ultrafiltration rates, and elevated pancreatic enzyme concentrations [21]. Finally, peritoneal dialysis (PD) can increase the risk of pancreatitis through mechanisms including hypertriglyceridemia, local hypercalcemia, elevated intra-abdominal pressure, and PD-related peritonitis [22,23,28,29,30,31,32,33,34].

Taken together, our findings are most consistent with a passive (non-inflammatory) elevation of serum fPL concentrations associated with declining renal function. In the absence of standardized assessment of clinical signs and pancreatic imaging findings, elevations in fPL alone have inherent limitations for diagnosing pancreatitis in cats with CKD. In this context, our study provides a practical interpretive message for clinicians. Specifically, when mild-to-moderate increases in fPL are observed in cats with CKD, these findings should not be interpreted in isolation but rather evaluated in conjunction with compatible clinical signs and, when feasible, pancreatic-focused ultrasonography.

This study has some limitations. First, histopathological confirmation of pancreatic or renal lesions was not available, which precluded definitive diagnosis of the underlying disease. Second, pancreatic ultrasonographic findings and standardized clinical signs compatible with pancreatitis were not consistently documented, and therefore pancreatitis could not be defined using uniform diagnostic criteria. Third, hydration status at the time of presentation was not consistently or objectively quantified, and USG and some laboratory parameters may have been influenced by this factor. Finally, the retrospective cross-sectional design and inclusion of cats undergoing same-day testing may have introduced selection bias and limited assessment of temporal relationships.

In summary, serum fPL concentrations were associated with CKD severity and renal biomarkers in cats. These results indicate that reduced renal function may contribute to higher measured fPL concentrations and emphasize that fPL should not be used in isolation to diagnose clinically significant pancreatitis in cats with CKD.

Footnotes

Funding: This research was supported by the Regional Innovation System & Education (RISE) program through the Jeju RISE center, funded by the Ministry of Education (MOE) and the Jeju Special Self-Governing Province, Republic of Korea (2025-RISE-17-001).

Conflict of Interest: The authors declare no conflicts of interest.

Author Contributions:

Conceptualization: Kim J, Yun Y.

Data curation: Kim J.

Formal analysis: Kim J, Yun Y.

Funding acquisition: Yun Y.

Investigation: Kim J.

Methodology: Kim J, Yun Y.

Project administration: Kim J, Yun Y.

Resources: Kim J.

Software: Kim J.

Supervision: Yun Y.

Validation: Kim J.

Visualization: Kim J, Yun Y.

Writing - original draft: Kim J.

Writing - review & editing: Yun Y.

References

1. O'Neill DG, Church DB, McGreevy PD, Thomson PC, Brodbelt DC. Prevalence of disorders recorded in cats attending primary-care veterinary practices in England. *Vet J*. 2014;202(2):286–291. doi: 10.1016/j.tvjl.2014.08.004. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
2. White JD, Norris JM, Baral RM, Malik R. Naturally-occurring chronic renal disease in Australian cats: a prospective study of 184 cases. *Aust Vet J*. 2006;84(6):188–194. doi: 10.1111/j.1751-0813.2006.tb12796.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
3. Marino CL, Lascelles BDX, Vaden SL, Gruen ME, Marks SL. Prevalence and classification of chronic kidney disease in cats randomly selected from four age groups and in cats recruited

for degenerative joint disease studies. *J Feline Med Surg*. 2014;16(6):465–472. doi: 10.1177/1098612X13511446. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

4. Lulich JP, Osborne CA, O'Brien TD, Polzin DJ. Feline renal failure: questions, answers, questions. *Compend Contin Educ Pract Vet*. 1992;14:127–152. [[Google Scholar](#)]

5. O'Neill DG, Church DB, McGreevy PD, Thomson PC, Brodbelt DC. Longevity and mortality of cats attending primary care veterinary practices in England. *J Feline Med Surg*. 2015;17(2):125–133. doi: 10.1177/1098612X14536176. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

6. Bartlett PC, Van Buren JW, Bartlett AD, Zhou C. Case-control study of risk factors associated with feline and canine chronic kidney disease. *Vet Med Int*. 2010;2010:957570. doi: 10.4061/2010/957570. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

7. Greene JP, Lefebvre SL, Wang M, Yang M, Lund EM, Polzin DJ. Risk factors associated with the development of chronic kidney disease in cats evaluated at primary care veterinary hospitals. *J Am Vet Med Assoc*. 2014;244(3):320–327. doi: 10.2460/javma.244.3.320. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

8. Hughes KL, Slater MR, Geller S, Burkholder WJ, Fitzgerald C. Diet and lifestyle variables as risk factors for chronic renal failure in pet cats. *Prev Vet Med*. 2002;55(1):1–15. doi: 10.1016/s0167-5877(02)00088-0. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

9. Jepson RE, Brodbelt D, Vallance C, Syme HM, Elliott J. Evaluation of predictors of the development of azotemia in cats. *J Vet Intern Med*. 2009;23(4):806–813. doi: 10.1111/j.1939-1676.2009.0339.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

10. Sparkes AH, Caney S, Chalhoub S, Elliott J, Finch N, Gajanayake I, et al. ISFM consensus guidelines on the diagnosis and management of feline chronic kidney disease. *J Feline Med Surg*. 2016;18(3):219–239. doi: 10.1177/1098612X16631234. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

11. De Cock HEV, Forman MA, Farver TB, Marks SL. Prevalence and histopathologic characteristics of pancreatitis in cats. *Vet Pathol*. 2007;44(1):39–49. doi: 10.1354/vp.44-1-39. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

12. Oppliger S, Hartnack S, Reusch CE, Kook PH. Agreement of serum feline pancreas-specific lipase and colorimetric lipase assays with pancreatic ultrasonographic findings in cats with

suspicion of pancreatitis: 161 cases (2008–2012) J Am Vet Med Assoc. 2014;244(9):1060–1065. doi: 10.2460/javma.244.9.1060. [\[DOI ↗\]](#) [\[PubMed\]](#) [\[Google Scholar ↗\]](#)

13. Rinderknecht H. In: The Pancreas: Biology, Pathobiology and Disease. 2nd ed. Go VLW, editor. Raven Press; 1993. Pancreatic secretory enzymes; pp. 219–251. [\[Google Scholar ↗\]](#)

14. Schnauß F, Hanisch F, Burgener IA. Diagnosis of feline pancreatitis with SNAP fPL and Spec fPL. J Feline Med Surg. 2019;21(8):700–707. doi: 10.1177/1098612X18796624. [\[DOI ↗\]](#) [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar ↗\]](#)

15. Forman MA, Marks SL, De Cock HE, Hergesell EJ, Wisner ER, Baker TW, et al. Evaluation of serum feline pancreatic lipase immunoreactivity and helical computed tomography versus conventional testing for the diagnosis of feline pancreatitis. J Vet Intern Med. 2004;18(6):807–815. doi: 10.1892/0891-6640(2004)18<807:eosfpl>2.0.co;2. [\[DOI ↗\]](#) [\[PubMed\]](#) [\[Google Scholar ↗\]](#)

16. Baggenstoss AH. The pancreas in uremia; a histopathologic study. Am J Pathol. 1948;24(5):1003–1017. [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar ↗\]](#)

17. Avram MM. High prevalence of pancreatic disease in chronic renal failure. Nephron J. 1977;18(1):68–71. [\[Google Scholar ↗\]](#)

18. Araki T, Ueda M, Ogawa K, Tsuji T. Histological pancreatitis in end-stage renal disease. Int J Pancreatol. 1992;12(3):263–269. doi: 10.1007/BF02924366. [\[DOI ↗\]](#) [\[PubMed\]](#) [\[Google Scholar ↗\]](#)

19. Vaziri ND, Dure-Smith B, Miller R, Mirahmadi M. Pancreatic pathology in chronic dialysis patients--an autopsy study of 78 cases. Nephron J. 1987;46(4):347–349. [\[Google Scholar ↗\]](#)

20. Hou SW, Lee YK, Hsu CY, Lee CC, Su YC. Increased risk of acute pancreatitis in patients with chronic hemodialysis: a 4-year follow-up study. PLoS One. 2013;8(8):e71801. doi: 10.1371/journal.pone.0071801. [\[DOI ↗\]](#) [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar ↗\]](#)

21. Lankisch PG, Weber-Dany B, Maisonneuve P, Lowenfels AB. Frequency and severity of acute pancreatitis in chronic dialysis patients. Nephrol Dial Transplant. 2008;23(4):1401–1405. doi: 10.1093/ndt/gfm769. [\[DOI ↗\]](#) [\[PubMed\]](#) [\[Google Scholar ↗\]](#)

22. Chen HJ, Wang JJ, Tsay WI, Her SH, Lin CH, Chien CC. Epidemiology and outcome of acute pancreatitis in end-stage renal disease dialysis patients: a 10-year national cohort study. Nephrol Dial Transplant. 2017;32(10):1731–1736. doi: 10.1093/ndt/gfw400. [\[DOI ↗\]](#) [\[PubMed\]](#) [\[Google Scholar ↗\]](#)

23. Owyang C, Miller LJ, DiMagno EP, Mitchell JC, 3rd, Go VL. Pancreatic exocrine function in severe human chronic renal failure. *Gut*. 1982;23(5):357–361. doi: 10.1136/gut.23.5.357. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
24. Jaturanratsamee K, Choisunirachon N, Soontornvipart K, Darawiroj D, Srisowanna N, Thanaboonnipat C. Ultrasonographic kidney length-to-abdominal aortic diameter for the diagnosis of feline chronic kidney disease: a preliminary study. *Vet World*. 2023;16(5):1114–1121. doi: 10.14202/vetworld.2023.1114-1121. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
25. Elliott J, Barber PJ. Feline chronic renal failure: clinical findings in 80 cases diagnosed between 1992 and 1995. *J Small Anim Pract*. 1998;39(2):78–85. doi: 10.1111/j.1748-5827.1998.tb03598.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
26. Elliott J, Syme HM, Reubens E, Markwell PJ. Assessment of acid-base status of cats with naturally occurring chronic renal failure. *J Small Anim Pract*. 2003;44(2):65–70. doi: 10.1111/j.1748-5827.2003.tb00122.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
27. Xenoulis PG, Moraiti KTR, R Finco D, Suchodolski JS, Steiner JM. Serum feline pancreatic lipase immunoreactivity and trypsin-like immunoreactivity concentrations in cats with experimentally induced chronic kidney disease. *J Vet Intern Med*. 2021;35(6):2821–2827. doi: 10.1111/jvim.16296. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
28. Barbara M, Tsen A, Rosenkranz L. Acute pancreatitis in chronic dialysis patients. *Pancreas*. 2018;47(8):946–951. doi: 10.1097/MPA.0000000000001119. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
29. Sakorafas GH, Tsiotos GG, Sarr MG. Ischemia/reperfusion-induced pancreatitis. *Dig Surg*. 2000;17(1):3–14. doi: 10.1159/000018793. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
30. Chen YH, Yang WC, Wang FM, Tarng DC, Chen JY, Ng YY, et al. Risk factors associated with elevated serum pancreatic amylase levels during hemodialysis. *Hemodial Int*. 2011;15(1):79–86. doi: 10.1111/j.1542-4758.2010.00519.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
31. Owyang C, Miller LJ, DiMagno EP, Brennan LA, Jr, Go VL. Gastrointestinal hormone profile in renal insufficiency. *Mayo Clin Proc*. 1979;54(12):769–773. [[PubMed](#)] [[Google Scholar](#)]

32. Bruno MJ, van Westerloo DJ, van Dorp WT, Dekker W, Ferwerda J, Tytgat GN, et al. Acute pancreatitis in peritoneal dialysis and haemodialysis: risk, clinical course, outcome, and possible aetiology. *Gut*. 2000;46(3):385–389. doi: 10.1136/gut.46.3.385. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
33. Quraishi ER, Goel S, Gupta M, Catanzaro A, Zasuwa G, Divine G. Acute pancreatitis in patients on chronic peritoneal dialysis: an increased risk? *Am J Gastroenterol*. 2005;100(10):2288–2293. doi: 10.1111/j.1572-0241.2005.41646.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
34. Kroner PT, Mareth K, Raimondo M, Lee DD, Alsaad A, Aslam N, et al. Acute pancreatitis in advanced chronic kidney disease and kidney transplant recipients: results of a US nationwide analysis. *Mayo Clin Proc Innov Qual Outcomes*. 2019;3(2):160–168. doi: 10.1016/j.mayocpiqo.2019.03.006. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
-

Articles from Journal of Veterinary Science are provided here courtesy of **The Korean Society of Veterinary Science**